

## Project Data Page:





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## **DESIGN CALCULATION**

*In Accordance with ASME Section VIII Division 1*

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\403-E-004.pvdb

Date of Analysis : Jul 4, 2026 12:39pm

**PV Elite 28, January 2026**

### Vessel Design Summary:

#### ASME Code, Section VIII Division 1, 2025

Diameter Spec : 490.000 mm. ID  
 Vessel Design Length, Tangent to Tangent 6803.30 mm.  
 Specified Datum Line Distance 50.00 mm.  
 Shell Side Design Temperature 127 °C  
 Channel Side Design Temperature 77 °C  
 Shell Side Design Pressure 25.511 bars  
 Channel Side Design Pressure 5.516 bars  
 Shop Shell Side Test Pressure 36.361 bars  
 Shop Channel Side Test Pressure 36.361 bars  
 Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)  
 Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

### Materials of Construction:

| Component Type | Material  | Class | Thickness | UNS #  | Material Treatment Options |
|----------------|-----------|-------|-----------|--------|----------------------------|
| Shell          | SA-516 70 | ...   | ...       | K02700 |                            |
| Head           | SA-516 70 | ...   | ...       | K02700 |                            |
| Flange         | SA-266 4  | ...   | ...       | K03017 |                            |
| Nozzle         | SA-106 B  | ...   | ...       | K03006 |                            |
| Re-Pad         | SA-516 70 | ...   | ...       | K02700 |                            |
| Nozzle Flg     | SA-105    | ...   | ...       | K03504 |                            |
| Tubes          | SA-179    | ...   | ...       | K01200 |                            |
| Tubesheet      | SA-266 4  | ...   | ...       | K03017 |                            |
| Flg Bolting    | SA-193 B7 | ...   | <= 64     | G41400 |                            |
| Hrz Bolting    | SA-193 B7 | ...   | <= 64     | G41400 |                            |
| Saddles        | SA-516 70 | ...   | ...       | K02700 |                            |

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized  
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

### Element Pressures and MAWP (bars & mm.):

| Element Description or Type | Design Pressure + Stat. head | Ext. Press. | Element M.A.W.P | Total Corrosion Allowance | Str. Flg. Gov. | In Creep Range |
|-----------------------------|------------------------------|-------------|-----------------|---------------------------|----------------|----------------|
| Fr. END COVER               | 5.564                        | 0.00        | 50.000          | 3.1750                    | No             | No             |
| CH.SHELL                    | 5.564                        | 0.00        | 48.000          | 3.1750                    | N/A            | No             |
| CH FLG                      | 5.564                        | 0.00        | 29.000          | 3.1750                    | N/A            | No             |
| Shell FLG                   | 25.559                       | 0.00        | 29.000          | 3.1750                    | N/A            | No             |
| Main shell                  | 25.559                       | 0.00        | 48.000          | 3.1750                    | N/A            | No             |
| Rear End Cover              | 25.559                       | 0.00        | 50.000          | 3.1750                    | No             | No             |

Liquid Level: 490.00 mm. Dens.: 0.001 kg./cm<sup>3</sup> Sp. Gr.: 1.000

### Element Types and Properties:

| Element Type | "To" Elev mm. | Element Length mm. | Nominal Thickness mm. | Finished Thickness mm. | Reqd Thk Internal mm. | Reqd Thk External mm. | Long Eff | Circ Eff |
|--------------|---------------|--------------------|-----------------------|------------------------|-----------------------|-----------------------|----------|----------|
| Ellipse      | 0.0           | 50.0               | 14.0                  | 12.0                   | 5.7                   | ...                   | 1.00     | 1.00     |
| Cylinder     | 400.0         | 400.0              | 12.0                  | 12.0                   | 5.7                   | ...                   | 1.00     | 1.00     |
| Body Flg     | 479.5         | 75.0               | 65.0                  | 60.0                   | 53.6                  | 53.6                  | 1.00     | 1.00     |
| Body Flg     | 606.5         | 75.0               | 65.0                  | 60.0                   | 53.6                  | 53.6                  | 1.00     | 1.00     |
| Cylinder     | 6702.5        | 6096.0             | 12.0                  | 12.0                   | 7.8                   | ...                   | 1.00     | 1.00     |



|         |        |      |      |      |     |     |      |      |
|---------|--------|------|------|------|-----|-----|------|------|
| Ellipse | 6753.3 | 50.8 | 14.0 | 12.0 | 7.7 | ... | 1.00 | 1.00 |
|---------|--------|------|------|------|-----|-----|------|------|

#### Local Stress Analysis Results:

| Description   | Analysis Type | Max Stress Ratio | High Stress Location | Pass Fail |
|---------------|---------------|------------------|----------------------|-----------|
| TS IN NOZ T1  | WRC-297       | 0.616            | Nozzle               | Passed    |
| TS OUT NOZ T2 | WRC-297       | 0.616            | Nozzle               | Passed    |
| SH IN NOZ S1  | WRC-297       | 0.988            | Shell                | Passed    |
| SH OUT NOZ S2 | WRC-297       | 0.988            | Shell                | Passed    |

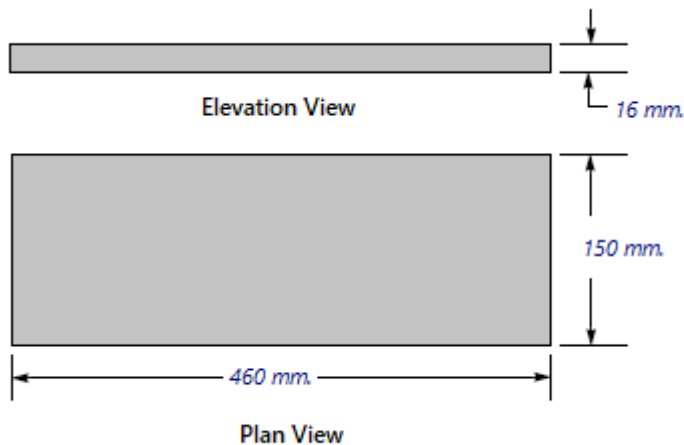
#### Foundation/Support Summary

##### Saddle Parameters:

|                                 |              |
|---------------------------------|--------------|
| Saddle Width                    | 150.000 mm.  |
| Saddle Bearing Angle            | 120.000 deg. |
| Centerline Dimension            | 500.000 mm.  |
| Wear Pad Width                  | 225.000 mm.  |
| Wear Pad Thickness              | 12.000 mm.   |
| Wear Pad Bearing Angle          | 132.000 deg. |
| Distance from Saddle to Tangent | 1220.000 mm. |

|                                         |             |
|-----------------------------------------|-------------|
| Baseplate Length                        | 460.000 mm. |
| Baseplate Thickness                     | 16.000 mm.  |
| Baseplate Width                         | 150.000 mm. |
| Number of Ribs (including outside ribs) | 2           |
| Rib Thickness                           | 12.000 mm.  |
| Web Thickness                           | 12.000 mm.  |
| Height of Center Web                    | 306.000 mm. |
| Number of Bolts in Baseplate            | 2           |

#### Baseplate Sketch



Maximum Tensile Bolt Load 87. Kg

#### Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

|                                        |            |
|----------------------------------------|------------|
| Maximum Vertical Saddle Load           | 4187.60 Kg |
| Maximum Transverse Saddle Shear Load   | 507.45 Kg  |
| Maximum Longitudinal Saddle Shear Load | 359.24 Kg  |

#### Summary of Maximum Saddle Loads, Pressure Test Case:

|                                        |            |
|----------------------------------------|------------|
| Maximum Vertical Saddle Load           | 3133.51 Kg |
| Maximum Transverse Saddle Shear Load   | 167.46 Kg  |
| Maximum Longitudinal Saddle Shear Load | 21.95 Kg   |

#### Weights:

|                                           |            |
|-------------------------------------------|------------|
| Fabricated - Bare w/o removable internals | 2985.5 kg. |
| Shop Test - Fabricated + water ( full )   | 4151.3 kg. |



|           |                                     |        |     |
|-----------|-------------------------------------|--------|-----|
| Shipping  | - Fab. + rem. intls.+ shipping app. | 3062.3 | kg. |
| Erected   | - Fab. + rem. intls.+ insul. (etc)  | 3062.3 | kg. |
| Empty     | - Fab. + intls. + details + wghts.  | 3062.3 | kg. |
| Operating | - Empty + operating liquid (no ca)  | 4227.3 | kg. |

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### Nozzle Calculation Summary (bars & mm.):

| Description   | MAWP | Ext | MAPNC | UG-45 | [tr] | Weld Path | Areas or Stresses |
|---------------|------|-----|-------|-------|------|-----------|-------------------|
| TS IN NOZ T1  | 48   | ... | ...   | OK    | 5.67 | OK        | Passed            |
| TS OUT NOZ T2 | 48   | ... | ...   | OK    | 5.67 | OK        | Passed            |
| SH IN NOZ S1  | 48   | ... | ...   | OK    | 7.81 | OK        | Passed            |
| SH OUT NOZ S2 | 48   | ... | ...   | OK    | 7.82 | OK        | Passed            |

### Nozzle MAWP Summary:

Minimum MAWP Nozzles : 48 Nozzle : SH IN NOZ S1 [Shellside]  
 Minimum MAWP Nozzles : 48 Nozzle : TS IN NOZ T1 [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

### Check the spatial relationship between the nozzles:

| From Node | Nozzle Description | X Coordinate mm. | Layout Angle deg | Dia. Limit mm. |
|-----------|--------------------|------------------|------------------|----------------|
| 20        | TS IN NOZ T1       | 250.000          | -90.000          | 305.359        |
| 20        | TS OUT NOZ T2      | 250.000          | 90.000           | 305.359        |
| 50        | SH IN NOZ S1       | 6502.499         | 90.000           | 400.050        |
| 50        | SH OUT NOZ S2      | 906.500          | -90.000          | 400.050        |

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.  
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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### Nozzle Schedule:

| Description   | Nominal or Actual Size | Schd or FVC Type | Flg Type | Nozzle O/Dia in | Wall Thk mm | Reinforcing Pad Diameter | Pad Thk mm | Cut Length mm | Flg Class |
|---------------|------------------------|------------------|----------|-----------------|-------------|--------------------------|------------|---------------|-----------|
| TS IN NOZ T1  | 6.000 in               | 80               | WNF      | 6.625           | 10.973      | 268.27                   | 12.00      | 176.90        | 300       |
| TS OUT NOZ T2 | 6.000 in               | 80               | WNF      | 6.625           | 10.973      | 268.27                   | 12.00      | 176.90        | 300       |
| SH IN NOZ S1  | 8.000 in               | 80               | WNF      | 8.625           | 12.700      | ...                      | ...        | 218.78        | 300       |
| SH OUT NOZ S2 | 8.000 in               | 80               | WNF      | 8.625           | 12.700      | ...                      | ...        | 218.78        | 300       |

### General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

### Nozzle Material and Weld Fillet Leg Size Details (mm.):

| Description | Material | Shl Grve Weld | Noz Shl/Pad Weld | Pad OD Weld | Pad Grve Weld | Inside Weld |
|-------------|----------|---------------|------------------|-------------|---------------|-------------|
| TS IN NOZ T | SA-106 B | ...           | 10.000           | 10.000      | 12.000        | ...         |
| TS OUT NOZ  | SA-106 B | ...           | 10.000           | 10.000      | 12.000        | ...         |
| SH IN NOZ S | SA-106 B | 12.000        | 10.000           | ...         | ...           | ...         |
| SH OUT NOZ  | SA-106 B | 12.000        | 10.000           | ...         | ...           | ...         |

Note: The Outside projections below do not include the flange thickness.

### Nozzle Miscellaneous Data:

| Description   | Elev/Distance From Datum mm. | Layout Angle deg | Proj Outside mm. | Proj Inside mm. | Installed in Component |
|---------------|------------------------------|------------------|------------------|-----------------|------------------------|
| TS IN NOZ T1  | 200.000                      | -90.0            | 150.00           | 0.00            | CH.SHELL               |
| TS OUT NOZ T2 | 200.000                      | 90.0             | 150.00           | 0.00            | CH.SHELL               |
| SH IN NOZ S1  | 6452.500                     | 90.0             | 150.00           | 0.00            | Main shell             |
| SH OUT NOZ S2 | 856.499                      | -90.0            | 150.00           | 0.00            | Main shell             |

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**Bill of Materials:**

| QTY | DESCRIPTION                                                   | MATERIAL  |
|-----|---------------------------------------------------------------|-----------|
| 1   | ELLIPTICAL HEAD: 2.0 X 1, 14.0mm. THK X 490.0mm. ID X 50.0mm. | SA-516 70 |
| 1   | CYLINDER: 12.0mm. THK X 490.0mm. ID X 400.0mm.                | SA-516 70 |
| 2   | WELD NECK FLANGE: 65.0mm. THK X 622.0mm. OD                   | SA-266 4  |
| 1   | CYLINDER: 12.0mm. THK X 490.0mm. ID X 6096.0mm.               | SA-516 70 |
| 1   | ELLIPTICAL HEAD: 2.0 X 1, 14.0mm. THK X 490.0mm. ID X 50.8mm. | SA-516 70 |
| 14  | BAFFLES: 484.0mm DIA X 6.0mm THK                              | SA-516 70 |
| 1   | TUBESHEET: 622.0mm X 52.0mm THK                               | SA-266 4  |
| 160 | TUBES: 6093.0mm X 19.0mm DIA X 2.8mm THK                      | SA-179    |
| 1   | INSULATION: 75mm X 45mm THK                                   | HOT       |
| 2   | SADDLE: 150.0mm X 120 DEG                                     | SA-516 70 |
| 1   | INSULATION: 6096mm X 45mm THK                                 | HOT       |
| 1   | INSULATION: 173mm X 45mm THK                                  | HOT       |
| 1   | GASKET: 518.0mm. OD X 498.0mm. ID                             | ...       |
| 24  | BODY FLANGE BOLTS: 19.0mm. DIA                                | SA-193 B7 |
| 48  | NUTS FOR BODY FLANGE BOLTS: 19.0mm. DIA                       | ...       |
| 1   | NAMEPLATE                                                     | ...       |



### Minimum Design Metal Temperature Results Summary (°C):

| Description       | Notes | Curve         | Basic MDMT | Reduced MDMT  | UG-20(f) MDMT | Thickness ratio | Gov Thk mm. | E*   | PWHT reqd |
|-------------------|-------|---------------|------------|---------------|---------------|-----------------|-------------|------|-----------|
| Shell FLG [11]    |       | B             | -23        | -47           | -29           | 0.582           | 12.000      | 1.00 | No        |
| Main shell [8]    |       | B             | -23        | -47           | -29           | 0.577           | 12.000      | 1.00 | No        |
| Rear End Cove[10] |       | B             | -23        | -48           | -29           | 0.562           | 12.000      | 1.00 | No        |
| Rear End Cover[7] |       | B             | -18        | -48           | -29           | 0.470           | 14.000      | 1.00 | No        |
| Left Saddle [24]  |       | B             | -23        | -47           | -29           | 0.577           | 12.000      | 1.00 | No        |
| RIGHT SADDLE [24] |       | B             | -23        | -47           | -29           | 0.577           | 12.000      | 1.00 | No        |
| SH IN NOZ S1 [1]  |       | B             | -23        | -48           | -29           | 0.526           | 12.000      | 1.00 | No        |
| Nozzle Flg [5]    |       | A             | -18        | -48           |               |                 |             |      |           |
| SH OUT NOZ S2 [1] |       | B             | -23        | -48           | -29           | 0.527           | 12.000      | 1.00 | No        |
| Nozzle Flg [5]    |       | A             | -18        | -48           |               |                 |             |      |           |
| Tubesheet: SS[13] |       | B             | -21        | -27           | -29           | 0.892           | 13.000      | 1.00 | No        |
| Warmest MDMT:     |       |               | -18        | -27           |               |                 |             |      |           |
| CH FLG [11]       |       | B             | -23        | -47           | -29           | 0.582           | 12.000      | 1.00 | No        |
| Fr. END COVER[10] |       | B             | -23        | -48           | -29           | 0.562           | 12.000      | 1.00 | No        |
| Fr. END COVER [7] |       | B             | -18        | -48           | -29           | 0.470           | 14.000      | 1.00 | No        |
| CH.SHELL [8]      |       | B             | -23        | -47           | -29           | 0.577           | 12.000      | 1.00 | No        |
| TS IN NOZ T1 [1]  |       | B             | -8         | -104          | -29           | 0.056           | 9.601       | 1.00 | No        |
| Nozzle Flg [5]    |       | A             | -18        | -104          |               |                 |             |      |           |
| TS OUT NOZ T2 [1] |       | B             | -8         | -104          | -29           | 0.056           | 9.601       | 1.00 | No        |
| Nozzle Flg [5]    |       | A             | -18        | -104          |               |                 |             |      |           |
| Tubesheet: CS[14] |       | B             | -21        | -27           | -29           | 0.892           | 13.000      | 1.00 | No        |
| Bolting [21]      |       |               | -48        |               |               |                 |             |      |           |
| Warmest MDMT:     |       |               | -8         | -27           |               |                 |             |      |           |
|                   |       |               |            |               |               |                 |             |      |           |
| Exchanger Side    |       | Computed MDMT |            | Required MDMT |               | Pass/Fail       |             |      |           |
| Shell             |       | -29.0         |            | -28.9         |               | Pass            |             |      |           |
| Channel/Tube      |       | -29.0         |            | -28.9         |               | Pass            |             |      |           |

### Notes:

- [ ! ] - This was an impact tested material.
- [ 1 ] - Governing Nozzle Weld.
- [ 4 ] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [ 5 ] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [ 6 ] - MDMT Calculations at the Shell/Head Joint.
- [ 7 ] - MDMT Calculations for the Straight Flange.
- [ 8 ] - Cylinder/Cone/Flange Junction MDMT.
- [ 9 ] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.  
 UCS-66(g) was not considered.  
 UCS-66(i) was not considered.

### Notes:

Impact test temps were not entered in and not considered in the analysis.



UCS-66(i) applies to impact tested materials not by specification and  
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).  
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

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Class From To: Check of Additional Element Data

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**Warning: CH FLG and Shell FLG**

*There are mating flanges in this model that have different design pressures. This will generate different bolt loads. If a tubesheet is in between these flanges, PV Elite will compute the mating flange loads for you. If there is not a tubesheet physically separating these flanges, you must enter in the mating flange loads into the flange dialog.*

**Warning:**

*There is a flange in this model connected to an element whose temperature is different. The bolt allowable stress is based on it's parent design temperature. You may need to manually adjust the operating bolt allowable stress to a value based upon a representative maximum temperature the bolts will be subject to.*

**PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.**

**Information Regarding Nozzle Loads**

*Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.*



*The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.*



*The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.*

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**Units used in this Analysis (Kgfcmm2mm):**

| Name           | System Unit | Constant   | User Unit           |
|----------------|-------------|------------|---------------------|
| Length         | Feet        | 304.8000   | mm.                 |
| Force          | Pounds      | 0.4536     | Kgf                 |
| Mass           | Pounds      | 0.4536     | kg.                 |
| Area           | sq. inches  | 6.4516     | cm <sup>2</sup>     |
| Moment         | ft. lbs.    | 0.1383     | Kg-m.               |
| Stress         | lbs./sq.in. | 0.0069     | N./mm <sup>2</sup>  |
| Temperature    | Degrees F   | 0.5556     | °C                  |
| Pressure       | psig        | 0.0689     | bars                |
| Elast. Modulus | lbs./sq.in. | 6.8946     | KPa.                |
| Pipe Density   | lbs./cu.in. | 0.0277     | kg./cm <sup>3</sup> |
| Ins. Density   | lbs./cu.ft. | 0.1600E-04 | kg./cm <sup>3</sup> |
| Fluid Density  | lbs./cu.ft. | 0.1600E-04 | kg./cm <sup>3</sup> |
| Wind Speed     | miles/hr    | 1.6093     | Km/hr               |
| Tray Weight    | lbs./sq.ft. | 0.0005     | Kgs/cm <sup>2</sup> |
| Inertia        | in.4        | 41.6231    | cm4                 |
| G Load         | G's         | 1.0000     | g's                 |
| Wind Load      | lbs./sq.ft. | 4.8825     | Kgs/m <sup>2</sup>  |
| Elevation      | Feet        | 304.8000   | mm.                 |
| Volume         | in.3        | 16.3900    | cm <sup>3</sup>     |
| Diameter       | inches      | 25.4000    | mm.                 |
| Thickness      | inches      | 25.4000    | mm.                 |

**PV Elite Vessel Analysis Program: Input Data**

**Exchanger Design Pressures and Temperatures**

|                                       |                 |      |
|---------------------------------------|-----------------|------|
| Shell Side Design Pressure            | 25.511          | bars |
| Channel Side Design Pressure          | 5.516           | bars |
| Shell Side Design Temperature         | 126.7           | °C   |
| Channel Side Design Temperature       | 76.7            | °C   |
| Radiography, Shell Side               | RT-1            |      |
| Radiography, Channel Side             | RT-1            |      |
| Service Type, Shell Side              | Sour            |      |
| Service Type, Channel Side            | Air/Water/Steam |      |
| MDMT (CET), Shell Side                | -28.9           | °C   |
| MDMT (CET), Tube Side                 | -28.9           | °C   |
| User defined MAWP, Shell Side         | 0               | bars |
| User defined MAWP, Channel Side       | 0               | bars |
| User defined MAPnc, Shell Side        | 0               | bars |
| User defined MAPnc, Channel Side      | 0               | bars |
| User defined Test Pres., Shell Side   | 0               | bars |
| User defined Test Pres., Channel Side | 0               | bars |

|                                         |        |     |
|-----------------------------------------|--------|-----|
| Projection of Nozzle from Vessel Top    | 0      | mm. |
| Projection of Nozzle from Vessel Bottom | 0      | mm. |
| Type of Construction                    | Welded |     |
| Use Higher Longitudinal Stresses (Flag) | Y      |     |
| Select t for Internal Pressure (Flag)   | N      |     |
| Select t for External Pressure (Flag)   | N      |     |
| Select t for Axial Stress (Flag)        | N      |     |
| Select Location for Stiff. Rings (Flag) | N      |     |

User Spectrum Data points specified for increments of Time Period

| Time Period | Sa/g  |
|-------------|-------|
| 0.010       | 0.070 |
| 0.050       | 0.117 |
| 0.100       | 0.175 |
| 0.130       | 0.175 |
| 0.200       | 0.175 |
| 0.270       | 0.175 |

|       |       |
|-------|-------|
| 0.300 | 0.175 |
| 0.400 | 0.175 |
| 0.500 | 0.140 |
| 0.550 | 0.127 |
| 0.600 | 0.117 |
| 0.670 | 0.105 |
| 0.800 | 0.087 |
| 1.000 | 0.070 |
| 1.200 | 0.058 |
| 1.400 | 0.050 |
| 1.600 | 0.044 |
| 1.800 | 0.039 |
| 2.000 | 0.035 |
| 2.400 | 0.029 |
| 0.000 | 0.000 |

Consider Vortex Shedding N

**Shop Pressure Test:**

|                                      |            |
|--------------------------------------|------------|
| Type of Pressure Test                | UG-99(b)   |
| Pressure Test Position               | Horizontal |
| Test Performed in Corroded Condition | Yes        |

|              |                |
|--------------|----------------|
| Load Case 1  | NP+EW+WI+FW+BW |
| Load Case 2  | NP+EW+EE+FS+BS |
| Load Case 3  | NP+OW+WI+FW+BW |
| Load Case 4  | NP+OW+EQ+FS+BS |
| Load Case 5  | NP+HW+HI       |
| Load Case 6  | NP+HW+HE       |
| Load Case 7  | IP+OW+WI+FW+BW |
| Load Case 8  | IP+OW+EQ+FS+BS |
| Load Case 9  | EP+OW+WI+FW+BW |
| Load Case 10 | EP+OW+EQ+FS+BS |
| Load Case 11 | HP+HW+HI       |
| Load Case 12 | HP+HW+HE       |
| Load Case 13 | IP+WE+EW       |
| Load Case 14 | IP+WF+CW       |
| Load Case 15 | IP+VO+OW       |
| Load Case 16 | IP+VE+EW       |
| Load Case 17 | NP+VO+OW       |
| Load Case 18 | FS+BS+IP+OW    |
| Load Case 19 | FS+BS+EP+OW    |
| Load Case 20 | BL+IP+OW       |

|                                         |                           |       |
|-----------------------------------------|---------------------------|-------|
| Wind Design Code                        | IS 875 (PART 3):2015-2020 |       |
| Basic Wind Speed for IS-875             | 180                       | Km/hr |
| Wind Zone Number                        | 2                         |       |
| Base Elevation                          | 300                       | mm.   |
| Percent Wind for Hydrotest              | 33.0                      |       |
| Risk Factor                             | [k1] 1.0                  |       |
| Terrain Category                        | 2                         |       |
| Equipment Class                         | Class A                   |       |
| Topography Factor                       | [k3] 1.0                  |       |
| Importance Factor                       | [k4] 1.15                 |       |
| Damping Factor (Beta) for Wind (Ope)    | 0.0100                    |       |
| Damping Factor (Beta) for Wind (Empty)  | 0.0000                    |       |
| Damping Factor (Beta) for Wind (Filled) | 0.0000                    |       |

|                                          |             |
|------------------------------------------|-------------|
| Seismic Design Code                      | IS 1893-RSM |
| Importance Factor for IS-1893            | 2.000       |
| Soil Factor                              | 1.000       |
| Zone Number                              | 2.000       |
| Percent Seismic for Hydrotest            | 0.000       |
| Period of Vibration ( User defined ) Sec | 0.000       |
| Damping Factor                           | 2.000       |

User Spectrum Data points specified for increments of Time Period

| Time Period | Sa/g |
|-------------|------|
|-------------|------|

|       |       |
|-------|-------|
| 0.010 | 0.070 |
| 0.050 | 0.117 |
| 0.100 | 0.175 |
| 0.130 | 0.175 |
| 0.200 | 0.175 |
| 0.270 | 0.175 |
| 0.300 | 0.175 |
| 0.400 | 0.175 |
| 0.500 | 0.140 |
| 0.550 | 0.127 |
| 0.600 | 0.117 |
| 0.670 | 0.105 |
| 0.800 | 0.087 |
| 1.000 | 0.070 |
| 1.200 | 0.058 |
| 1.400 | 0.050 |
| 1.600 | 0.044 |
| 1.800 | 0.039 |
| 2.000 | 0.035 |
| 2.400 | 0.029 |
| 0.000 | 0.000 |

Consider Vertical Acceleration Yes

Design Pressure + Static Head Y

Consider MAP New and Cold in Noz. Design N

Consider External Loads for Nozzle Des. Y

Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No

Material Database Year Current w/Addenda or Code Year

#### Configuration Directives:

|                                                          |     |
|----------------------------------------------------------|-----|
| g0 is taken as min( g0, attached shell thk )             | Yes |
| Do not use Nozzle MDMT Interpretation VIII-1 01-37       | No  |
| Use Table G instead of exact equation for "A"            | Yes |
| Shell Head Joints are Tapered                            | Yes |
| Compute "K" in corroded condition                        | Yes |
| Use Code Case 2286                                       | No  |
| Use the MAWP to compute the MDMT                         | Yes |
| For thickness ratios <= 0.35, MDMT will be -155F (-104C) | Yes |
| For PWHT & P1 Materials the MDMT can be < -55F (-48C)    | No  |

|                                                    |     |
|----------------------------------------------------|-----|
| Using Metric Material Databases, ASME II D         | Yes |
| Calculate B31.3 type stress for Nozzles with Loads | Yes |
| Reduce the MDMT due to lower membrane stress       | Yes |
| Consider Longitudinal Stress in MDMT Calculations  | Yes |

#### Complete Listing of Vessel Elements and Details:

|                                      |                        |
|--------------------------------------|------------------------|
| Element#                             | 1/6                    |
| Element From Node                    | 10                     |
| Element To Node                      | 20                     |
| Element Type                         | Elliptical             |
| Description                          | Fr. END COVER          |
| Distance "FROM" to "TO"              | 50 mm.                 |
| Inside Diameter                      | 490 mm.                |
| Element Thickness                    | 12 mm.                 |
| Internal Corrosion Allowance         | 3.175 mm.              |
| Nominal Thickness                    | 14 mm.                 |
| External Corrosion Allowance         | 0 mm.                  |
| Design Internal Pressure             | 5.516 bars             |
| Design Temperature Internal Pressure | 77 °C                  |
| Design External Pressure             | 0 bars                 |
| Design Temperature External Pressure | 77 °C                  |
| Effective Diameter Multiplier        | 1.2                    |
| Material Name                        | SA-516 70              |
| Allowable Stress, Ambient            | 138 N./mm <sup>2</sup> |
| Allowable Stress, Operating          | 138 N./mm <sup>2</sup> |



|                                  |         |                     |
|----------------------------------|---------|---------------------|
| Allowable Stress, Hydrotest      | 179.4   | N./mm <sup>2</sup>  |
| Material Density                 | 0.00775 | kg./cm <sup>3</sup> |
| P Number Thickness               | 32.004  | mm.                 |
| Yield Stress, Operating          | 243.68  | N./mm <sup>2</sup>  |
| UCS-66 Chart Curve Designation   | B       |                     |
| External Pressure Chart Name     | CS-2    |                     |
| UNS Number                       | K02700  |                     |
| Product Form                     | Plate   |                     |
| Efficiency, Longitudinal Seam    | 1.0     |                     |
| Efficiency, Circumferential Seam | 1.0     |                     |
| Elliptical Head Factor           | 2.0     |                     |
| Weld is pre-Heated               | No      |                     |

|                                      |           |                     |
|--------------------------------------|-----------|---------------------|
| Element From Node                    | 10        |                     |
| Detail Type                          | Liquid    |                     |
| Detail ID                            | Liquid 10 |                     |
| Dist. from "FROM" Node / Offset dist | 0         | mm.                 |
| Height/Length of Liquid              | 490       | mm.                 |
| Liquid Density                       | 0.0009984 | kg./cm <sup>3</sup> |

---

|                                      |           |      |
|--------------------------------------|-----------|------|
| Element#                             | 2/6       |      |
| Element From Node                    | 20        |      |
| Element To Node                      | 30        |      |
| Element Type                         | Cylinder  |      |
| Description                          | CH.SHELL  |      |
| Distance "FROM" to "TO"              | 400       | mm.  |
| Inside Diameter                      | 490       | mm.  |
| Element Thickness                    | 12        | mm.  |
| Internal Corrosion Allowance         | 3.175     | mm.  |
| Nominal Thickness                    | 12        | mm.  |
| External Corrosion Allowance         | 0         | mm.  |
| Design Internal Pressure             | 5.516     | bars |
| Design Temperature Internal Pressure | 77        | °C   |
| Design External Pressure             | 0         | bars |
| Design Temperature External Pressure | 77        | °C   |
| Effective Diameter Multiplier        | 1.2       |      |
| Material Name                        | SA-516 70 |      |
| Efficiency, Longitudinal Seam        | 1.0       |      |
| Efficiency, Circumferential Seam     | 1.0       |      |
| Weld is pre-Heated                   | No        |      |

|                                      |           |                     |
|--------------------------------------|-----------|---------------------|
| Element From Node                    | 20        |                     |
| Detail Type                          | Liquid    |                     |
| Detail ID                            | Liquid 20 |                     |
| Dist. from "FROM" Node / Offset dist | 0         | mm.                 |
| Height/Length of Liquid              | 490       | mm.                 |
| Liquid Density                       | 0.0009984 | kg./cm <sup>3</sup> |

|                                      |              |     |
|--------------------------------------|--------------|-----|
| Element From Node                    | 20           |     |
| Detail Type                          | Nozzle       |     |
| Detail ID                            | TS IN NOZ T1 |     |
| Dist. from "FROM" Node / Offset dist | 200          | mm. |
| Nozzle Diameter                      | 6            | in. |
| Nozzle Schedule                      | 80           |     |
| Nozzle Class                         | 300          |     |
| Layout Angle                         | -90.0        |     |
| Blind Flange (Y/N)                   | N            |     |
| Weight of Nozzle ( Used if > 0 )     | 31.17        | Kgf |
| Grade of Attached Flange             | GR 1.1       |     |
| Nozzle Matl                          | SA-106 B     |     |

|                                      |               |     |
|--------------------------------------|---------------|-----|
| Element From Node                    | 20            |     |
| Detail Type                          | Nozzle        |     |
| Detail ID                            | TS OUT NOZ T2 |     |
| Dist. from "FROM" Node / Offset dist | 200           | mm. |
| Nozzle Diameter                      | 6             | in. |
| Nozzle Schedule                      | 80            |     |
| Nozzle Class                         | 300           |     |



|                                  |          |     |
|----------------------------------|----------|-----|
| Layout Angle                     | 90.0     |     |
| Blind Flange (Y/N)               | N        |     |
| Weight of Nozzle ( Used if > 0 ) | 31.17    | Kgf |
| Grade of Attached Flange         | GR 1.1   |     |
| Nozzle Matl                      | SA-106 B |     |

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|                                         |          |                     |
|-----------------------------------------|----------|---------------------|
| Element#                                | 3/6      |                     |
| Element From Node                       | 30       |                     |
| Element To Node                         | 40       |                     |
| Element Type                            | Flange   |                     |
| Description                             | CH FLG   |                     |
| Distance "FROM" to "TO"                 | 75       | mm.                 |
| Flange Inside Diameter                  | 490      | mm.                 |
| Element Thickness                       | 60       | mm.                 |
| Internal Corrosion Allowance            | 3.175    | mm.                 |
| Nominal Thickness                       | 65       | mm.                 |
| External Corrosion Allowance            | 0        | mm.                 |
| Design Internal Pressure                | 5.516    | bars                |
| Design Temperature Internal Pressure    | 77       | °C                  |
| Design External Pressure                | 0        | bars                |
| Design Temperature External Pressure    | 77       | °C                  |
| Effective Diameter Multiplier           | 1.2      |                     |
| Material Name                           | SA-266 4 |                     |
| Allowable Stress, Ambient               | 138      | N./mm <sup>2</sup>  |
| Allowable Stress, Operating             | 138      | N./mm <sup>2</sup>  |
| Allowable Stress, Hydrotest             | 179.4    | N./mm <sup>2</sup>  |
| Material Density                        | 0.00775  | kg./cm <sup>3</sup> |
| P Number Thickness                      | 32.004   | mm.                 |
| Yield Stress, Operating                 | 231      | N./mm <sup>2</sup>  |
| UCS-66 Chart Curve Designation          | B        |                     |
| External Pressure Chart Name            | CS-2     |                     |
| UNS Number                              | K03017   |                     |
| Product Form                            | Forgings |                     |
| Perform Flange Stress Calculation (Y/N) | Y        |                     |
| Weld is pre-Heated                      | No       |                     |

|                                      |           |                     |
|--------------------------------------|-----------|---------------------|
| Element From Node                    | 30        |                     |
| Detail Type                          | Liquid    |                     |
| Detail ID                            | Liquid 30 |                     |
| Dist. from "FROM" Node / Offset dist | 0         | mm.                 |
| Height/Length of Liquid              | 490       | mm.                 |
| Liquid Density                       | 0.0009984 | kg./cm <sup>3</sup> |

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|                                         |           |      |
|-----------------------------------------|-----------|------|
| Element#                                | 4/6       |      |
| Element From Node                       | 40        |      |
| Element To Node                         | 50        |      |
| Element Type                            | Flange    |      |
| Description                             | Shell FLG |      |
| Distance "FROM" to "TO"                 | 75        | mm.  |
| Flange Inside Diameter                  | 490       | mm.  |
| Element Thickness                       | 60        | mm.  |
| Internal Corrosion Allowance            | 3.175     | mm.  |
| Nominal Thickness                       | 65        | mm.  |
| External Corrosion Allowance            | 0         | mm.  |
| Design Internal Pressure                | 25.511    | bars |
| Design Temperature Internal Pressure    | 127       | °C   |
| Design External Pressure                | 0         | bars |
| Design Temperature External Pressure    | 127       | °C   |
| Effective Diameter Multiplier           | 1.2       |      |
| Material Name                           | SA-266 4  |      |
| Perform Flange Stress Calculation (Y/N) | Y         |      |
| Weld is pre-Heated                      | No        |      |

|                   |           |  |
|-------------------|-----------|--|
| Element From Node | 40        |  |
| Detail Type       | Liquid    |  |
| Detail ID         | Liquid 40 |  |



Dist. from "FROM" Node / Offset dist 0 mm.  
 Height/Length of Liquid 490 mm.  
 Liquid Density 0.0009984 kg./cm<sup>3</sup>

Element From Node 40  
 Detail Type Insulation  
 Detail ID HOT INSULATION  
 Dist. from "FROM" Node / Offset dist 0 mm.  
 Height/Length of Insulation 75 mm.  
 Thickness of Insulation 45 mm.  
 Density 0.00015 kg./cm<sup>3</sup>

Element# 5/6  
 Element From Node 50  
 Element To Node 60  
 Element Type Cylinder  
 Description Main shell  
 Distance "FROM" to "TO" 6096 mm.  
 Inside Diameter 490 mm.  
 Element Thickness 12 mm.  
 Internal Corrosion Allowance 3.175 mm.  
 Nominal Thickness 12 mm.  
 External Corrosion Allowance 0 mm.  
 Design Internal Pressure 25.511 bars  
 Design Temperature Internal Pressure 127 °C  
 Design External Pressure 0 bars  
 Design Temperature External Pressure 127 °C  
 Effective Diameter Multiplier 1.2  
 Material Name SA-516 70  
 Allowable Stress, Ambient 138 N./mm<sup>2</sup>  
 Allowable Stress, Operating 138 N./mm<sup>2</sup>  
 Allowable Stress, Hydrotest 179.4 N./mm<sup>2</sup>  
 Material Density 0.00775 kg./cm<sup>3</sup>  
 P Number Thickness 32.004 mm.  
 Yield Stress, Operating 234.8 N./mm<sup>2</sup>  
 UCS-66 Chart Curve Designation B  
 External Pressure Chart Name CS-2  
 UNS Number K02700  
 Product Form Plate  
 Efficiency, Longitudinal Seam 1.0  
 Efficiency, Circumferential Seam 1.0  
 Weld is pre-Heated No

Element From Node 50  
 Detail Type Saddle  
 Detail ID Left Saddle  
 Dist. from "FROM" Node / Offset dist 1220 mm.  
 Width of Saddle 150 mm.  
 Height of Saddle at Bottom 500 mm.  
 Saddle Contact Angle 120.0  
 Height of Composite Ring Stiffener 0 mm.  
 Width of Wear Plate 225 mm.  
 Thickness of Wear Plate 12 mm.  
 Contact Angle, Wear Plate (degrees) 132.0  
 Friction coefficient 0.44999999  
 Moment Factor 3.0  
 Dimension E at base (optional) 0 mm.  
 Circumferential Eff. over Saddle 1.0  
 Circumferential Eff. at Midspan 1.0  
 Tangent to Tangent dist. (optional) 0 mm.

Element From Node 50  
 Detail Type Saddle  
 Detail ID RIGHT SADDLE  
 Dist. from "FROM" Node / Offset dist 4876 mm.  
 Width of Saddle 150 mm.  
 Height of Saddle at Bottom 500 mm.  
 Saddle Contact Angle 120.0



|                                     |            |     |
|-------------------------------------|------------|-----|
| Height of Composite Ring Stiffener  | 0          | mm. |
| Width of Wear Plate                 | 225        | mm. |
| Thickness of Wear Plate             | 12         | mm. |
| Contact Angle, Wear Plate (degrees) | 132.0      |     |
| Friction coefficient                | 0.44999999 |     |
| Moment Factor                       | 3.0        |     |
| Dimension E at base (optional)      | 0          | mm. |
| Circumferential Eff. over Saddle    | 1.0        |     |
| Circumferential Eff. at Midspan     | 1.0        |     |
| Tangent to Tangent dist. (optional) | 0          | mm. |

|                                      |           |                     |
|--------------------------------------|-----------|---------------------|
| Element From Node                    | 50        |                     |
| Detail Type                          | Liquid    |                     |
| Detail ID                            | Liquid 50 |                     |
| Dist. from "FROM" Node / Offset dist | 0         | mm.                 |
| Height/Length of Liquid              | 490       | mm.                 |
| Liquid Density                       | 0.0009984 | kg./cm <sup>3</sup> |

|                                      |                |                     |
|--------------------------------------|----------------|---------------------|
| Element From Node                    | 50             |                     |
| Detail Type                          | Insulation     |                     |
| Detail ID                            | HOT INSULATION |                     |
| Dist. from "FROM" Node / Offset dist | 0              | mm.                 |
| Height/Length of Insulation          | 6096           | mm.                 |
| Thickness of Insulation              | 45             | mm.                 |
| Density                              | 0.00015        | kg./cm <sup>3</sup> |

|                                      |              |     |
|--------------------------------------|--------------|-----|
| Element From Node                    | 50           |     |
| Detail Type                          | Nozzle       |     |
| Detail ID                            | SH IN NOZ S1 |     |
| Dist. from "FROM" Node / Offset dist | 5846         | mm. |
| Nozzle Diameter                      | 8            | in. |
| Nozzle Schedule                      | 80           |     |
| Nozzle Class                         | 300          |     |
| Layout Angle                         | 90.0         |     |
| Blind Flange (Y/N)                   | N            |     |
| Weight of Nozzle ( Used if > 0 )     | 71.292       | Kgf |
| Grade of Attached Flange             | GR 1.1       |     |
| Nozzle Matl                          | SA-106 B     |     |

|                                      |               |     |
|--------------------------------------|---------------|-----|
| Element From Node                    | 50            |     |
| Detail Type                          | Nozzle        |     |
| Detail ID                            | SH OUT NOZ S2 |     |
| Dist. from "FROM" Node / Offset dist | 250           | mm. |
| Nozzle Diameter                      | 8             | in. |
| Nozzle Schedule                      | 80            |     |
| Nozzle Class                         | 300           |     |
| Layout Angle                         | -90.0         |     |
| Blind Flange (Y/N)                   | N             |     |
| Weight of Nozzle ( Used if > 0 )     | 46.732        | Kgf |
| Grade of Attached Flange             | GR 1.1        |     |
| Nozzle Matl                          | SA-106 B      |     |

|                                      |                |      |
|--------------------------------------|----------------|------|
| Element#                             | 6/6            |      |
| Element From Node                    | 60             |      |
| Element To Node                      | 70             |      |
| Element Type                         | Elliptical     |      |
| Description                          | Rear End Cover |      |
| Distance "FROM" to "TO"              | 50.8           | mm.  |
| Inside Diameter                      | 490            | mm.  |
| Element Thickness                    | 12             | mm.  |
| Internal Corrosion Allowance         | 3.175          | mm.  |
| Nominal Thickness                    | 14             | mm.  |
| External Corrosion Allowance         | 0              | mm.  |
| Design Internal Pressure             | 25.511         | bars |
| Design Temperature Internal Pressure | 127            | °C   |
| Design External Pressure             | 0              | bars |
| Design Temperature External Pressure | 128            | °C   |
| Effective Diameter Multiplier        | 1.2            |      |

|                                      |                               |
|--------------------------------------|-------------------------------|
| Material Name                        | SA-516 70                     |
| Efficiency, Longitudinal Seam        | 1.0                           |
| Efficiency, Circumferential Seam     | 1.0                           |
| Elliptical Head Factor               | 2.0                           |
| Weld is pre-Heated                   | No                            |
|                                      |                               |
| Element From Node                    | 60                            |
| Detail Type                          | Liquid                        |
| Detail ID                            | Liquid 60                     |
| Dist. from "FROM" Node / Offset dist | 0 mm.                         |
| Height/Length of Liquid              | 490 mm.                       |
| Liquid Density                       | 0.0009984 kg./cm <sup>3</sup> |
|                                      |                               |
| Element From Node                    | 60                            |
| Detail Type                          | Insulation                    |
| Detail ID                            | HOT INSULATION                |
| Dist. from "FROM" Node / Offset dist | 0 mm.                         |
| Height/Length of Insulation          | 173.3 mm.                     |
| Thickness of Insulation              | 45 mm.                        |
| Density                              | 0.00015 kg./cm <sup>3</sup>   |



**XY Coordinate Calculations:**

| From           | To | X (Horiz.)<br>mm. | Y (Vert.)<br>mm. | DX (Horiz.)<br>mm. | DY (Vert.)<br>mm. |
|----------------|----|-------------------|------------------|--------------------|-------------------|
| Fr. END COVER  |    | 50                | ...              | 50                 | ...               |
| CH.SHELL       |    | 450               | ...              | 400                | ...               |
| CH FLG         |    | 529.5             | ...              | 75                 | ...               |
| Shell FLG      |    | 656.5             | ...              | 75                 | ...               |
| Main shell     |    | 6752.5            | ...              | 6096               | ...               |
| Rear End Cover |    | 6803.3            | ...              | 50.8               | ...               |

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

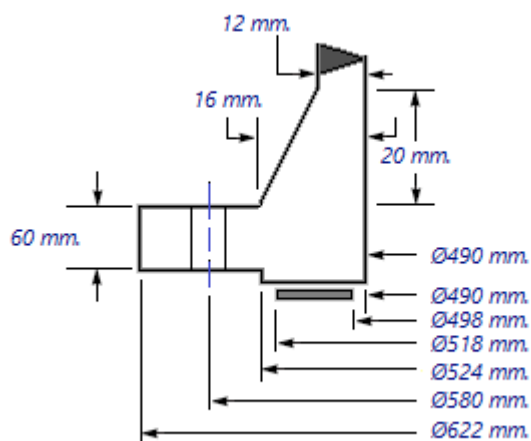
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## Flange Input Data Values

Description: FLANGE :

### CH FLG

|                                             |                    |           |                    |
|---------------------------------------------|--------------------|-----------|--------------------|
| Flange Type                                 | Integral Weld Neck |           |                    |
| Design Pressure                             | P                  | 5.56      | bars               |
| Design Temperature                          |                    | 77        | °C                 |
| Internal Corrosion Allowance                | ci                 | 3.1750    | mm.                |
| External Corrosion Allowance                | ce                 | 0.0000    | mm.                |
| Use Corrosion Allowance in Thickness Calcs. |                    | No        |                    |
| Flange Inside Diameter                      | B                  | 490.000   | mm.                |
| Flange Outside Diameter                     | A                  | 622.000   | mm.                |
| Flange Thickness                            | t                  | 60.0000   | mm.                |
| Thickness of Hub at Small End               | go                 | 12.0000   | mm.                |
| Thickness of Hub at Large End               | gl                 | 16.0000   | mm.                |
| Length of Hub                               | h                  | 20.0000   | mm.                |
| Flange Material                             |                    | SA-266 4  |                    |
| Flange Material UNS number                  |                    | K03017    |                    |
| Flange Allowable Stress At Temperature      | Sfo                | 138.00    | N./mm <sup>2</sup> |
| Flange Allowable Stress At Ambient          | Sfa                | 138.00    | N./mm <sup>2</sup> |
| Bolt Material                               |                    | SA-193 B7 |                    |
| Bolt Allowable Stress At Temperature        | Sb                 | 172.00    | N./mm <sup>2</sup> |
| Bolt Allowable Stress At Ambient            | Sa                 | 172.00    | N./mm <sup>2</sup> |
| Diameter of Bolt Circle                     | C                  | 580.000   | mm.                |
| Nominal Bolt Diameter                       | a                  | 19.0500   | mm.                |
| Type of Threads                             | TEMA Thread Series |           |                    |
| Number of Bolts                             |                    | 24        |                    |
| Flange Face Outside Diameter                | Fod                | 524.000   | mm.                |
| Flange Face Inside Diameter                 | Fid                | 490.000   | mm.                |
| Flange Facing Sketch                        |                    | Sketch 1a |                    |
| Gasket Outside Diameter                     | Go                 | 518.000   | mm.                |
| Gasket Inside Diameter                      | Gi                 | 498.000   | mm.                |
| Gasket Factor                               | m                  | 3.7500    |                    |
| Gasket Design Seating Stress                | y                  | 62.06     | N./mm <sup>2</sup> |
| Gasket Thickness                            | tg                 | 4.5000    | mm.                |
| Raised face depth                           | Fd                 | 0.0000    | mm.                |
| Length of Partition Gasket                  | lp                 | 498.0000  | mm.                |
| Width of Partition Gasket                   | tp                 | 10.0000   | mm.                |
| Partition Gasket Factor                     | mPart              | 3.7500    |                    |
| Partition Gasket Design Seating Stress      | yPart              | 52.40     | N./mm <sup>2</sup> |





**ASME Code, Section VIII Division 1, 2025**

Hub Small End Required Thickness due to Internal Pressure:

$$\begin{aligned} &= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= (5.56 \cdot (490.0/2 + 3.175)) / (138.0 \cdot 1.0 - 0.6 \cdot 5.56) + Ca \\ &= 4.1781 \text{ mm.} \end{aligned}$$

Small End Hub MAWP:

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\ &= (138.0 \cdot 1.0 \cdot 8.825) / (248.175 + 0.6 \cdot 8.825) \\ &= 48.043 \text{ bars} \end{aligned}$$

Hub Small End Hub MAPnc:

$$\begin{aligned} &= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\ &= (138.0 \cdot 1.0 \cdot 12.0) / (245.0 + 0.6 \cdot 12.0) \\ &= 65.657 \text{ bars} \end{aligned}$$

|                           |                        |             |
|---------------------------|------------------------|-------------|
| Corroded Flange ID,       | Bcor = B+2*ci          | 496.350 mm. |
| Corroded Large Hub,       | g1Cor = g1-ci          | 12.825 mm.  |
| Corroded Small Hub,       | g0Cor = go-ci          | 8.825 mm.   |
| Code R Dimension,         | R = ((C-Bcor)/2)-g1cor | 29.000 mm.  |
| Gasket Contact Width,     | N = (Go - Gi)/2        | 10.000 mm.  |
| Basic Gasket Width,       | bo = N/2               | 5.000 mm.   |
| Effective Gasket Width,   | b = bo                 | 5.000 mm.   |
| Gasket Reaction Diameter, | G = (Go + Gi)/2        | 508.000 mm. |

**Basic Flange and Bolt Loads:**

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$\begin{aligned} &= \pi/4 \cdot G^2 \cdot P \\ &= \pi/4 \cdot 508.0^2 \cdot 5.564 \\ &= 11500.146 \text{ Kg} \end{aligned}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$\begin{aligned} &= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot lp \cdot bPart \cdot mPart \cdot P \\ &= 2 \cdot 5.0 \cdot \pi \cdot 508.0 \cdot 3.75 \cdot 5.56 \\ &\quad + 2 \cdot 498.0 \cdot 5.0 \cdot 3.75 \cdot 5.564 \\ &= 4455.322 \text{ Kg} \end{aligned}$$

Hydrostatic End Load at Flange ID [Hd]:

$$\begin{aligned} &= \pi/4 \cdot Bcor^2 \cdot P \\ &= \pi/4 \cdot 496.35^2 \cdot 5.564 \\ &= 10978.726 \text{ Kg} \end{aligned}$$

Pressure Force on Flange Face [Ht]:

$$\begin{aligned} &= H - Hd \\ &= 11500 - 10979 \\ &= 521.419 \text{ Kg} \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned} &= \max( Wo,1 + Wo,2 + H'p, 0 ) \\ &= \max( 11500 + 4455 + 0, 0 ) \\ &= 15955.468 \text{ Kg} \\ &= 73293.320 \text{ Kg, Mating Flange Load Governs} \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned} &= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp \\ &= 62.06 \cdot 5.0 \cdot \pi \cdot 508.0 + 52.4 \cdot 5.0 \cdot 498.0 \\ &= 63798.156 \text{ Kg} \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned} &= \max( ( Wo + Fa + 4Me/G ) / Sb, Wg/Sa ) \\ &= \max( ( 73293 + 0 + 4 \cdot 0 / 508.0 ) / 172, 63798 / 172 ) \\ &= 41.790 \text{ cm}^2 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$\begin{aligned} &= 2a + 6t / (m + 0.5) \\ &= 2 \cdot 19.05 + 6 \cdot 60.0 / (3.75 + 0.5) \\ &= 122.806 \text{ mm.} \end{aligned}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 580.0 \cdot \sin(\pi / 24)$$

$$= 75.705 \text{ mm.}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{75.705 / (2 \cdot 19.05 + 60.0)}, 1)$$

$$= 1.0000$$

#### Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

|                                         | Minimum | Actual | Maximum |
|-----------------------------------------|---------|--------|---------|
| Bolt Area:                              | 41.790  | 46.761 |         |
| Radial Distance between Hub and Bolts:  | 28.575  | 29.000 |         |
| Radial Distance between Bolts and Edge: | 20.637  | 21.000 |         |
| Circ. Spacing between the Bolts:        | 44.450  | 75.705 | 122.806 |

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 46.761 \cdot 172.0 / (62.06 \cdot \pi(518.0 + 498.0))$$

$$= 4.061 \text{ mm.}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 46.7612$$

$$= 82012.02 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 73293 - 11500$$

$$= 61793.18 \text{ Kg}$$

#### Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (580.0 - 508.0) / 2$$

$$= 36.0000 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((580.0 - 496.35) / 2 + 36.0)$$

$$= 38.9125 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (580.0 - 496.35 - 12.825) / 2$$

$$= 35.4125 \text{ mm.}$$

#### Summary of Moments for Internal Pressure: (Kg-m.)

| Loading              | Force  | Distance | Bolt Corr | Moment |
|----------------------|--------|----------|-----------|--------|
| End Pressure, Md     | 10979. | 35.4125  | 1.0000    | 389.   |
| Face Pressure, Mt    | 521.   | 38.9125  | 1.0000    | 20.    |
| Gasket Load, Mg      | 61793. | 36.0000  | 1.0000    | 2225.  |
| Gasket Seating, Matm | 82012. | 36.0000  | 1.0000    | 2952.  |

Total Moment for Operation, Mo 2634. Kg-m.  
 Total Moment for Gasket seating, Mg 2952. Kg-m.

Effective Hub Length, ho = sqrt(Bcor\*goCor) 66.184 mm.  
 Hub Ratio, h/h0 = HL / H0 0.302  
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.453

Flange Factors for Integral Flange:

|                     |       |
|---------------------|-------|
| Factor F per 4.16.5 | 0.883 |
| Factor V per 4.16.5 | 0.379 |
| Factor f per 4.16.5 | 1.094 |

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 622.0/496.35$$

$$= 1.253$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.253^2 \cdot \log_{10}(1.253))) / (1.253^2 - 1)) / (1.253 - 1)$$

$$= 8.734$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.253^2(1 + 8.55246 \cdot \log_{10}(1.253)) - 1) / ((1.0472 + 1.9448 \cdot 1.253^2) \cdot (1.253 - 1))$$

$$= 1.817$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.253^2(1 + 8.55246 \cdot \log_{10}(1.253)) - 1) / (1.36136(1.253^2 - 1) \cdot (1.253 - 1))$$

$$= 9.598$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.253^2 + 1) / (1.253^2 - 1)$$

$$= 4.506$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 9.598 \cdot 8.825^2 \cdot 66.184 / 0.379$$

$$= 130467.039 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.883/66.184$$

$$= 0.013 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (60.0 \cdot 0.013 + 1) / 1.817 + 60.0^3/130467.039$$

$$= 2.646$$

## Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.0937 \cdot 2634/496.35) / (2.6464 \cdot 12.825^2)$$

$$= 130.74 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.0937 \cdot 2952/496.35) / (2.6464 \cdot 12.825^2)$$

$$= 146.57 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.0673 \cdot 2634/496.35) / (2.6464 \cdot 60.0^2)$$

$$= 11.29 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.0673 \cdot 2952/496.35) / (2.6464 \cdot 60.0^2)$$

$$= 12.66 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (8.734 \cdot 2634 / (60.0^2 \cdot 496.35)) - 4.5064 \cdot 11$$

$$= 75.36 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:  
 $= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$   
 $= (8.734 \cdot 2952 / (60.0^2 \cdot 496.35)) - 4.5064 \cdot 13$   
 $= 84.48 \text{ N./mm}^2$

Average Flange Stress, Operating [SAo]:  
 $= (SHo + \max(SRo, STo)) / 2$   
 $= (131 + \max(11, 75)) / 2$   
 $= 103.05 \text{ N./mm}^2$

Average Flange Stress, Seating [SAa]:  
 $= (SHA + \max(SRa, STa)) / 2$   
 $= (147 + \max(13, 84)) / 2$   
 $= 115.52 \text{ N./mm}^2$

Bolt Stress, Operating [BSo]:  
 $= Wm1 / Ab$   
 $= 73293 / 46.7612$   
 $= 153.71 \text{ N./mm}^2$

Bolt Stress, Seating [BSa]:  
 $= (Wm2 / Ab)$   
 $= (63798 / 46.7612)$   
 $= 133.80 \text{ N./mm}^2$

#### Flange Stress Analysis Results: N./mm²

|                   | Actual | Operating Allowed | Gasket Seating Actual Allowed |
|-------------------|--------|-------------------|-------------------------------|
| Longitudinal Hub  | 130.74 | 207.00            | 146.57 207.00                 |
| Radial Flange     | 11.29  | 138.00            | 12.66 138.00                  |
| Tangential Flange | 75.36  | 138.00            | 84.48 138.00                  |
| Maximum Average   | 103.05 | 138.00            | 115.52 138.00                 |
| Bolting           | 153.71 | 172.00            | 133.80 172.00                 |

Minimum Required Flange Thickness 53.594 mm.

Estimated M.A.W.P. ( Operating ) 29 bars

Estimated Finished Weight of Flange at given Thk. 57.0 kg.

Estimated Unfinished Weight of Forging at given Thk 71.5 kg.

#### Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:  
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$   
 $= 52.14 \cdot 2952.5 / 1.0 \cdot 9806.637 \cdot 0.379 / (2.123 \cdot 202713 \cdot 8.825^2 \cdot 66.184 \cdot 0.3)$   
 $= 0.860 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:  
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$   
 $= 52.14 \cdot 2633.7 / 1.0 \cdot 9806.637 \cdot 0.379 / (2.123 \cdot 199531 \cdot 8.825^2 \cdot 66.184 \cdot 0.3)$   
 $= 0.779 \text{ (should be } \leq 1)$

#### Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:  
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$   
 $= 52.14 \cdot 2952.5 / 1.0 \cdot 9806.637 \cdot 0.379 / (2.646 \cdot 202713 \cdot 8.825^2 \cdot 66.184 \cdot 0.3)$   
 $= 0.690 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:  
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$   
 $= 52.14 \cdot 2633.7 / 1.0 \cdot 9806.637 \cdot 0.379 / (2.646 \cdot 199531 \cdot 8.825^2 \cdot 66.184 \cdot 0.3)$   
 $= 0.625 \text{ (should be } \leq 1)$

#### Minimum Design Metal Temperature Results:

Thickness Ratio = 0.582, Temperature Reduction per Fig. UCS 66.1 = 24 °C



Min Metal Temp. w/o impact per UCS-66, Curve B      -23 °C  
Min Metal Temp. at Required thickness (UCS 66.1)      -47 °C  
Min Metal Temp. w/o impact per UG-20(f)      -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

**Flange Stress Analysis Results: N./mm<sup>2</sup>**

|                   | Operating |         | Gasket Seating |         |
|-------------------|-----------|---------|----------------|---------|
|                   | Actual    | Allowed | Actual         | Allowed |
| Longitudinal Hub  | 115.66    | 207.00  | 128.72         | 207.00  |
| Radial Flange     | 15.47     | 138.00  | 17.22          | 138.00  |
| Tangential Flange | 55.96     | 138.00  | 62.28          | 138.00  |
| Maximum Average   | 85.81     | 138.00  | 95.50          | 138.00  |
| Bolting           | 153.71    | 172.00  | 133.80         | 172.00  |

Minimum Req'd Thickness, ambient, no Corrosion: 42.469 mm.

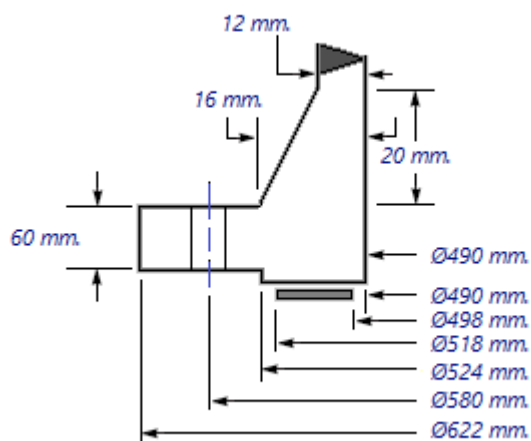
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## Flange Input Data Values

Description: Shell FLG :

### Shell FLG

|                                             |                    |           |                    |
|---------------------------------------------|--------------------|-----------|--------------------|
| Flange Type                                 | Integral Weld Neck |           |                    |
| Design Pressure                             | P                  | 25.56     | bars               |
| Design Temperature                          |                    | 127       | °C                 |
| Internal Corrosion Allowance                | ci                 | 3.1750    | mm.                |
| External Corrosion Allowance                | ce                 | 0.0000    | mm.                |
| Use Corrosion Allowance in Thickness Calcs. |                    | No        |                    |
| Flange Inside Diameter                      | B                  | 490.000   | mm.                |
| Flange Outside Diameter                     | A                  | 622.000   | mm.                |
| Flange Thickness                            | t                  | 60.0000   | mm.                |
| Thickness of Hub at Small End               | go                 | 12.0000   | mm.                |
| Thickness of Hub at Large End               | gl                 | 16.0000   | mm.                |
| Length of Hub                               | h                  | 20.0000   | mm.                |
| Flange Material                             |                    | SA-266 4  |                    |
| Flange Material UNS number                  |                    | K03017    |                    |
| Flange Allowable Stress At Temperature      | Sfo                | 138.00    | N./mm <sup>2</sup> |
| Flange Allowable Stress At Ambient          | Sfa                | 138.00    | N./mm <sup>2</sup> |
| Bolt Material                               |                    | SA-193 B7 |                    |
| Bolt Allowable Stress At Temperature        | Sb                 | 172.00    | N./mm <sup>2</sup> |
| Bolt Allowable Stress At Ambient            | Sa                 | 172.00    | N./mm <sup>2</sup> |
| Diameter of Bolt Circle                     | C                  | 580.000   | mm.                |
| Nominal Bolt Diameter                       | a                  | 19.0500   | mm.                |
| Type of Threads                             | TEMA Thread Series |           |                    |
| Number of Bolts                             |                    | 24        |                    |
| Flange Face Outside Diameter                | Fod                | 524.000   | mm.                |
| Flange Face Inside Diameter                 | Fid                | 490.000   | mm.                |
| Flange Facing Sketch                        |                    | Sketch 1a |                    |
| Gasket Outside Diameter                     | Go                 | 518.000   | mm.                |
| Gasket Inside Diameter                      | Gi                 | 498.000   | mm.                |
| Gasket Factor                               | m                  | 3.7500    |                    |
| Gasket Design Seating Stress                | y                  | 62.06     | N./mm <sup>2</sup> |
| Gasket Thickness                            | tg                 | 4.5000    | mm.                |
| Raised face depth                           | Fd                 | 0.0000    | mm.                |
| Length of Partition Gasket                  | lp                 | 498.0000  | mm.                |
| Width of Partition Gasket                   | tp                 | 10.0000   | mm.                |
| Partition Gasket Factor                     | mPart              | 3.7500    |                    |
| Partition Gasket Design Seating Stress      | yPart              | 52.40     | N./mm <sup>2</sup> |





**ASME Code, Section VIII Division 1, 2025**

Hub Small End Required Thickness due to Internal Pressure:

$$\begin{aligned} &= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= (25.56 \cdot (490.0/2 + 3.175)) / (138.0 \cdot 1.0 - 0.6 \cdot 25.56) + Ca \\ &= 7.8235 \text{ mm.} \end{aligned}$$

Small End Hub MAWP:

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\ &= (138.0 \cdot 1.0 \cdot 8.825) / (248.175 + 0.6 \cdot 8.825) \\ &= 48.043 \text{ bars} \end{aligned}$$

Hub Small End Hub MAPnc:

$$\begin{aligned} &= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\ &= (138.0 \cdot 1.0 \cdot 12.0) / (245.0 + 0.6 \cdot 12.0) \\ &= 65.657 \text{ bars} \end{aligned}$$

|                           |                        |             |
|---------------------------|------------------------|-------------|
| Corroded Flange ID,       | Bcor = B+2*ci          | 496.350 mm. |
| Corroded Large Hub,       | g1Cor = g1-ci          | 12.825 mm.  |
| Corroded Small Hub,       | g0Cor = go-ci          | 8.825 mm.   |
| Code R Dimension,         | R = ((C-Bcor)/2)-g1cor | 29.000 mm.  |
| Gasket Contact Width,     | N = (Go - Gi)/2        | 10.000 mm.  |
| Basic Gasket Width,       | bo = N/2               | 5.000 mm.   |
| Effective Gasket Width,   | b = bo                 | 5.000 mm.   |
| Gasket Reaction Diameter, | G = (Go + Gi)/2        | 508.000 mm. |

**Basic Flange and Bolt Loads:**

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$\begin{aligned} &= \pi/4 \cdot G^2 \cdot P \\ &= \pi/4 \cdot 508.0^2 \cdot 25.559 \\ &= 52827.273 \text{ Kgf} \end{aligned}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$\begin{aligned} &= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot lp \cdot bPart \cdot mPart \cdot P \\ &= 2 \cdot 5.0 \cdot \pi \cdot 508.0 \cdot 3.75 \cdot 25.56 \\ &\quad + 2 \cdot 498.0 \cdot 5.0 \cdot 3.75 \cdot 25.559 \\ &= 20466.047 \text{ Kgf} \end{aligned}$$

Hydrostatic End Load at Flange ID [Hd]:

$$\begin{aligned} &= \pi/4 \cdot Bcor^2 \cdot P \\ &= \pi/4 \cdot 496.35^2 \cdot 25.559 \\ &= 50432.074 \text{ Kgf} \end{aligned}$$

Pressure Force on Flange Face [Ht]:

$$\begin{aligned} &= H - Hd \\ &= 52827 - 50432 \\ &= 2395.199 \text{ Kgf} \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned} &= \max( Wo,1 + Wo,2 + H'p, 0 ) \\ &= \max( 52827 + 20466 + 0, 0 ) \\ &= 73293.320 \text{ Kgf} \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned} &= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp \\ &= 62.06 \cdot 5.0 \cdot \pi \cdot 508.0 + 52.4 \cdot 5.0 \cdot 498.0 \\ &= 63798.156 \text{ Kgf} \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned} &= \max( ( Wo + Fa + 4Me/G ) / Sb, Wg/Sa ) \\ &= \max( ( 73293 + 0 + 4 \cdot 0 / 508.0 ) / 172, 63798 / 172 ) \\ &= 41.790 \text{ cm}^2 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$\begin{aligned} &= 2a + 6t/(m + 0.5) \\ &= 2 \cdot 19.05 + 6 \cdot 60.0/(3.75 + 0.5) \\ &= 122.806 \text{ mm.} \end{aligned}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 580.0 \cdot \sin(\pi / 24)$$

$$= 75.705 \text{ mm.}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{75.705 / (2 \cdot 19.05 + 60.0)}, 1)$$

$$= 1.0000$$

#### Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

|                                         | Minimum | Actual | Maximum |
|-----------------------------------------|---------|--------|---------|
| Bolt Area:                              | 41.790  | 46.761 |         |
| Radial Distance between Hub and Bolts:  | 28.575  | 29.000 |         |
| Radial Distance between Bolts and Edge: | 20.637  | 21.000 |         |
| Circ. Spacing between the Bolts:        | 44.450  | 75.705 | 122.806 |

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 46.761 \cdot 172.0 / (62.06 \cdot \pi(518.0 + 498.0))$$

$$= 4.061 \text{ mm.}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 46.7612$$

$$= 82012.02 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 73293 - 52827$$

$$= 20466.05 \text{ Kgf}$$

#### Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (580.0 - 508.0) / 2$$

$$= 36.0000 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((580.0 - 496.35) / 2 + 36.0)$$

$$= 38.9125 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (580.0 - 496.35 - 12.825) / 2$$

$$= 35.4125 \text{ mm.}$$

#### Summary of Moments for Internal Pressure: (Kg-m.)

| Loading              | Force  | Distance | Bolt Corr | Moment |
|----------------------|--------|----------|-----------|--------|
| End Pressure, Md     | 50432. | 35.4125  | 1.0000    | 1786.  |
| Face Pressure, Mt    | 2395.  | 38.9125  | 1.0000    | 93.    |
| Gasket Load, Mg      | 20466. | 36.0000  | 1.0000    | 737.   |
| Gasket Seating, Matm | 82012. | 36.0000  | 1.0000    | 2952.  |

Total Moment for Operation, Mo 2616. Kg-m.  
 Total Moment for Gasket seating, Mg 2952. Kg-m.

Effective Hub Length, ho = sqrt(Bcor\*goCor) 66.184 mm.  
 Hub Ratio, h/h0 = HL / H0 0.302  
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.453

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.883

Factor V per 4.16.5 0.379  
Factor f per 4.16.5 1.094

Stress factor from Table 4.16.4 [K]:

$$\begin{aligned} &= A / B \\ &= 622.0/496.35 \\ &= 1.253 \end{aligned}$$

Stress factor from Table 4.16.4 [Y]:

$$\begin{aligned} &= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1) \\ &= (0.66845 + (5.7169(1.253^2 \cdot \log_{10}(1.253))) / (1.253^2 - 1)) / (1.253 - 1) \\ &= 8.734 \end{aligned}$$

Stress factor from Table 4.16.4 [T]:

$$\begin{aligned} &= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1)) \\ &= (1.253^2(1 + 8.55246 \cdot \log_{10}(1.253)) - 1) / ((1.0472 + 1.9448 \cdot 1.253^2) \cdot (1.253 - 1)) \\ &= 1.817 \end{aligned}$$

Stress factor from Table 4.16.4 [U]:

$$\begin{aligned} &= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1)) \\ &= (1.253^2(1 + 8.55246 \cdot \log_{10}(1.253)) - 1) / (1.36136(1.253^2 - 1) \cdot (1.253 - 1)) \\ &= 9.598 \end{aligned}$$

Stress factor from Table 4.16.4 [Z]:

$$\begin{aligned} &= (K^2 + 1) / (K^2 - 1) \\ &= (1.253^2 + 1) / (1.253^2 - 1) \\ &= 4.506 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned} &= U \cdot g_0^2 \cdot h_0 / V \\ &= 9.598 \cdot 8.825^2 \cdot 66.184 / 0.379 \\ &= 130467.039 \text{ mm}^3 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned} &= F / h_0 \\ &= 0.883/66.184 \\ &= 0.013 \text{ 1/mm} \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned} &= (t \cdot e + 1) / T + t^3/d \\ &= (60.0 \cdot 0.013 + 1) / 1.817 + 60.0^3/130467.039 \\ &= 2.646 \end{aligned}$$

## Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned} &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\ &= (1.0937 \cdot 2616 / 496.35) / (2.6464 \cdot 12.825^2) \\ &= 129.86 \text{ N./mm}^2 \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned} &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\ &= (1.0937 \cdot 2952 / 496.35) / (2.6464 \cdot 12.825^2) \\ &= 146.57 \text{ N./mm}^2 \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned} &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\ &= (2.0673 \cdot 2616 / 496.35) / (2.6464 \cdot 60.0^2) \\ &= 11.22 \text{ N./mm}^2 \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned} &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\ &= (2.0673 \cdot 2952 / 496.35) / (2.6464 \cdot 60.0^2) \\ &= 12.66 \text{ N./mm}^2 \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned} &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\ &= (8.734 \cdot 2616 / (60.0^2 \cdot 496.35)) - 4.5064 \cdot 11 \\ &= 74.85 \text{ N./mm}^2 \end{aligned}$$



Tangential Flange Stress, Seating [STa]:  
 $= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$   
 $= (8.734 \cdot 2952 / (60.0^2 \cdot 496.35)) - 4.5064 \cdot 13$   
 $= 84.48 \text{ N./mm}^2$

Average Flange Stress, Operating [SAo]:  
 $= (SHo + \max(SRo, STo)) / 2$   
 $= (130 + \max(11, 75)) / 2$   
 $= 102.36 \text{ N./mm}^2$

Average Flange Stress, Seating [SAa]:  
 $= (SHA + \max(SRa, STa)) / 2$   
 $= (147 + \max(13, 84)) / 2$   
 $= 115.52 \text{ N./mm}^2$

Bolt Stress, Operating [BSo]:  
 $= Wm1 / Ab$   
 $= 73293 / 46.7612$   
 $= 153.71 \text{ N./mm}^2$

Bolt Stress, Seating [BSa]:  
 $= (Wm2 / Ab)$   
 $= (63798 / 46.7612)$   
 $= 133.80 \text{ N./mm}^2$

#### Flange Stress Analysis Results: N./mm²

|                   | Actual | Operating Allowed | Gasket Seating Actual Allowed |
|-------------------|--------|-------------------|-------------------------------|
| Longitudinal Hub  | 129.86 | 207.00            | 146.57 207.00                 |
| Radial Flange     | 11.22  | 138.00            | 12.66 138.00                  |
| Tangential Flange | 74.85  | 138.00            | 84.48 138.00                  |
| Maximum Average   | 102.36 | 138.00            | 115.52 138.00                 |
| Bolting           | 153.71 | 172.00            | 133.80 172.00                 |

Minimum Required Flange Thickness 53.594 mm.

Estimated M.A.W.P. ( Operating ) 29 bars  
 Estimated Finished Weight of Flange at given Thk. 57.0 kg.  
 Estimated Unfinished Weight of Forging at given Thk 71.5 kg.

#### Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:  
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$   
 $= 52.14 \cdot 2952.5 / 1.0 \cdot 9806.637 \cdot 0.379 / (2.123 \cdot 202713 \cdot 8.825^2 \cdot 66.184 \cdot 0.3)$   
 $= 0.860 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:  
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$   
 $= 52.14 \cdot 2616.0 / 1.0 \cdot 9806.637 \cdot 0.379 / (2.123 \cdot 196508 \cdot 8.825^2 \cdot 66.184 \cdot 0.3)$   
 $= 0.786 \text{ (should be } \leq 1)$

#### Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:  
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$   
 $= 52.14 \cdot 2952.5 / 1.0 \cdot 9806.637 \cdot 0.379 / (2.646 \cdot 202713 \cdot 8.825^2 \cdot 66.184 \cdot 0.3)$   
 $= 0.690 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:  
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$   
 $= 52.14 \cdot 2616.0 / 1.0 \cdot 9806.637 \cdot 0.379 / (2.646 \cdot 196508 \cdot 8.825^2 \cdot 66.184 \cdot 0.3)$   
 $= 0.630 \text{ (should be } \leq 1)$

#### Minimum Design Metal Temperature Results:

Thickness Ratio = 0.582, Temperature Reduction per Fig. UCS 66.1 = 24 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C



Min Metal Temp. at Required thickness (UCS 66.1)      -47 °C  
Min Metal Temp. w/o impact per UG-20(f)      -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

**Flange Stress Analysis Results: N./mm<sup>2</sup>**

|                   | Operating |         | Gasket Seating |         |
|-------------------|-----------|---------|----------------|---------|
|                   | Actual    | Allowed | Actual         | Allowed |
| Longitudinal Hub  | 117.90    | 207.00  | 128.72         | 207.00  |
| Radial Flange     | 15.77     | 138.00  | 17.22          | 138.00  |
| Tangential Flange | 57.05     | 138.00  | 62.28          | 138.00  |
| Maximum Average   | 87.47     | 138.00  | 95.50          | 138.00  |
| Bolting           | 153.71    | 172.00  | 133.80         | 172.00  |

Minimum Req'd Thickness, ambient, no Corrosion: 42.469 mm.

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### Internal Pressure Results Summary:

#### Element Thickness, Pressure, Diameter and Allowable Stress :

| From           | To | Int. Press<br>+ Liq. Hd<br>bars | Nominal<br>Thickness<br>mm. | Total Corr<br>Allowance<br>mm. | Element<br>Diameter<br>mm. | Allowable<br>Stress(SE)<br>N./mm <sup>2</sup> | Element# |
|----------------|----|---------------------------------|-----------------------------|--------------------------------|----------------------------|-----------------------------------------------|----------|
| Fr. END COVER  |    | 5.5643                          | 14                          | 3.175                          | 490                        | 138                                           | 1        |
| CH.SHELL       |    | 5.5643                          | 12                          | 3.175                          | 490                        | 138                                           | 2        |
| CH FLG         |    | 5.564                           | 65                          | 3.175                          | 490                        | 138                                           | 3        |
| Shell FLG      |    | 25.559                          | 65                          | 3.175                          | 490                        | 138                                           | 4        |
| Main shell     |    | 25.559                          | 12                          | 3.175                          | 490                        | 138                                           | 5        |
| Rear End Cover |    | 25.559                          | 14                          | 3.175                          | 490                        | 138                                           | 6        |

#### Element Required Thickness and MAWP :

| From           | To | Design<br>Pressure<br>bars | M.A.W.P.<br>Corroded<br>bars | M.A.P.<br>New & Cold<br>bars | Minimum<br>Thickness<br>mm. | Required<br>Thickness<br>mm. | Element# |
|----------------|----|----------------------------|------------------------------|------------------------------|-----------------------------|------------------------------|----------|
| Fr. END COVER  |    | 5.516                      | 49.6755                      | 67.2573                      | 12                          | 5.675                        | 1        |
| CH.SHELL       |    | 5.516                      | 47.9951                      | 65.6572                      | 12                          | 5.675                        | 2        |
| CH FLG         |    | 5.516                      | 28.5514                      | 28.5994                      | 60                          | 53.594                       | 3        |
| Shell FLG      |    | 25.511                     | 28.5514                      | 28.5994                      | 60                          | 53.594                       | 4        |
| Main shell     |    | 25.511                     | 47.9951                      | 65.6572                      | 12                          | 7.82353                      | 5        |
| Rear End Cover |    | 25.511                     | 49.6755                      | 67.2573                      | 12                          | 7.70332                      | 6        |

#### Summary of Heat Exchanger Maximum Allowable Working Pressures :

##### Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

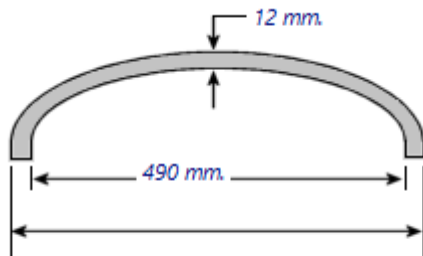
Shell Side MAWP = 28 bars  
 Shell Side MAPnc = 29 bars  
 Channel Side MAWP = 28 bars  
 Channel Side MAPnc = 29 bars

Elements Suitable for Design Internal Pressure.

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



#### Internal Pressure Results for: Fr. END COVER

| Elliptical Head Design Information:               |    |                         |                         |    |                          |
|---------------------------------------------------|----|-------------------------|-------------------------|----|--------------------------|
| Design Pressure                                   | P  | 5.564 bars              | Int. Design Temperature |    | 77 °C                    |
| Material                                          |    | SA-516 70               | Ext. Design Temperature |    | 77 °C                    |
| Length                                            |    | 50.000 mm.              | External Pressure Chart |    | CS-2                     |
|                                                   |    |                         | UNS Number              |    | K02700                   |
| Int. Corr. All.                                   | c  | 3.1750 mm.              | Allowable Stress (ope)  | S  | 138.0 N./mm <sup>2</sup> |
| Ext. Corr. All.                                   | ce | 0.0000 mm.              | Allowable Stress (amb)  | Sa | 138.0 N./mm <sup>2</sup> |
| Inside Diameter                                   | Di | 490.000 mm.             | Actual Stress           |    | 15.4 N./mm <sup>2</sup>  |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 12.0000 mm.             | Spec. Nominal Thk.      |    | 14.000 mm.               |
| Surface Area                                      |    | 3722.3 cm <sup>2</sup>  | SG of Contents          |    | 1.000                    |
| Element Volume                                    |    | 24833.4 cm <sup>3</sup> | Weight of Contents      |    | 24.8 Kg <sub>f</sub>     |
| Empty Weight                                      |    | 41.6 Kg <sub>f</sub>    | Operating Weight        |    | 66.4 Kg <sub>f</sub>     |
|                                                   |    |                         | Comp. All. Ope. Stress  |    | 141.0 N./mm <sup>2</sup> |
|                                                   |    |                         | Aspect Ratio            |    | 2.00                     |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

#### Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.125 mm.  
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.062 mm.

Required Thickness due to Internal Pressure [t<sub>r</sub>]:  

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (5.564 \cdot 496.35 \cdot 0.983) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 5.564)$$

$$= 0.9844 + 3.1750 = 4.1594 \text{ mm.}$$

*Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 2.5000 mm. per UG-16 will be used.*

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.048 bars  

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 8.825) / (0.983 \cdot 496.35 + 0.2 \cdot 8.825)$$

$$= 50 - 0 = 50 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 12.0) / (1.0 \cdot 490.0 + 0.2 \cdot 12.0)$$

$$= 67 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [S<sub>act</sub>]:

$$= (P \cdot (K \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (5.564 \cdot (0.983 \cdot 496.35 + 0.2 \cdot 8.825)) / (2 \cdot 1.0 \cdot 8.825) \\ = 15.443 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)} \\ = (5.564 \cdot 248.175) / (138.0 \cdot 1.0 - 0.6 \cdot 5.564) + 3.175 \\ = 4.178 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 0.048 bars](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\ = (138.0 \cdot 1.0 \cdot 10.825) / (248.175 + 0.6 \cdot 10.825) \\ = 59 - 0 = 59 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6 \\ = (2 + (496.35 / (2 \cdot 125.675))^2) / 6 \\ = 0.983264$$

Percent Elong. per UCS-79, VIII-1-01-57  $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$  11.628 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

#### MDMT Calculations in the Knuckle Portion:

Govrn. thk,  $t_g = 12.0$ ,  $t_r = 4.956$ ,  $c = 3.175 \text{ mm.}$ ,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.562$ , Temp. Reduction = 26 °C

|                                                  |        |
|--------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -23 °C |
| Min Metal Temp. at Required thickness (UCS 66.1) | -48 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C |

#### MDMT Calculations in the Head Straight Flange:

Govrn. thk,  $t_g = 14.0$ ,  $t_r = 5.092$ ,  $c = 3.175 \text{ mm.}$ ,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.47$ , Temp. Reduction = 37 °C

|                                                  |        |
|--------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -18 °C |
| Min Metal Temp. at Required thickness (UCS 66.1) | -48 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C |

| MDMT Design Information: |            |                        |             |
|--------------------------|------------|------------------------|-------------|
| MDMT Curve               | B          | Press. at MDMT         | 27.970 bars |
| UCS-66(b) red.           | Yes        | Min. Des. Mtl. Temp.   | -29 °C      |
| Governing Thk.           | 12.000 mm. | Basic MDMT             | -23 °C      |
| Stress/Thk. Ratio        | 0.562      | Temperature Difference | 25 °C       |
| UCS-66(c) red.           | No         | Computed Min. Temp.    | -48 °C      |
|                          |            | MDMT per UG-20(f)      | -29 °C      |
|                          |            | Joint Efficiency $E^*$ | 100 %       |

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

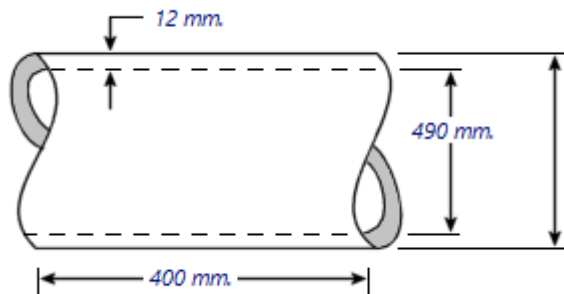
#### Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.125 mm.  
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.062 mm.

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



#### Internal Pressure Results for: CH.SHELL

| Cylindrical Shell Design Information:             |    |                         |                         |    |                          |
|---------------------------------------------------|----|-------------------------|-------------------------|----|--------------------------|
| Design Pressure                                   | P  | 5.564 bars              | Int. Design Temperature |    | 77 °C                    |
| Material                                          |    | SA-516 70               | Ext. Design Temperature |    | 77 °C                    |
| Length                                            |    | 400.000 mm.             | External Pressure Chart |    | CS-2                     |
|                                                   |    |                         | UNS Number              |    | K02700                   |
| Int. Corr. All.                                   | c  | 3.1750 mm.              | Allowable Stress (ope)  | S  | 138.0 N./mm <sup>2</sup> |
| Ext. Corr. All.                                   | ce | 0.0000 mm.              | Allowable Stress (amb)  | Sa | 138.0 N./mm <sup>2</sup> |
| Inside Diameter                                   | Di | 490.000 mm.             | Actual Stress           |    | 16.0 N./mm <sup>2</sup>  |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 12.0000 mm.             | Spec. Nominal Thk.      |    | 12.000 mm.               |
| Surface Area                                      |    | 6459.1 cm <sup>2</sup>  | SG of Contents          |    | 1.000                    |
| Element Volume                                    |    | 75443.1 cm <sup>3</sup> | Weight of Contents      |    | 75.4 Kg                  |
| Empty Weight                                      |    | 58.7 Kg                 | Operating Weight        |    | 134.1 Kg                 |
|                                                   |    |                         | Comp. All. Ope. Stress  |    | 141.0 N./mm <sup>2</sup> |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (5.564 \cdot 248.175) / (138.0 \cdot 1.0 \cdot 0.6 \cdot 5.564) \\
 &= 1.0032 + 3.1750 = 4.1782 \text{ mm.}
 \end{aligned}$$

*Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 2.5000 mm. per UG-16 will be used.*

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &\text{Less Operating Hydrostatic Head Pressure of 0.048 bars} \\
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 8.825) / (248.175 + 0.6 \cdot 8.825) \\
 &= 48 - 0 = 48 \text{ bars}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 12.0) / (245.0 + 0.6 \cdot 12.0) \\
 &= 66 \text{ bars}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (5.564 \cdot (248.175 + 0.6 \cdot 8.825)) / (1.0 \cdot 8.825) \\
 &= 15.983 \text{ N./mm}^2
 \end{aligned}$$

% Elongation per Table UG-79-1 (50 \* t<sub>nom</sub> / R<sub>f</sub> \* (1 - R<sub>f</sub> / R<sub>o</sub>)) 2.390 %



### Minimum Design Metal Temperature Results:

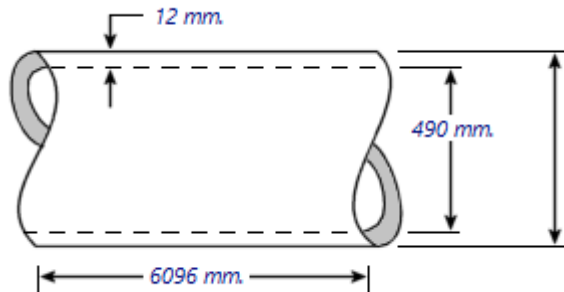
Govrn. thk, tg = 12.0, tr = 5.092, c = 3.175 mm., E\* = 1.0  
Thickness Ratio =  $tr * E^* / (tg - c) = 0.577$ , Temp. Reduction = 24 °C

Min Metal Temp. w/o impact per UCS-66, Curve B                      -23 °C  
Min Metal Temp. at Required thickness (UCS 66.1)                      -47 °C  
Min Metal Temp. w/o impact per UG-20(f)                                  -29 °C

| MDMT Design Information: |            |                        |             |
|--------------------------|------------|------------------------|-------------|
| MDMT Curve               | B          | Press. at MDMT         | 27.970 bars |
| UCS-66(b) red.           | Yes        | Min. Des. Mtl. Temp.   | -29 °C      |
| Governing Thk.           | 12.000 mm. | Basic MDMT             | -23 °C      |
| Stress/Thk. Ratio        | 0.577      | Temperature Difference | 24 °C       |
| UCS-66(c) red.           | No         | Computed Min. Temp.    | -47 °C      |
|                          |            | MDMT per UG-20(f)      | -29 °C      |
|                          |            | Joint Efficiency E*    | 100 %       |

### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



#### Internal Pressure Results for: Main shell

| Cylindrical Shell Design Information: |    |                          |                         |    |                          |
|---------------------------------------|----|--------------------------|-------------------------|----|--------------------------|
| Design Pressure                       | P  | 25.559 bars              | Int. Design Temperature |    | 127 °C                   |
| Material                              |    | SA-516 70                | Ext. Design Temperature |    | 127 °C                   |
| Length                                |    | 6096.000 mm.             | External Pressure Chart |    | CS-2                     |
|                                       |    |                          | UNS Number              |    | K02700                   |
| Int. Corr. All.                       | c  | 3.1750 mm.               | Allowable Stress (ope)  | S  | 138.0 N./mm <sup>2</sup> |
| Ext. Corr. All.                       | ce | 0.0000 mm.               | Allowable Stress (amb)  | Sa | 138.0 N./mm <sup>2</sup> |
| Inside Diameter                       | Di | 490.000 mm.              | Actual Stress           |    | 73.4 N./mm <sup>2</sup>  |
| Spec. Min. Thk t,e, or sa             |    | 12.0000 mm.              | Spec. Nominal Thk.      |    | 12.000 mm.               |
| Surface Area                          |    | 98436.9 cm <sup>2</sup>  | SG of Contents          |    | 1.000                    |
| Element Volume                        |    | 859996.3 cm <sup>3</sup> | Weight of Contents      |    | 859.5 Kg                 |
| Empty Weight                          |    | 894.1 Kg                 | Operating Weight        |    | 1753.6 Kg                |
|                                       |    |                          | Comp. All. Ope. Stress  |    | 141.0 N./mm <sup>2</sup> |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (25.559 \cdot 248.175) / (138.0 \cdot 1.0 - 0.6 \cdot 25.559)$$

$$= 4.6485 + 3.1750 = 7.8235 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.048 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 8.825) / (248.175 + 0.6 \cdot 8.825)$$

$$= 48 - 0 = 48 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (245.0 + 0.6 \cdot 12.0)$$

$$= 66 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (25.559 \cdot (248.175 + 0.6 \cdot 8.825)) / (1.0 \cdot 8.825)$$

$$= 73.415 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 \* tnom / Rf \* (1 - Rf / Ro)) 2.390 %

#### Minimum Design Metal Temperature Results:

Govern. thk, tg = 12.0, tr = 5.092, c = 3.175 mm., E\* = 1.0



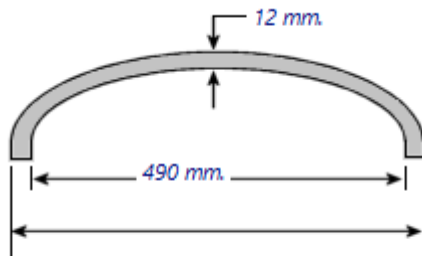
Thickness Ratio =  $tr * E^* / (tg - c) = 0.577$ , Temp. Reduction = 24 °C

Min Metal Temp. w/o impact per UCS-66, Curve B      -23 °C  
Min Metal Temp. at Required thickness (UCS 66.1)      -47 °C  
Min Metal Temp. w/o impact per UG-20(f)      -29 °C

| MDMT Design Information: |            |                        |             |
|--------------------------|------------|------------------------|-------------|
| MDMT Curve               | B          | Press. at MDMT         | 27.970 bars |
| UCS-66(b) red.           | Yes        | Min. Des. Mtl. Temp.   | -29 °C      |
| Governing Thk.           | 12.000 mm. | Basic MDMT             | -23 °C      |
| Stress/Thk. Ratio        | 0.577      | Temperature Difference | 24 °C       |
| UCS-66(c) red.           | No         | Computed Min. Temp.    | -47 °C      |
|                          |            | MDMT per UG-20(f)      | -29 °C      |
|                          |            | Joint Efficiency E*    | 100 %       |

### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



#### Internal Pressure Results for: Rear End Cover

| Elliptical Head Design Information: |    |                         |                         |    |                          |
|-------------------------------------|----|-------------------------|-------------------------|----|--------------------------|
| Design Pressure                     | P  | 25.559 bars             | Int. Design Temperature |    | 127 °C                   |
| Material                            |    | SA-516 70               | Ext. Design Temperature |    | 128 °C                   |
| Length                              |    | 50.800 mm.              | External Pressure Chart |    | CS-2                     |
|                                     |    |                         | UNS Number              |    | K02700                   |
| Int. Corr. All.                     | c  | 3.1750 mm.              | Allowable Stress (ope)  | S  | 138.0 N./mm <sup>2</sup> |
| Ext. Corr. All.                     | ce | 0.0000 mm.              | Allowable Stress (amb)  | Sa | 138.0 N./mm <sup>2</sup> |
| Inside Diameter                     | Di | 490.000 mm.             | Actual Stress           |    | 70.9 N./mm <sup>2</sup>  |
| Spec. Min. Thk t,e, or sa           |    | 12.0000 mm.             | Spec. Nominal Thk.      |    | 14.000 mm.               |
| Surface Area                        |    | 3735.3 cm <sup>2</sup>  | SG of Contents          |    | 1.000                    |
| Element Volume                      |    | 24984.3 cm <sup>3</sup> | Weight of Contents      |    | 25.0 Kg <sub>f</sub>     |
| Empty Weight                        |    | 41.7 Kg <sub>f</sub>    | Operating Weight        |    | 66.7 Kg <sub>f</sub>     |
|                                     |    |                         | Comp. All. Ope. Stress  |    | 141.0 N./mm <sup>2</sup> |
|                                     |    |                         | Aspect Ratio            |    | 2.00                     |

| Radiography Information: |       |                  |              |
|--------------------------|-------|------------------|--------------|
| User Defined             |       | User Defined     |              |
| Circ. Joint Efficiency   | 100 % | Long. Joint Eff. | E or v 100 % |

#### Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.125 mm.  
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.062 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (25.559 \cdot 496.35 \cdot 0.983) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 25.559)$$

$$= 4.5283 + 3.1750 = 7.7033 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.048 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 8.825) / (0.983 \cdot 496.35 + 0.2 \cdot 8.825)$$

$$= 50 - 0 = 50 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 12.0) / (1.0 \cdot 490.0 + 0.2 \cdot 12.0)$$

$$= 67 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (25.559 \cdot (0.983 \cdot 496.35 + 0.2 \cdot 8.825)) / (2 \cdot 1.0 \cdot 8.825)$$

$$= 70.934 \text{ N./mm}^2$$

Straight Flange Required Thickness:  
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$  per UG-27 (c)(1)  
 $= (25.559 \cdot 248.175) / (138.0 \cdot 1.0 - 0.6 \cdot 25.559) + 3.175$   
 $= 7.824 \text{ mm.}$

Straight Flange Maximum Allowable Working Pressure:  
[Less Operating Hydrostatic Head Pressure of 0.048 bars](#)  
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$  per UG-27 (c)(1)  
 $= (138.0 \cdot 1.0 \cdot 10.825) / (248.175 + 0.6 \cdot 10.825)$   
 $= 59 - 0 = 59 \text{ bars}$

Factor K, corroded condition [Kcor]:  
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$   
 $= (2 + (496.35 / (2 \cdot 125.675))^2) / 6$   
 $= 0.983264$

Percent Elong. per UCS-79, VIII-1-01-57  $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$  11.628 %  
[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

#### MDMT Calculations in the Knuckle Portion:

Govrn. thk,  $t_g = 12.0$ ,  $t_r = 4.956$ ,  $c = 3.175 \text{ mm.}$ ,  $E^* = 1.0$   
Thickness Ratio  $= t_r \cdot E^* / (t_g - c) = 0.562$ , Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

#### MDMT Calculations in the Head Straight Flange:

Govrn. thk,  $t_g = 14.0$ ,  $t_r = 5.092$ ,  $c = 3.175 \text{ mm.}$ ,  $E^* = 1.0$   
Thickness Ratio  $= t_r \cdot E^* / (t_g - c) = 0.47$ , Temp. Reduction = 37 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

| MDMT Design Information: |            |                        |             |
|--------------------------|------------|------------------------|-------------|
| MDMT Curve               | B          | Press. at MDMT         | 27.970 bars |
| UCS-66(b) red.           | Yes        | Min. Des. Mtl. Temp.   | -29 °C      |
| Governing Thk.           | 12.000 mm. | Basic MDMT             | -23 °C      |
| Stress/Thk. Ratio        | 0.562      | Temperature Difference | 25 °C       |
| UCS-66(c) red.           | No         | Computed Min. Temp.    | -48 °C      |
|                          |            | MDMT per UG-20(f)      | -29 °C      |
|                          |            | Joint Efficiency $E^*$ | 100 %       |

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

#### Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.125 mm.  
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.062 mm.

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### Hydrostatic Test Pressure Results:

#### Exchanger Shell Side Hydrostatic Test Pressures:

|                 |                               |             |
|-----------------|-------------------------------|-------------|
| UG99(b)         | = 1.30 * M.A.W.P. * Sa/S      | 36.361 bars |
| UG99b(b)(35)    | = 1.30 * Design Pres * Sa/S   | 33.164 bars |
| UG99(c)         | = 1.30 * M.A.P. - Head(Hyd)   | 37.130 bars |
| UG100           | = 1.10 * M.A.W.P. * Sa/S      | 30.767 bars |
| PED             | = max(1.43*DP, 1.25*DP*ratio) | 36.481 bars |
| PED&UG99(b)(35) | = max(1.43*DP, 1.3*DP*ratio)  | 36.481 bars |
| App 27-4        | = M.A.W.P.                    | 27.970 bars |

#### Exchanger Channel Side Hydrostatic Test Pressures:

|                 |                               |             |
|-----------------|-------------------------------|-------------|
| UG99(b)         | = 1.30 * M.A.W.P. * Sa/S      | 36.361 bars |
| UG99b(b)(35)    | = 1.30 * Design Pres * Sa/S   | 7.171 bars  |
| UG99(c)         | = 1.30 * M.A.P. - Head(Hyd)   | 37.131 bars |
| UG100           | = 1.10 * M.A.W.P. * Sa/S      | 30.767 bars |
| PED             | = max(1.43*DP, 1.25*DP*ratio) | 7.888 bars  |
| PED&UG99(b)(35) | = max(1.43*DP, 1.3*DP*ratio)  | 7.888 bars  |
| App 27-4        | = M.A.W.P.                    | 27.970 bars |

#### Horizontal Test performed per: UG-99b

*Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.*

#### Stresses on Elements due to Test Pressure (N./mm<sup>2</sup> & bars):

| From To        | Stress | Allowable | Ratio | Pressure |
|----------------|--------|-----------|-------|----------|
| Fr. END COVER  | 101.0  | 179.4     | 0.563 | 36.41    |
| CH.SHELL       | 104.6  | 179.4     | 0.583 | 36.41    |
| Main shell     | 104.6  | 179.4     | 0.583 | 36.41    |
| Rear End Cover | 101.0  | 179.4     | 0.563 | 36.41    |

#### Stress ratios for Nozzle and Pad Materials (N./mm<sup>2</sup>):

| Description   | Pad/Nozzle | Ambient | Operating | Ratio |
|---------------|------------|---------|-----------|-------|
| TS IN NOZ T1  | Nozzle     | 118.00  | 118.00    | 1.000 |
| TS IN NOZ T1  | Pad        | 138.00  | 138.00    | 1.000 |
| TS OUT NOZ T2 | Nozzle     | 118.00  | 118.00    | 1.000 |
| TS OUT NOZ T2 | Pad        | 138.00  | 138.00    | 1.000 |
| SH IN NOZ S1  | Nozzle     | 118.00  | 118.00    | 1.000 |
| SH OUT NOZ S2 | Nozzle     | 118.00  | 118.00    | 1.000 |

Minimum 1.000

#### Stress ratios for Pressurized Vessel Elements (N./mm<sup>2</sup>):

| Description    | Ambient | Operating | Ratio |
|----------------|---------|-----------|-------|
| Fr. END COVER  | 138.00  | 138.00    | 1.000 |
| CH.SHELL       | 138.00  | 138.00    | 1.000 |
| CH FLG         | 138.00  | 138.00    | 1.000 |
| Shell FLG      | 138.00  | 138.00    | 1.000 |
| Main shell     | 138.00  | 138.00    | 1.000 |
| Rear End Cover | 138.00  | 138.00    | 1.000 |

Minimum 1.000

#### Stress ratios for Exchanger Materials (N./mm<sup>2</sup>):

| Description   | Ambient | Operating | Ratio |
|---------------|---------|-----------|-------|
| Tube Material | 92.40   | 92.40     | 1.000 |



|                    |        |        |       |
|--------------------|--------|--------|-------|
| Tubesheet Material | 138.00 | 138.00 | 1.000 |
|--------------------|--------|--------|-------|

Minimum 1.000

**Hoop Stress in Nozzle Wall during Pressure Test (N./mm<sup>2</sup>):**

| Description   | Ambient | Operating | Ratio |
|---------------|---------|-----------|-------|
| TS IN NOZ T1  | 37.83   | 153.40    | 0.247 |
| TS OUT NOZ T2 | 37.83   | 153.40    | 0.247 |
| SH IN NOZ S1  | 40.42   | 153.40    | 0.263 |
| SH OUT NOZ S2 | 40.42   | 153.40    | 0.263 |

Elements Suitable for Test Pressure.

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**Input Echo, Tubesheet Number 1, Description: U-Tubesheet**

**Shell Data:**

**Main Shell Description: Main shell**

|                                         |        |             |
|-----------------------------------------|--------|-------------|
| Shell Test Pressure                     |        | 36.36 bars  |
| Shell Thickness                         | ts     | 12.0000 mm. |
| Shell Internal Corrosion Allowance      | cas    | 0.0000 mm.  |
| Shell External Corrosion Allowance      | casext | 0.0000 mm.  |
| Inside Diameter of Shell                | Ds     | 490.000 mm. |
| Shell Temperature for Internal Pressure | Ts     | 21.11 °C    |
| Shell Material                          |        | SA-516 70   |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                       |    |                           |
|---------------------------------------|----|---------------------------|
| Shell Material UNS Number             |    | K02700                    |
| Shell Allowable Stress at Temperature | Ss | 138.00 N./mm <sup>2</sup> |
| Shell Allowable Stress at Ambient     |    | 138.00 N./mm <sup>2</sup> |

**Channel Data:**

**Channel Description: CH.SHELL**

|                                      |        |             |
|--------------------------------------|--------|-------------|
| Channel Type:                        |        | Cylinder    |
| Channel Test Pressure                |        | 36.36 bars  |
| Channel Thickness                    | tc     | 12.0000 mm. |
| Channel Internal Corrosion Allowance | cac    | 0.0000 mm.  |
| Channel External Corrosion Allowance | cacext | 0.0000 mm.  |
| Inside Diameter of Channel           | Dc     | 490.000 mm. |
| Channel Design Temperature           | TEMPC  | 21.11 °C    |
| Channel Flange Matl                  |        | SA-516 70   |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                         |    |                           |
|-----------------------------------------|----|---------------------------|
| Channel Material UNS Number             |    | K02700                    |
| Channel Allowable Stress at Temperature | Sc | 138.00 N./mm <sup>2</sup> |
| Channel Allowable Stress at Ambient     |    | 138.00 N./mm <sup>2</sup> |

**Tube Data:**

|                                                  |            |                           |
|--------------------------------------------------|------------|---------------------------|
| Number of Tube Holes                             | Nt         | 160                       |
| Tube Wall Thickness                              | et         | 2.7700 mm.                |
| Tube Outside Diameter                            | D          | 19.0500 mm.               |
| Total Straight Tube Length                       | Lt         | 6093.00 mm.               |
| Straight Tube Length (bet. inner tubsht faces) L |            | 6041.00 mm.               |
| Design Temperature of the Tubes                  |            | 21.11 °C                  |
| Tube Material                                    |            | SA-179                    |
| Tube Material UNS Number                         |            | K01200                    |
| Is this a Welded Tube                            |            | No                        |
| Tube Material Specification used                 | Smls. tube |                           |
| Tube Allowable Stress at Temperature             |            | 92.40 N./mm <sup>2</sup>  |
| Tube Allowable Stress At Ambient                 |            | 92.40 N./mm <sup>2</sup>  |
| Tube Yield Stress At design Temperature          | Syt        | 180.00 N./mm <sup>2</sup> |
| Tube Pitch (Center to Center Spacing)            | P          | 31.7500 mm.               |
| Tube Layout Pattern                              |            | Triangular                |
| Fillet Weld Leg                                  | af         | 2.7700 mm.                |
| Groove Weld Leg                                  | ag         | 2.7700 mm.                |
| Tube-Tubesheet Joint Weld Type                   |            | Full Strength             |
| Method for Tube-Tubesheet Jt. Allowable          |            | UW-20.3                   |
| Is the Tube Inset                                |            | No                        |
| Radius to Outermost Tube Hole Center             | ro         | 228.665 mm.               |
| Largest Center-to-Center Tube Distance           | Ul         | 35.0000 mm.               |
| Length of Expanded Portion of Tube               | ltx        | 0.0000 mm.                |
| Tube-side pass partition groove depth            | hg         | 6.0000 mm.                |

**Tubesheet Data:**

**Tubesheet TYPE: U-tube, Gasketed both Sides, Conf. d**

|                                    |   |          |
|------------------------------------|---|----------|
| Tubesheet Design Metal Temperature | T | 21.11 °C |
| Tubesheet Material                 |   | SA-266 4 |



See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                           |      |                           |
|-------------------------------------------|------|---------------------------|
| Tubesheet Material UNS Number             |      | K03017                    |
| Tubesheet Allowable Stress at Temperature | S    | 138.00 N./mm <sup>2</sup> |
| Tubesheet Allowable Stress at Ambient     | Tt   | 138.00 N./mm <sup>2</sup> |
| Thickness of Tubesheet                    | h    | 52.0000 mm.               |
| Tubesheet Corr. Allowance (Shell side)    | Cats | 0.0000 mm.                |
| Tubesheet Corr. Allowance (Channel side)  | Catc | 0.0000 mm.                |
| Tubesheet Outside Diameter                | A    | 622.000 mm.               |

#### Additional Data for Stepped Tubesheets:

|                                                 |     |            |
|-------------------------------------------------|-----|------------|
| Is the Tubesheet Stepped?                       |     | Yes        |
| Is the Tubesheet Flat on Tubeside?              |     | No         |
| Step 1 Diameter on the Tubeside                 | dt1 | 521.00 mm. |
| Step 1 Depth on the Tubeside                    | ht1 | 6.00 mm.   |
| Step 2 Diameter on the Tubeside                 | dt2 | 487.00 mm. |
| Step 2 Depth on the Tubeside                    | ht2 | 6.00 mm.   |
| Is the Tubesheet Flat on Shellside?             |     | No         |
| Step 1 Diameter on the Shellside                | ds1 | 521.00 mm. |
| Step 1 Depth on the Shellside                   | hs1 | 6.00 mm.   |
| Step 2 Diameter on the Shellside                | ds2 | 487.00 mm. |
| Step 2 Depth on the Shellside                   | hs2 | 6.00 mm.   |
| Calculated Tubesheet Diameter per Div 2, 4.18.4 |     | 487.00 mm. |

Note: Tubesheet diameter is now: 487.000 mm. Div 2, 4.18.4.

|                                  |    |                       |
|----------------------------------|----|-----------------------|
| Dimension G for the Channel Side | Gc | 508.000 mm.           |
| Area of the Untubed Lanes        | AL | 253.0 cm <sup>2</sup> |

|                                      |                             |
|--------------------------------------|-----------------------------|
| Junction Stress Reduction option     | Perform Plastic Calculation |
| Perform Differential Pressure Design | No                          |
| Run Multiple Load Cases              | Yes                         |

#### Additional Data for Gasketed Tubesheets:

|                                            |                 |                          |
|--------------------------------------------|-----------------|--------------------------|
| Tubesheet Gasket on which Side             |                 | Both                     |
| Flange Outside Diameter                    | A               | 622.000 mm.              |
| Flange Inside Diameter                     | B               | 490.000 mm.              |
| Flange Face Outside Diameter               | Fod             | 524.000 mm.              |
| Flange Face Inside Diameter                | Fid             | 490.000 mm.              |
| Gasket Outside Diameter                    | Go              | 518.000 mm.              |
| Gasket Inside Diameter                     | Gi              | 498.000 mm.              |
| Small end Hub thk.                         | g0              | 12.0000 mm.              |
| Large end Hub thk.                         | g1              | 16.0000 mm.              |
| Gasket Factor,                             | m               | 3.75                     |
| Gasket Design Seating Stress               | y               | 62.06 N./mm <sup>2</sup> |
| Flange Facing Sketch                       | Code Sketch 1a  |                          |
| Gasket Thickness                           | tg              | 4.5000 mm.               |
| Full face Gasket Flange Option             | Program Selects |                          |
| Length of Partition Gasket                 | lp              | 498.000 mm.              |
| Width of Partition Gasket                  | wp              | 10.0000 mm.              |
| Partition Gasket Factor,                   | mPart           | 3.7500                   |
| Partition Gasket Design Seating Stress     | yPart           | 52.40 N./mm <sup>2</sup> |
| Partition Gasket Facing Sketch             | Code Sketch 1a  |                          |
| Partition Gasket Column for Gasket Seating | Code Column II  |                          |

#### Bolting Information:

|                                       |                    |                           |
|---------------------------------------|--------------------|---------------------------|
| Diameter of Bolt Circle               | C                  | 580.000 mm.               |
| Nominal Bolt Diameter                 | dB                 | 19.0500 mm.               |
| Thread Series                         | TEMA Thread Series |                           |
| Number of Bolts                       | n                  | 24                        |
| Bolt Material                         | SA-193 B7          |                           |
| Bolt Allowable Stress At Temperature  | Sb                 | 172.00 N./mm <sup>2</sup> |
| Bolt Allowable Stress At Ambient      | Sa                 | 172.00 N./mm <sup>2</sup> |
| Weld between Flange and Shell/Channel |                    | 0.0000 mm.                |

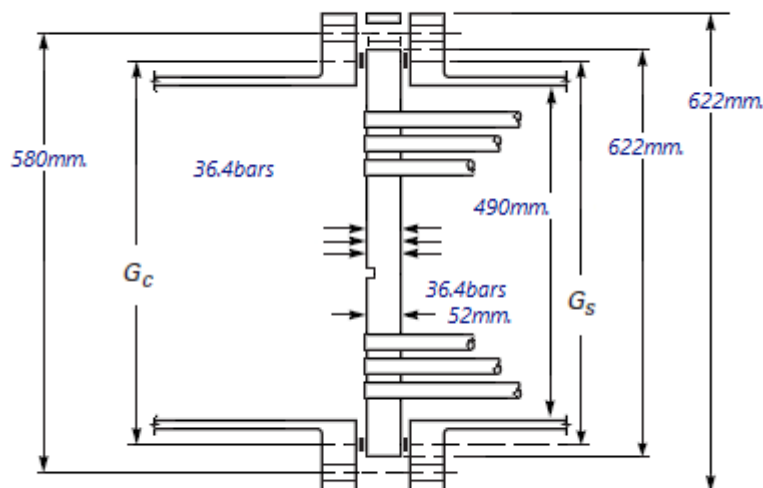
|                                           |              |
|-------------------------------------------|--------------|
| Alternate Flange Operating Bolt Load, Wm1 | 104268.09 Kg |
| Alternate Flange Seating Bolt Load, Wm2   | 63798.16 Kg  |
| Alternate Flange Design Bolt Load, W      | 82012.02 Kg  |



Note: The full bolt load  $S_a \cdot A_b$ , was used for all load cases.

|                                            |                |
|--------------------------------------------|----------------|
| Tubesheet Integral with                    | None           |
| Tubesheet Extended as Flange               | Yes            |
| Thickness of Extended Portion of Tubesheet | Tf 32.0000 mm. |
| Is Bolt Load Transferred to the Tubesheet  | Yes            |

### Tubesheet Gasketed w/Shell and Channel



Configuration d:

### ASME TubeSheet Results per Part UHX/4.18, 2025

#### Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with  $C \leq 0.3\%$

|                                     |         |                  |
|-------------------------------------|---------|------------------|
| Elastic Mod. at Design Temperature  | 21.1 °C | 0.20270E+09 KPa. |
| Elastic Mod. at Ambient Temperature | 21.1 °C | 0.20270E+09 KPa. |

Channel - TM-1 Carbon Steels with  $C \leq 0.3\%$

|                                     |         |                  |
|-------------------------------------|---------|------------------|
| Elastic Mod. at Design Temperature  | 21.1 °C | 0.20270E+09 KPa. |
| Elastic Mod. at Ambient Temperature | 21.1 °C | 0.20270E+09 KPa. |

Tubes - TM-1 Carbon Steels with  $C \leq 0.3\%$

|                                      |         |                  |
|--------------------------------------|---------|------------------|
| Elastic Mod. at Tubsht. Design Temp. | 21.1 °C | 0.20270E+09 KPa. |
| Elastic Mod. at Ambient Temperature  | 21.1 °C | 0.20270E+09 KPa. |

TubeSheet - TM-1 Carbon Steels with  $C \leq 0.3\%$

|                                     |         |                  |
|-------------------------------------|---------|------------------|
| Elastic Mod. at Design Temperature  | 21.1 °C | 0.20270E+09 KPa. |
| Elastic Mod. at Ambient Temperature | 21.1 °C | 0.20270E+09 KPa. |

#### Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:  
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= (36.36 \cdot (19.05/2 - 0)) / (92.4 \cdot 1.0 + 0.4 \cdot 36.36)$   
 $= 0.3690 + 0.0000 = 0.3690 \text{ mm.}$

#### Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 21.11 °C  
 Elastic Modulus for Material 199943392.00 KPa.



Results for Max. Allowable External Pressure (Emawp):

| TCA                                                    | ODCA  | SLEN    | D/T  | L/D     | Factor A  | B     |
|--------------------------------------------------------|-------|---------|------|---------|-----------|-------|
| 2.7700                                                 | 19.05 | 6269.67 | 6.88 | 50.0000 | 0.0232575 | 95.84 |
| EMAWP = $(2.167 / (D/T) - 0.0833) * B = 222.1421$ bars |       |         |      |         |           |       |

Results for Req'd Thickness for Ext. Pressure (Tca):

| TCA                                                                       | ODCA  | SLEN    | D/T   | L/D     | Factor A  | B     |
|---------------------------------------------------------------------------|-------|---------|-------|---------|-----------|-------|
| 0.6458                                                                    | 19.05 | 6269.67 | 29.50 | 50.0000 | 0.0012643 | 80.45 |
| EMAWP = $(4*B) / (3*(D/T)) = (4 * 80.4497) / (3 * 29.4962) = 36.364$ bars |       |         |       |         |           |       |

**Summary of Tube Required Thickness Results:**

|                                                   |             |
|---------------------------------------------------|-------------|
| Total Required Thickness including Corrosion all. | 0.6458 mm.  |
| Allowable Internal Pressure at Corroded thickness | 304.06 bars |
| Required Internal Design Pressure                 | 36.36 bars  |
| Allowable External Pressure at Corroded thickness | 222.14 bars |
| Required External Design Pressure                 | 36.36 bars  |
| Required Thickness due to Shell Side pressure     | 0.6458 mm.  |

**Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)**

**Intermediate Calculations For Tubesheets Extended As Flanges:**

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|                           |                       |             |
|---------------------------|-----------------------|-------------|
| Gasket Contact Width,     | $N = (Goc - Gic) / 2$ | 10.000 mm.  |
| Basic Gasket Width,       | $b0 = N / 2$          | 5.000 mm.   |
| Effective Gasket Width,   | $b = b0$              | 5.000 mm.   |
| Gasket Reaction Diameter, | $G = (Go + Gi) / 2$   | 508.000 mm. |

**Warning:** Available bolt area, 46.761 cm<sup>2</sup> for then shell flange is < required 59.451 cm<sup>2</sup>

**Note that for the pressure test case, ASME VIII-2 paragraph 4.18.13 (c)(9) does not require that the bolt area requirement be met.**

**Results for ASME U-tube Tubesheet Calculations for Configuration d:**

**Per Edition 2025, Original Thickness :**

**ASME Section VIII-2 Part 4.18.7.4 Step 1:**

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot ro + dt \\ &= 2 \cdot 228.665 + 19.05 \\ &= 476.380 \text{ mm.} \end{aligned}$$

Basic ligament efficiency for shear [ $\mu$ ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (31.75 - 19.05) / 31.75 \\ &= 0.400 \end{aligned}$$

**ASME Section VIII-2 Part 4.18.7.4 Step 2:**

Ratio [ $\rho_s$ ]:

$$\begin{aligned} &= Gs / Do \text{ (Configurations d, e, f)} \\ &= 508.0 / 476.38 \\ &= 1.0664 \end{aligned}$$

Ratio [ $\rho_c$ ]:

$$\begin{aligned} &= Gc / Do \text{ (Configurations d)} \\ &= 508.0 / 476.38 \\ &= 1.0664 \end{aligned}$$

**ASME Section VIII-2 Part 4.18.7.4 Step 3:**

Ratio [ $\rho$ ]:

$$= ltx / h$$



$$= 0/52.0$$
$$= 0.000 \text{ ( must be } 0 \leq \rho \leq 1 \text{ )}$$

Effective tube hole Diameter [d\*]:

$$= \max( dt - 2tt( EtT/E ) \cdot ( StT/S ) \cdot ( \rho ); dt - 2tt )$$
$$= \max( 19.05 - 2 \cdot 2.77 ( 0.20270E+09/.20270E+09 ) \cdot$$
$$( 92/137 ) \cdot ( 0 ); 19.05 - 2 \cdot 2.77 )$$
$$= 19.0500 \text{ mm.}$$

Effective tube pitch [p\*]:

$$= p / \sqrt{ ( 1 - 4 \cdot \min( A1; 4 \cdot Do \cdot p ) / ( \pi \cdot Do^2 ) ) }$$
$$= 31.75 / \sqrt{ ( 1 - 4 \cdot \min( 252.98 ; 4 \cdot 476.38 \cdot 31.75 ) / ( \pi \cdot 476.38^2 ) ) }$$
$$= 34.2755 \text{ mm.}$$

Effective ligament efficiency for bending [ $\mu^*$ ]:

$$= ( p^* - d^* ) / p^*$$
$$= ( 34.2755 - 19.05 ) / 34.2755$$
$$= 0.4442$$

$$h/p = 1.637795 ; \mu^* = 0.444209 ; E^* = 98626944 \text{ KPa.}$$
$$E^*/E = 0.486563 ; \nu^* = 0.297378$$

**Skip Step 4 for Configuration d :**

**ASME Section VIII-2 Part 4.18.7.4 Step 5:**

Diameter ratio [K]:

$$= A / Do$$
$$= 487.0/476.38$$
$$= 1.0223$$

Coefficient [F]:

$$= ( 1 - \nu^* ) / E^* \cdot ( E \cdot \ln(K) )$$
$$= ( 1 - 0.3 ) / 98626944 \cdot ( 0.20270E+09 \cdot \ln(1.02) )$$
$$= 0.0318$$

**ASME Section VIII-2 Part 4.18.7.4 Step 6:**

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= Do^2/16 \cdot [ (ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt ]$$
$$= 476.38^2/16 \cdot [ (1.066 - 1) \cdot (1.066^2 + 1) \cdot 36.361 -$$
$$(1.066 - 1) \cdot (1.066^2 + 1) \cdot 36.361 ]$$
$$= 0.0000 \text{ bars}\cdot\text{mm.}^2$$

Moment acting on unperforated tubesheet rim [M\*]:

$$= Mts + W^* \cdot ( Gc - Gs ) / ( 2 \cdot \pi \cdot Do )$$
$$= 0 + 104268.1 \cdot ( 508.0 - 508.0 ) / ( 2 \cdot \pi \cdot 476.38 )$$
$$= 0.0000 \text{ bars}\cdot\text{mm.}^2$$

*Note: W\* is the maximum of the bolt loads between the shell and channel sides.*

**ASME Section VIII-2 Part 4.18.7.4 Step 7:**

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ( (M^*) - Do^2/32 \cdot F( Ps - Pt ) ) / ( 1 + F )$$
$$= ( (0) - 476.38^2/32 \cdot 0.032 ( 36.36 - 36.36 ) ) / ( 1 + 0.03 )$$
$$= 0.0000 \text{ bars}\cdot\text{mm.}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot ( 3 + \nu^* ) \cdot ( Ps - Pt )$$
$$= 0 + 476.38^2/64 \cdot ( 3 + 0.297 ) \cdot ( 36.36 - 36.36 )$$
$$= 0.0000 \text{ bars}\cdot\text{mm.}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max( | Mp | ; | Mo | )$$
$$= \max( | 0 | ; | 0 | )$$
$$= 0.0000 \text{ bars}\cdot\text{mm.}^2$$

**ASME Section VIII-2 Part 4.18.7.4 Step 8:**



Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 0 / (0.4442 (52.0 - 6.0)^2) \\ &= 0.0000 \text{ N./mm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [ $\sigma_{All}$ ]:

$$\begin{aligned} &= 2 \cdot \beta T \cdot S_y \\ &= 2 \cdot 0.95 \cdot 250.0 \\ &= 474.99 \text{ N./mm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [ $\sigma$ ]:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 0 / (0.4442 (0.508 - 6.0)^2) \\ &= 0.0000 \text{ N./mm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [ $H_{reqB}$ ]:

$$\begin{aligned} &= h + C_{ats} + C_{atc} \\ &= 0.508 + 0 + 0 \\ &= 0.5080 \text{ mm.} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [ $H_{req}$ ]:

$$\begin{aligned} &= \max(H_{reqB}; H_{reqS}) \\ &= \max(0.508; 0) \\ &= 0.5080 \text{ mm.} \end{aligned}$$

#### ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [ $|P_s - P_t|$ ]:

$$\begin{aligned} &= |P_s - P_t| \\ &= |36.36 - 36.36| \\ &= 0.000 \text{ bars} \end{aligned}$$

Shear Stress check [ $\tau_{limit}$ ]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / D_o \cdot 0.8 \cdot \beta T \cdot S_y \\ &= 4 \cdot 0.4 \cdot 52.0 / 476.38 \cdot 0.8 \cdot 0.95 \cdot 249.995 \\ &= 33.183 \text{ N./mm}^2 \end{aligned}$$

$|P_s - P_t|$  is  $\leq$  the Shear Stress limit ( $\tau_{limit}$ ). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [ $H_{reqS}$ ]:

$$\begin{aligned} &= 1 / (4 \cdot \mu) \cdot (D_o / (0.8 \cdot \beta T \cdot S_y)) \cdot |P_s - P_t| + C_{ats} + C_{atc} \\ &= 1 / (4 \cdot 0.4) \cdot (476.38 / 189.997) \cdot |36.36 - 36.36| + 0 \\ &= 0.0000 \text{ mm.} \end{aligned}$$

Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [ $A_t$ ]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (19.05 - 2.77) 2.77 \\ &= 1.417 \text{ cm}^2 \end{aligned}$$

#### Tube Weld Size Results per UW-20.3:

Tube Strength [ $F_t$ ]:

$$\begin{aligned} &= \pi \cdot t (d_o - t) S \\ &= \pi \cdot 2.77 (19.05 - 2.77) 170.997 \\ &= 2470.266 \text{ Kgf} \end{aligned}$$

Groove Weld Strength [ $F_g$ ]:

$$\begin{aligned} &= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t) \\ &= 0.85 \cdot \pi \cdot 2.77 (19.05 + 0.67 \cdot 2.77) 171.0 \\ &= 2470.2656 \text{ Kgf} \end{aligned}$$

Weld Strength Factor [ $f_w$ ]:

$$\begin{aligned} &= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\ &= 170.997 / 170.997 \\ &= 1.000 \end{aligned}$$

Design Strength Ratio [ $f_d$ ]:

= 1 for full strength welds

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 2.77 (19.05 + 0.67 \cdot 2.77) 171.0$$

$$= 1744.7003 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left( \sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd} - 0.75 \cdot do \right)$$

$$= 2 \left( \sqrt{(0.75 \cdot 19.05)^2 + 1.07 \cdot 2.77 \cdot (19.05 - 2.77) 1.0 \cdot 1.0} - 0.75 \cdot 19.05 \right)$$

$$= 3.198 \text{ mm.}$$

Minimum Required Fillet Weld Leg [afr]: 1.6620 mm.  
 Minimum Required Groove Weld Leg [agr]: 1.6620 mm.

Tube-Tubesheet Jt allowable, 2470.27 is >= tube strength 2470.27 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

#### Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

| Stress Description     | Actual | Allowable                | Pass/Fail |
|------------------------|--------|--------------------------|-----------|
| Tubesheet bend. stress | 0.0 <= | 475.0 N./mm <sup>2</sup> | Ok        |
| Tubesheet shear stress | 0.0 <= | 190.0 N./mm <sup>2</sup> | Ok        |

#### Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

| Thickness (mm.)                  | Required | Actual | P/F |
|----------------------------------|----------|--------|-----|
| Tubesheet Thickness :            | 0.508    | 52.000 | Ok  |
| Tube-Tubesheet Fillet Weld Leg : | 1.662    | 2.770  | Ok  |
| Tube-Tubesheet Groove Weld Leg : | 1.662    | 2.770  | Ok  |

#### U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

##### Results for 3 Load Cases:

| Case# | --Reqd. Thk. + CA<br>Tbsht Extnsn | ----       | Tubesheet<br>Bend | Stresses<br>Allwd | Shear<br>Allwd | Case<br>Type | Pass/<br>Fail |
|-------|-----------------------------------|------------|-------------------|-------------------|----------------|--------------|---------------|
| D1uc  | 43.244                            | ...        | 312               | 475               | 21             | 190          | Ps+Pt D1 Ok   |
| D2uc  | 43.244                            | 24.077     | 312               | 475               | 21             | 190          | Ps+Pt D2 Ok   |
| D3uc  | 0.508                             | ...        | ...               | 475               | ...            | 190          | Ps+Pt D3 Ok   |
| Max:  | 43.2435                           | 24.077 mm. | 0.656             | 0.110             | (Str. Ratio)   |              |               |

#### Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures  
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the  
 Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.



**Summary of Thickness Comparisons for 3 Load Cases:**

| Thickness (mm.)                         | Required | Actual | P/F |
|-----------------------------------------|----------|--------|-----|
| Tubesheet thickness :                   | 43.244   | 52.000 | Ok  |
| Tubesheet thickness flanged extension : | 24.077   | 32.000 | Ok  |
| Tube thickness :                        | 0.646    | 2.770  | Ok  |
| Tube-tubesheet fillet weld leg :        | 1.662    | 2.770  | Ok  |
| Tube-tubesheet groove weld leg :        | 1.662    | 2.770  | Ok  |

Note: This is a full strength Tube to Tubesheet Joint.

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## External Pressure Calculation Results Summary:

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#### External Pressure Calculations:

| From | To | Section Length<br>mm. | Outside Diameter<br>mm. | Corroded Thickness<br>mm. | Factor A  | Factor B<br>N./mm <sup>2</sup> |
|------|----|-----------------------|-------------------------|---------------------------|-----------|--------------------------------|
| 10   | 20 | No Calc               | 514                     | 8.825                     | 0.0023846 | 107.531                        |
| 20   | 30 | 490.833               | 514                     | 8.825                     | 0.0032165 | 113.11                         |
| 30   | 40 | No Calc               | ...                     | 56.825                    | No Calc   | No Calc                        |
| 40   | 50 | No Calc               | ...                     | 56.825                    | No Calc   | No Calc                        |
| 50   | 60 | 6187.63               | 514                     | 8.825                     | 0.0003364 | 33.6329                        |
| 60   | 70 | No Calc               | 514                     | 8.825                     | 0.0023846 | 107.531                        |

#### External Pressure Calculations:

| From | To | External Actual T.<br>mm. | External Required T.<br>mm. | External Design Pressure<br>bars | External M.A.W.P.<br>bars |
|------|----|---------------------------|-----------------------------|----------------------------------|---------------------------|
| 10   | 20 | 12                        | No Calc                     | ...                              | 20.5125                   |
| 20   | 30 | 12                        | No Calc                     | ...                              | 25.892                    |
| 30   | 40 | 60                        | 53.594                      | ...                              | No Calc                   |
| 40   | 50 | 60                        | 53.594                      | ...                              | No Calc                   |
| 50   | 60 | 12                        | No Calc                     | ...                              | 7.69892                   |
| 60   | 70 | 12                        | No Calc                     | ...                              | 20.5125                   |

Minimum

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#### External Pressure Calculations:

| From | To | Actual Length<br>Bet. Stiffeners<br>mm. | Allowable Length<br>Bet. Stiffeners<br>mm. | Ring Inertia<br>Required<br>cm <sup>4</sup> | Ring Inertia<br>Available<br>cm <sup>4</sup> |
|------|----|-----------------------------------------|--------------------------------------------|---------------------------------------------|----------------------------------------------|
| 10   | 20 | No Calc                                 | No Calc                                    | No Calc                                     | No Calc                                      |
| 20   | 30 | 490.833                                 | No Calc                                    | No Calc                                     | No Calc                                      |
| 30   | 40 | No Calc                                 | No Calc                                    | No Calc                                     | No Calc                                      |
| 40   | 50 | No Calc                                 | No Calc                                    | No Calc                                     | No Calc                                      |
| 50   | 60 | 6187.63                                 | No Calc                                    | No Calc                                     | No Calc                                      |
| 60   | 70 | No Calc                                 | No Calc                                    | No Calc                                     | No Calc                                      |

Elements Suitable for External Pressure.

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### Element and Detail Weights:

| From  | To | Element<br>Metal Wgt.<br>kg. | Element<br>ID Volume<br>cm <sup>3</sup> | Corroded<br>Metal Wgt.<br>kg. | Corroded<br>ID Volume<br>cm <sup>3</sup> | Extra due<br>Misc %<br>kg. |
|-------|----|------------------------------|-----------------------------------------|-------------------------------|------------------------------------------|----------------------------|
| 10    | 20 | 41.5786                      | 24833.4                                 | 32.1492                       | 25686                                    | 4.15786                    |
| 20    | 30 | 58.6711                      | 75443.1                                 | 43.4206                       | 77411.2                                  | 5.86711                    |
| 30    | 40 | 61.5142                      | 16351.5                                 | 58.2734                       | 16449.9                                  | 6.15142                    |
| 40    | 50 | 61.5142                      | 16351.5                                 | 58.2734                       | 16449.9                                  | 6.15142                    |
| 50    | 60 | 894.148                      | 859996                                  | 661.729                       | 889989                                   | 89.4148                    |
| 60    | 70 | 41.716                       | 24984.3                                 | 32.2554                       | 25840.8                                  | 4.1716                     |
| Total |    | 1159                         | 1017960.06                              | 886                           | 1051827.00                               | 115                        |



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

### Weight of Details:

| From | Type | Weight of<br>Detail<br>kg. | X Offset,<br>Dtl. Cent.<br>mm. | Y Offset,<br>Dtl. Cent.<br>mm. | Z Offset<br>Dtl. Cent.<br>mm. | Description    |
|------|------|----------------------------|--------------------------------|--------------------------------|-------------------------------|----------------|
| 10   | Liqd | 24.8182                    | -40.8333                       | ...                            | ...                           | Liquid 10      |
| 20   | Liqd | 75.3971                    | 200                            | ...                            | ...                           | Liquid 20      |
| 20   | Nozl | 34.2868                    | 200                            | -318.165                       | ...                           | TS IN NOZ T1   |
| 20   | Nozl | 34.2868                    | 200                            | 318.165                        | ...                           | TS OUT NOZ T2  |
| 30   | Liqd | 16.3415                    | 37.5                           | ...                            | ...                           | Liquid 30      |
| 40   | Liqd | 16.3415                    | 37.5                           | ...                            | ...                           | Liquid 40      |
| 40   | Insl | 1.17105                    | 37.5                           | ...                            | ...                           | HOT INSULATION |
| 50   | Sadl | 49.723                     | 1220                           | 360.5                          | ...                           | Left Saddle    |
| 50   | Sadl | 49.723                     | 4876                           | 360.5                          | ...                           | RIGHT SADDLE   |
| 50   | Liqd | 859.472                    | 3048                           | ...                            | ...                           | Liquid 50      |
| 50   | Insl | 72.3467                    | 3048                           | ...                            | ...                           | HOT INSULATION |
| 50   | Nozl | 78.4214                    | 5846                           | 341.837                        | ...                           | SH IN NOZ S1   |
| 50   | Nozl | 51.4053                    | 250                            | -341.837                       | ...                           | SH OUT NOZ S2  |
| 60   | Liqd | 24.969                     | 91.6333                        | 0.18167E-04                    | ...                           | Liquid 60      |
| 60   | Insl | 3.3753                     | 86.65                          | ...                            | ...                           | HOT INSULATION |
| 30   | FTsh | 114.492                    | 110.5                          | ...                            | ...                           | U-Tubesheet    |
| 30   | Tube | 1224.9                     | 3131                           | ...                            | ...                           | Tubes          |
| 40   | Baf1 | 5.22512                    | 466                            | ...                            | ...                           | Baffles        |
| 40   | Baf1 | 5.22512                    | 891                            | ...                            | ...                           | Baffles        |
| 40   | Baf1 | 5.22512                    | 1316                           | ...                            | ...                           | Baffles        |
| 40   | Baf1 | 5.22512                    | 1741                           | ...                            | ...                           | Baffles        |
| 40   | Baf1 | 5.22512                    | 2166                           | ...                            | ...                           | Baffles        |
| 40   | Baf1 | 5.22512                    | 2591                           | ...                            | ...                           | Baffles        |
| 40   | Baf1 | 5.22512                    | 3016                           | ...                            | ...                           | Baffles        |
| 40   | Baf1 | 5.22512                    | 3441                           | ...                            | ...                           | Baffles        |
| 40   | Baf1 | 5.22512                    | 3866                           | ...                            | ...                           | Baffles        |
| 40   | Baf1 | 5.22512                    | 4291                           | ...                            | ...                           | Baffles        |
| 40   | Baf1 | 5.22512                    | 4716                           | ...                            | ...                           | Baffles        |
| 40   | Baf1 | 5.22512                    | 5141                           | ...                            | ...                           | Baffles        |
| 40   | Baf1 | 5.22512                    | 5566                           | ...                            | ...                           | Baffles        |
| 40   | Baf1 | 5.22512                    | 5991                           | ...                            | ...                           | Baffles        |

### Total Weight of Each Detail Type:

|                           |            |
|---------------------------|------------|
| Saddles                   | 99.4       |
| Liquid                    | 1017.3     |
| Insulation                | 76.9       |
| Nozzles                   | 198.4      |
| Exchanger Components      | 1412.5     |
| Liquid in Tubes           | 147.7      |
| Sum of the Detail Weights | 2952.3 kg. |



### Weight Summation Results: (kg.)

|               | Fabricated | Shop Test | Shipping | Erected | Empty  | Operating |
|---------------|------------|-----------|----------|---------|--------|-----------|
| Main Elements | 1275.1     | 1275.1    | 1275.1   | 1275.1  | 1275.1 | 1275.1    |
| Saddles       | 99.4       | 99.4      | 99.4     | 99.4    | 99.4   | 99.4      |
| Nozzles       | 198.4      | 198.4     | 198.4    | 198.4   | 198.4  | 198.4     |
| Exchanger     | 1412.5     | 1412.5    | 1412.5   | 1412.5  | 1412.5 | 1412.5    |
| Insulation    | ...        | ...       | 76.9     | 76.9    | 76.9   | 76.9      |
| Ope. Liquid   | ...        | ...       | ...      | ...     | ...    | 1017.3    |
| Tube Ope Lqd  | ...        | ...       | ...      | ...     | ...    | 147.7     |
| Test Liquid   | ...        | 1017.3    | ...      | ...     | ...    | ...       |
| Tube Tst Lqd  | ...        | 148.6     | ...      | ...     | ...    | ...       |
| Totals        | 2985.5     | 4151.3    | 3062.3   | 3062.3  | 3062.3 | 4227.3    |

### Field Installation Options:

#### Miscellaneous Weight Percent: 10.0 %

*Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.*

### Weight Summary:

|                 |                                                     |            |
|-----------------|-----------------------------------------------------|------------|
| Fabricated Wt.  | - Bare weight without removable internals           | 2985.5 kg. |
| Shop Test Wt.   | - Fabricated weight + water ( full )                | 4151.3 kg. |
| Shipping Wt.    | - Fab. weight + removable intls.+ shipping app.     | 3062.3 kg. |
| Erected Wt.     | - Fab. wt + or - loose items (trays,platforms etc.) | 3062.3 kg. |
| Ope. Wt. no Liq | - Fab. weight + internals. + details + weights      | 3062.3 kg. |
| Operating Wt.   | - Empty weight + operating liq. uncorroded          | 4227.3 kg. |
| Oper. Wt. + CA  | - Corr wt. + operating liquid                       | 3927.0 kg. |

### Exchanger Tube Data

|                              |                          |
|------------------------------|--------------------------|
| Volume of Exchanger tubes :  | 148645.5 cm <sup>3</sup> |
| Weight of Ope Liq in tubes : | 147.7 kg.                |
| Weight of Water in tubes :   | 148.6 kg.                |

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Field Test weight is computed in the corroded condition.

### Surface Areas of Elements:

| From | To | Outside Surface Area cm <sup>2</sup> | Inside Surface Area cm <sup>2</sup> |
|------|----|--------------------------------------|-------------------------------------|
| 10   | 20 | 3722.26                              | 3372.34                             |
| 20   | 30 | 6459.11                              | 6157.52                             |
| 30   | 40 | 2750.97                              | 1308.47                             |
| 40   | 50 | 2750.97                              | 1308.47                             |
| 50   | 60 | 98436.9                              | 93840.6                             |
| 60   | 70 | 3735.28                              | 3384.65                             |

|       |          |                          |
|-------|----------|--------------------------|
| Total | 117855.5 | 109372.1 cm <sup>2</sup> |
| Total | 126.9    | 117.7 ft <sup>2</sup>    |

### Nozzle Flange MAWP Results: (bars & °C)

| Nozzle Description | Flange Rating |         | Des. Temp | Class | Grade/Group | Equiv. Press | Max Pressure |        |        |
|--------------------|---------------|---------|-----------|-------|-------------|--------------|--------------|--------|--------|
|                    | Ope.          | Ambient |           |       |             |              | UG-44        | 50%    | DNV    |
| TS IN NOZ T1       | 48.15         | 51.00   | 77        | 300   | GR 1.1      | 31.710       | 40.511       | 48.147 | 40.511 |
| TS OUT NOZ T2      | 48.15         | 51.00   | 77        | 300   | GR 1.1      | 31.710       | 40.511       | 48.147 | 40.511 |
| SH IN NOZ S1       | 45.85         | 51.00   | 127       | 300   | GR 1.1      | 27.642       | 41.129       | 45.847 | 41.129 |
| SH OUT NOZ S2      | 45.85         | 51.00   | 127       | 300   | GR 1.1      | 27.642       | 41.129       | 45.847 | 41.129 |

### Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 41.129 bars  
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 51.000 bars

### Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide ]: 40.511 bars  
 Lowest Flange Pressure Rating was (Amb)[TubeSide ]: 51.000 bars

### Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

| Nozzle Description | Gasket Dimension G mm. | Fm Value | Resolved Moment ME Kg-m. | Axial Force FE Kg | MAWP PD bars | Maximum Pressure Pmax bars | Pass Fail |
|--------------------|------------------------|----------|--------------------------|-------------------|--------------|----------------------------|-----------|
| TS IN NOZ T1       | 196.088                | 0.50     | 459                      | 408               | 27.970       | 40.511                     | Pass      |
| TS OUT NOZ T2      | 196.088                | 0.50     | 459                      | 408               | 27.970       | 40.511                     | Pass      |
| SH IN NOZ S1       | 248.539                | 0.50     | 816                      | 551               | 27.970       | 41.129                     | Pass      |
| SH OUT NOZ S2      | 248.539                | 0.50     | 816                      | 551               | 27.970       | 41.129                     | Pass      |

Evaluating Flange on Nozzle [TS IN NOZ T1]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(459) + 4 \cdot \text{abs}(408) \cdot 196.088 \leq \pi \cdot 196.088^3 \cdot ((48.147 - 27.970) + .500 \cdot 48.147)$$

$$= 0.76596E+04 \leq 0.10689E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle TS IN NOZ T1 [Pmax]:

$$= \min( PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3) )$$

$$= \min( 48.147, 0.500 \cdot 48.147 + 48.147 - 7659.6 / (\pi \cdot 196.088^3) )$$

$$= \min( 48.147, 40.511 )$$

$$= 40.511 \text{ bars}$$

Evaluating Flange on Nozzle [TS OUT NOZ T2]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(459) + 4 \cdot \text{abs}(408) \cdot 196.088 \leq \pi \cdot 196.088^3 \cdot ((48.147 - 27.970) + .500 \cdot 48.147)$$

$$= 0.76596E+04 \leq 0.10689E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle TS OUT NOZ T2 [Pmax]:

$$= \min( PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3) )$$

$$= \min( 48.147, 0.500 \cdot 48.147 + 48.147 - 7659.6 / (\pi \cdot 196.088^3) )$$

$$= \min( 48.147, 40.511 )$$

$$= 40.511 \text{ bars}$$

Evaluating Flange on Nozzle [SH IN NOZ S1]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(816) + 4 \cdot \text{abs}(551) \cdot 248.539 \leq \pi \cdot 248.539^3 \cdot ((45.847 - 27.970) + .500 \cdot 45.847)$$

$$= 0.13596E+05 \leq 0.20068E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle SH IN NOZ S1 [Pmax]:



$$\begin{aligned} &= \min( PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3) ) \\ &= \min( 45.847, 0.500 \cdot 45.847 + 45.847 - 13595.7 / (\pi \cdot 248.539^3) ) \\ &= \min( 45.847, 41.129 ) \\ &= 41.129 \text{ bars} \end{aligned}$$

Evaluating Flange on Nozzle [SH OUT NOZ S2]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [ (PR - PD) + FM \cdot PR ] \\ &= 16(816) + 4 \cdot \text{abs}(551) 248.539 \leq \pi \cdot 248.539^3 [ (45.847 - 27.970) + .500 \cdot 45.847 ] \\ &= 0.13596E+05 \leq 0.20068E+05 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle SH OUT NOZ S2 [Pmax]:

$$\begin{aligned} &= \min( PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3) ) \\ &= \min( 45.847, 0.500 \cdot 45.847 + 45.847 - 13595.7 / (\pi \cdot 248.539^3) ) \\ &= \min( 45.847, 41.129 ) \\ &= 41.129 \text{ bars} \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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## Wind Load Calculation Results:

### Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

#### Input Values:

|                                             |                           |
|---------------------------------------------|---------------------------|
| Wind Design Code                            | IS 875 (PART 3):2015-2020 |
| Basic Wind Speed for IS-875                 | 180 Km/hr                 |
| Wind Zone Number                            | 2                         |
| Base Elevation                              | 300 mm.                   |
| Percent Wind for Hydrotest                  | 33.0                      |
| Risk Factor                                 | [k1] 1.0                  |
| Terrain Category                            | 2                         |
| Equipment Class                             | Class A                   |
| Topography Factor                           | [k3] 1.0                  |
| Importance Factor                           | [k4] 1.15                 |
| Damping Factor (Beta) for Wind (Ope)        | 0.0100                    |
| Damping Factor (Beta) for Wind (Empty)      | 0.0000                    |
| Damping Factor (Beta) for Wind (Filled)     | 0.0000                    |
| Use Gust Response Factor (Dynamic Analysis) | No                        |

#### Calculated Values:

Vessel Length to Diameter ratio: 11.805

Design Wind Speed [Vz]:

= Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)

= 50.0 • 1.0 • 1.0 • 1.0 • 1.15

= 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:

= max( 0.7 • pz, Kd • Ka • Kc • 0.6 • Vz<sup>2</sup>)

= max( 0.7 • 1983.503, 1.0 • 1.0 • 1.0 • 0.6 • 57.5<sup>2</sup>)

= 1983.5 N/m<sup>2</sup>

Element Wind Load (operating condition):

= Wind Area • Cf • pd

= 0.1 • 0.918 • 1983.503

= 155.0 N

| From | Height<br>mm. | k1     | k2     | k3     | Vz<br>m/sec | Cf     |
|------|---------------|--------|--------|--------|-------------|--------|
| 10   | 800.00        | 1.0000 | 1.0000 | 1.0000 | 57.50       | 0.9181 |
| 20   | 800.00        | 1.0000 | 1.0000 | 1.0000 | 57.50       | 0.9181 |
| 30   | 800.00        | 1.0000 | 1.0000 | 1.0000 | 57.50       | 0.9181 |
| 40   | 800.00        | 1.0000 | 1.0000 | 1.0000 | 57.50       | 0.9181 |
| 50   | 800.00        | 1.0000 | 1.0000 | 1.0000 | 57.50       | 0.9181 |
| 60   | 800.00        | 1.0000 | 1.0000 | 1.0000 | 57.50       | 0.9181 |

#### Wind Load Calculation:

| From | To | Wind<br>Height<br>mm. | Wind<br>Diameter<br>mm. | Wind<br>Area<br>cm <sup>2</sup> | Wind<br>Pressure<br>Kgs/m <sup>2</sup> | Element<br>Wind Load<br>Kgf |
|------|----|-----------------------|-------------------------|---------------------------------|----------------------------------------|-----------------------------|
| 10   | 20 | 800                   | 616.8                   | 851.369                         | 202.347                                | 17.3907                     |
| 20   | 30 | 800                   | 616.8                   | 2467.2                          | 202.347                                | 50.397                      |
| 30   | 40 | 800                   | 588                     | 441                             | 202.347                                | 9.00822                     |
| 40   | 50 | 800                   | 696                     | 522                             | 202.347                                | 10.6628                     |
| 50   | 60 | 800                   | 724.8                   | 44183.8                         | 202.347                                | 902.533                     |
| 60   | 70 | 800                   | 724.8                   | 1219.71                         | 202.347                                | 24.9148                     |

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

PV Elite 28      Licensee: SL User  
FileName : 403-E-004  
Wind Load Calculation:

Step: 14 12:39pm

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Jul 4, 2026



**HEXAGON**

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### Earthquake Load Calculation Results:

#### Seismic Analysis Results per IS 1893 (Part 4): 2024

##### Input Data:

Importance Factor as Entered by User I 2.0000  
Zone Number 2  
  
Soil Type Rocky or Hard Soil  
Response reduction factor (Force factor) R 2.0000  
Time Period of Vibration ( program or user ) 0.0303 Seconds  
Value of Sa/g from user Spectrum Data 0.0939  
Percent Damping 2.0 %  
Considering MCE for load determination No

##### Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:  
= ( Sa/g )ss • DampVal ) • ( I / R )  
= ( 0.0939 • 1.0 ) • 1.0  
= 0.094

User Spectrum Data points specified for increments of Time Period

| Time Period | Sa/g  |
|-------------|-------|
| 0.010       | 0.070 |
| 0.050       | 0.117 |
| 0.100       | 0.175 |
| 0.130       | 0.175 |
| 0.200       | 0.175 |
| 0.270       | 0.175 |
| 0.300       | 0.175 |
| 0.400       | 0.175 |
| 0.500       | 0.140 |
| 0.550       | 0.127 |
| 0.600       | 0.117 |
| 0.670       | 0.105 |
| 0.800       | 0.087 |
| 1.000       | 0.070 |
| 1.200       | 0.058 |
| 1.400       | 0.050 |
| 1.600       | 0.044 |
| 1.800       | 0.039 |
| 2.000       | 0.035 |
| 2.400       | 0.029 |
| 0.000       | 0.000 |

Actual Time Period = 0.0303 secs  
Computed Sa/g = 0.0939

##### Note:

It was indicated to ignore the damping factor, assuming that the Site Spectrum specified is per the actual damping value of this vessel.  
The value of "DampVal" has been set to 1 (not used).

Earthquake Element Load, for the first Element:  
= Earthquake Weight • Ah • Scalar  
= 478.44 • 0.0939 • 1.0  
= 44.9 Kgf

Vertical load per 8.4:  
= 2/3 • Ah • W • Scalar  
= 2/3 • 0.0939 • 3828 • 1.0  
= 239.5 Kgf

##### Earthquake Load Calculation:

|            |            |         |
|------------|------------|---------|
| Earthquake | Earthquake | Element |
|------------|------------|---------|



| From | To   | Height<br>mm. | Weight<br>Kgf | Ope Load<br>Kgf |
|------|------|---------------|---------------|-----------------|
| 10   | 20   | 245           | 478.444       | 44.9049         |
| 20   | 30   | 245           | 478.444       | 44.9049         |
| 30   | 40   | 245           | 478.444       | 44.9049         |
| 40   | 50   | 245           | 478.444       | 44.9049         |
| 50   | Sad1 | 245           | 478.444       | 44.9049         |
| Sad1 | 60   | 245           | 478.444       | 44.9049         |
| 50   | 60   | 245           | 478.444       | 44.9049         |
| 60   | 70   | 245           | 478.444       | 44.9049         |

The Earthquake Weight does not include the weight of the saddles.

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**Shop/Field Installation Options :**

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

|                                                    |              |
|----------------------------------------------------|--------------|
| Center of Gravity of the Saddles                   | 3704.500 mm. |
| Center of Gravity of the Liquid                    | 3333.003 mm. |
| Center of Gravity of the Insulation                | 3795.107 mm. |
| Center of Gravity of the Nozzles                   | 2891.516 mm. |
| Center of Gravity of the Tubesheet(s)              | 560.500 mm.  |
| Center of Gravity of the Tubes                     | 3581.000 mm. |
| Center of Gravity of the Baffles                   | 3810.000 mm. |
|                                                    |              |
| Center of Gravity of Bare Shell New and Cold       | 3174.049 mm. |
| Center of Gravity of Bare Shell Corroded           | 3099.425 mm. |
|                                                    |              |
| Vessel CG in the Operating Condition               | 3285.912 mm. |
| Vessel CG in the Fabricated (Shop/Empty) Condition | 3268.819 mm. |
| Vessel CG in the Test Condition                    | 3295.228 mm. |

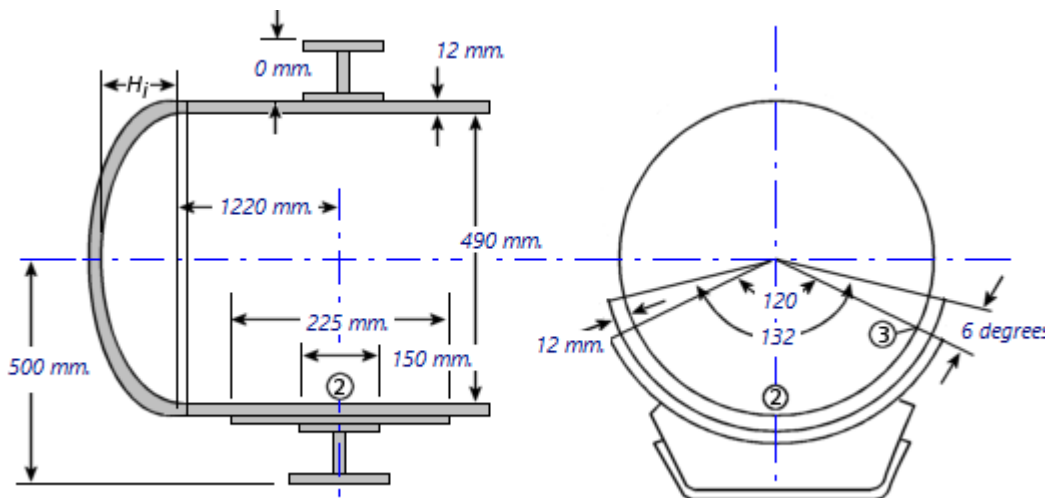
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### ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

#### Input and Calculated Values:

|                                               |                  |         |                    |
|-----------------------------------------------|------------------|---------|--------------------|
| Vessel Mean Radius                            | Rm               | 252.59  | mm.                |
| Shell Thickness used in this Case             | t                | 12.000  | mm.                |
| Stiffened Vessel Length per 4.15.6            | L                | 6221.80 | mm.                |
| Distance from Saddle to Vessel tangent        | a1 or a          | 1220.00 | mm.                |
| Saddle Width                                  | b1 or b          | 150.00  | mm.                |
| Saddle Bearing Angle                          | delta or theta   | 120.00  | degrees            |
| Wear Plate Width                              | b2 or b1         | 225.00  | mm.                |
| Wear Plate Bearing Angle                      | delta2 or theta1 | 132.00  | degrees            |
| Wear Plate Thickness                          | e2 or tr         | 12.0    | mm.                |
| Wear Plate Allowable Stress                   | fw or Sr         | 138.00  | N./mm <sup>2</sup> |
| Shell Allowable Stress used in Calculation    |                  | 138.00  | N./mm <sup>2</sup> |
| Head Allowable Stress used in Calculation     |                  | 138.00  | N./mm <sup>2</sup> |
| Circumferential Efficiency in Plane of Saddle |                  | 1.00    |                    |
| Circumferential Efficiency at Mid-Span        |                  | 1.00    |                    |
| Distance from Saddle Base to Centerline       | B                | 500.00  | mm.                |
| Coefficient of Friction                       | mu               | 0.45    |                    |



#### Results for Horizontal Vessel Analysis:

|                                |   |         |      |
|--------------------------------|---|---------|------|
| Saddle Force Q, Operating Case |   | 4137.88 | Kgf  |
| Pressure used in Analysis      | P | 25.535  | bars |

| Horizontal Vessel Analysis Results: | Actual<br>N./mm <sup>2</sup> | Allowable<br>N./mm <sup>2</sup> |
|-------------------------------------|------------------------------|---------------------------------|
| Long. Stress at Top of Midspan      | 28.73                        | 138.00                          |
| Long. Stress at Bottom of Midspan   | 44.36                        | 138.00                          |
| Long. Stress at Top of Saddles      | 86.92                        | 138.00                          |
| Long. Stress at Bottom of Saddles   | 8.62                         | 138.00                          |
| Tangential Shear in Shell           | 12.95                        | 110.40                          |
| Circ. Stress at Horn of Saddle      | 9.60                         | 172.50                          |
| Circ. Compressive Stress in Shell   | 0.66                         | 138.00                          |

#### Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:



$$= \text{Ftr}(\text{Ft}/\text{Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(1014.9/2 + 0) \cdot 500.0/460.0$$

$$= 1654.7 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(66.51, 0) \cdot 500.0/3656.0$$

$$= 9.1 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(359.24, 0) \cdot 500.0/3656.0$$

$$= 49.1 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft}/\text{Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(359/2 + 0) \cdot 500.0/460.0$$

$$= 585.7 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 2483 + \max(9, 1655, 49, 586)$$

$$= 4137.9 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1984.306(0.0 + \pi/4(0.514 \cdot 1.2)^2)$$

$$= 652.203 \text{ N}$$

#### Summary of Loads at the Base of this Saddle:

|                                         |         |     |
|-----------------------------------------|---------|-----|
| Vertical Load (including saddle weight) | 4187.60 | Kgf |
| Transverse Shear Load                   | 507.45  | Kgf |
| Longitudinal Shear Load                 | 359.24  | Kgf |

Note: Wear Plate is Welded to the Shell,  $k = 0.1$

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 460.000 \text{ mm.}$$

#### The Computed K values from Table 4.15.1:

|              |              |              |              |
|--------------|--------------|--------------|--------------|
| K1 = 0.1066  | K2 = 1.1707  | K3 = 0.8799  | K4 = 0.4011  |
| K5 = 0.7603  | K6 = 0.0529  | K7 = 0.0529  | K8 = 0.3405  |
| K9 = 0.2711  | K10 = 0.0581 | K1* = 0.1923 | K6p = 0.0434 |
| K7p = 0.0434 |              |              |              |

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to  $Rm/2$ .

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -4138 \cdot 1220.0 [1 - (1 - 1220.0/6221.8 + (252.587^2 - 0.0^2)/(2 \cdot 1220.0 \cdot 6221.8))/(1 + (4 \cdot 0.0)/(3 \cdot 6221.8))]$$

$$= -968.7 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 4138 \cdot 6222/4((1 + 2(253^2 - 0^2)/(6222^2))/(1 + (4 \cdot 0)/(3 \cdot 6222)) - 4 \cdot 1220/6222)$$

$$= 1409.3 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [ $\sigma_1$ ]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 25.535 \cdot 252.587/(2 \cdot 8.825) - 1409.3/(\pi \cdot 252.6^2 \cdot 8.825)$$

$$= 28.73 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [ $\sigma_2$ ]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 25.535 \cdot 252.587/(2 \cdot 8.825) + 1409.3/(\pi \cdot 252.6^2 \cdot 8.825)$$



$$= 44.36 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [ $\sigma^*3$ ]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 25.535 \cdot 252.587 / (2 \cdot 8.825) - 968.7 / (0.1066 \cdot \pi \cdot 252.6^2 \cdot 8.825)$$

$$= 86.92 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [ $\sigma^*4$ ]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 25.535 \cdot 252.587 / (2 \cdot 8.825) + 968.7 / (0.1923 \cdot \pi \cdot 252.6^2 \cdot 8.825)$$

$$= 8.62 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot hm / 3))$$

$$= 4138(6221.8 - 2 \cdot 1220.0) / (6221.8 + (4 \cdot 0.0 / 3))$$

$$= 2515.1 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [ $\tau2$ ]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 2515.13 / (252.5875 \cdot 8.825)$$

$$= 12.95 \text{ N./mm}^2$$

Decay Length (4.15.20) [ $x1, x2$ ]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(252.587 \cdot 8.825)}$$

$$= 36.826 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{(252.587 \cdot 8.825)}, 2 \cdot 1220.0)$$

$$= 223.65 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(Sr/S, 1)$$

$$= \min(137.997/137.997, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [ $\sigma6, r$ ]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 4138 \cdot 0.1 / (223.653(8.825 + 1 \cdot 12.0))$$

$$= -0.66 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle,  $L \geq 8Rm$  (4.15.28) [ $\sigma7, r$ ]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -4138 / (4(8.825 + 1 \cdot 12.0)223.653) - 3 \cdot 0.053 \cdot 4138 / (2(8.825 + 1 \cdot 12.0)^2)$$

$$= -9.60 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 3656.0 \text{ mm.}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot Ls(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000012 \cdot 3656.0(126.7 - 21.1)$$

$$= 4.737 \text{ mm.}$$

#### Results for Vessel Ribs, Web and Base:

|                                      |        |          |                    |
|--------------------------------------|--------|----------|--------------------|
| Baseplate Length                     | Bplen  | 460.0000 | mm.                |
| Baseplate Thickness                  | Bpthk  | 16.0000  | mm.                |
| Baseplate Width                      | Bpwid  | 150.0000 | mm.                |
| Number of Ribs ( inc. outside ribs ) | Nribs  | 2        |                    |
| Rib Thickness                        | Ribtk  | 12.0000  | mm.                |
| Web Thickness                        | Webtk  | 12.0000  | mm.                |
| Web Location                         | Webloc | Side     |                    |
| Saddle Design Temperature            |        | 21.1     | °C                 |
| Saddle Yield Stress                  | Sy     | 234.8    | N./mm <sup>2</sup> |
| Height of Web at Center              | Hw,c   | 306.0    | mm.                |
| Friction Coefficient                 | mu     | 0.450    |                    |

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

**Inertia of Shell [ShellInertia]:**

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(248.175 \cdot 8.825)} + 225.0)^3) / 12$$

$$= 1.707 \text{ cm}^4$$

**Resolved Inertia for the Shell [I,shell]:**

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 1.707 + 26.299 (93.88 - 4.412)^2$$

$$= 2106.781 \text{ cm}^4$$

|           | B<br>mm. | D<br>mm. | Y<br>mm. | A<br>cm <sup>2</sup> | AY<br>mm. <sup>3</sup> | I + AD <sup>(2)</sup><br>cm <sup>4</sup> |
|-----------|----------|----------|----------|----------------------|------------------------|------------------------------------------|
| Shell     | 298.0    | 8.8      | 4.4      | 26.3                 | 11604.5                | 0.211E+04                                |
| Wearplate | 225.0    | 12.0     | 14.8     | 27.0                 | 40027.5                | 0.169E+04                                |
| Web       | 12.0     | 215.0    | 128.3    | 25.8                 | 331078.6               | 0.130E+04                                |
| BasePlate | 150.0    | 16.0     | 243.8    | 24.0                 | 585180.1               | 0.540E+04                                |
| Totals    | ...      | ...      | ...      | 103.1                | 967890.6               | 0.105E+05                                |

**Distance to Centroid [C1]:**

$$= AY / A$$

$$= 967891 / 103.099$$

$$= 93.880 \text{ mm.}$$

**Angle [β]:**

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

**Saddle Splitting Coefficient [K1]:**

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

**Saddle Splitting Force [Fh]:**

$$= K1 \cdot Q$$

$$= 0.204 \cdot 4137.878$$

$$= 842.1480 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.0754 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 140.8543 \text{ N./mm}^2$$

**Saddle Splitting Dimension [d]:**

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 500.0 - 248.17 \cdot \sin(120.0/2) / 1.0472$$

$$= 294.761 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 248.2373 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 2.1769 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 156.5048 \text{ N./mm}^2$$

**Minimum Thickness of Baseplate per Moss:**

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(4138 + 50) \cdot 150.0) / (2 \cdot 460.0 \cdot 156.505)}$$

$$= 11.329 \text{ mm.}$$

**Calculation of Axial Load, Intermediate Values and Compressive Stress:**

**Web Length Dimension [Web Length]:**

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (245.0 + 12.0 + 12.0)$$

$$= 465.922 \text{ mm.}$$

**Distance between Ribs [e]:**

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 465.9216 / (2 - 1)$$

$$= 465.922 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 465.9216 \cdot 150.0 / 2$$

$$= 349.441 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= ( Q + SaddleWt ) / ( BasePlateLength \cdot BasePlateWidth )$$

$$= 4187.602 / ( 460.0 \cdot 150.0 )$$

$$= 6.069 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 349.4 \cdot 6.07$$

$$= 2120.754 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 16.56 + 27.955$$

$$= 44.515 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 2120.8 / 44.5153$$

$$= 4.672 \text{ N./mm}^2$$

#### Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

|        | B     | D     | Y    | A    | AY       | Io   |
|--------|-------|-------|------|------|----------|------|
| Rib    | 12.0  | 138.0 | 81.0 | 16.6 | 134136.0 | 630. |
| Web    | 233.0 | 12.0  | 6.0  | 28.0 | 16773.2  | 221. |
| Totals | ...   | ...   | ...  | 44.5 | 150909.2 | 851. |

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 150.0 - 12.0$$

$$= 138.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 150909.172 / 44.515$$

$$= 33.901 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max( ytot, Saddle \text{ Width} - ytot )$$

$$= \max( 33.901, 150.0 - 33.901 )$$

$$= 116.099 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ ( Total \text{ Inertia} / Total \text{ Area} ) }$$

$$= \sqrt{ ( 851.1 / 44.515 ) }$$

$$= 43.727 \text{ mm.}$$

Length of Outer Rib, step 1, [Lv]:

$$= Saddle \text{ Height} - \cos( saddle \text{ bearing angle} / 2 )( radius + shlthk + wpdthk ) - bpthk$$

$$= 500.0 - \cos( 120.0 / 2 )( 245.0 + 12.0 + 12.0 ) - 16.0$$

$$= 349.500 \text{ mm.}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin( saddle \text{ bearing angle} / 2 )( radius + shlthk + wpdthk )$$

$$= 2 \cdot \sin( 120.0 / 2 )( 245.0 + 12.0 + 12.0 )$$

$$= 465.922 \text{ mm.}$$

Length of Outer Rib, step 3, [Bd]:

$$= Baseplate \text{ length} - clearance$$

$$= 460.0 - 50.8$$

$$= 409.200 \text{ mm.}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (465.922 - 409.2) / 2$$

$$= 28.361 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{349.5^2 + 28.361^2}$$

$$= 350.649 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19648\text{E}+09 / 234.8)}$$

$$= 128.536$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 350.649 / 43.727$$

$$= 8.019$$

Bending Moment [Rm]:

$$= Fl / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 359.2 / (2 \cdot 460.0) \cdot 465.922 \cdot 350.65 / 2$$

$$= 31.898 \text{ Kg-m.}$$

Compressive Allowable,  $KL/r < Cc$  (  $8.0191 < 128.5358$  ) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (8.02)^2 / (2 \cdot 128.54^2)) 235 /$$

$$(5/3 + 3 \cdot (8.02) / (8 \cdot 128.54) - (8.02^3) / (8 \cdot 128.54^3))$$

$$= 138.6 \text{ N./mm}^2$$

**AISC Unity Check of Outside Ribs ( must be  $\leq 1$  )**

$$= Sc / Sca + (Rm \cdot C1 / I) / Sba$$

$$= 4.67 / 138.64 + (31.9 \cdot 116.099 / 8511374) / 156.5$$

$$= 0.061$$

#### **Input Data for Base Plate Bolting Calculations:**

|                                            |         |                           |
|--------------------------------------------|---------|---------------------------|
| Total Number of Bolts per BasePlate        | Nbolts  | 2                         |
| Total Number of Bolts in Tension/Baseplate | Nbt     | 1                         |
| Bolt Design Temperature                    |         | 21.1 °C                   |
| Bolt Material Specification                |         | SA-193 B7                 |
| Bolt Allowable Stress                      | Stba    | 137.30 N./mm <sup>2</sup> |
| Bolt Corrosion Allowance                   | Bca     | 1.5 mm.                   |
| Distance from Bolts to Edge                | Edgedis | 50.8 mm.                  |
| Nominal Bolt Diameter                      | Bnd     | 24.0000 mm.               |
| Thread Series                              | Series  | TEMA Metric               |
| BasePlate Allowable Stress                 | S       | 138.00 N./mm <sup>2</sup> |
| Area Available in a Single Bolt            | BltArea | 2.2586 cm <sup>2</sup>    |
| Saddle Load Q0 (Weight)                    | Q0      | 2532.9 Kg                 |
| Saddle Load QL (Wind/Seismic contribution) | QL      | 49.1 Kg                   |
| Maximum Transverse Force                   | Ft      | 507.5 Kg                  |
| Maximum Longitudinal Force                 | Fl      | 359.2 Kg                  |
| Saddle Bolted to Steel Foundation          |         | No                        |

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 359.2 / (2.259 \cdot 2)$$

$$= 7.8 \text{ N./mm}^2. \text{ Must be less than } 107.9 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 507.5 / (2.259 \cdot 2)$$

$$= 11.0 \text{ N./mm}^2. \text{ Must be less than } 107.9 \text{ N./mm}^2.$$

#### **Bolt Area Calculation per Dennis R. Moss**

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:  
=  $F_l / (BoltShearAllowable \cdot Nbolts)$   
=  $359.24 / (107.87 \cdot 2.0)$   
=  $0.1633 \text{ cm}^2$

**Bolt Area due to Transverse Load:**

Moment on Baseplate Due to Transverse Load [Rmom]:  
=  $B \cdot F_t + \text{Sum of X Moments}$   
=  $500.0 \cdot 507.45 + 0.0$   
=  $253.73 \text{ Kg-m.}$

Eccentricity (e):  
=  $Rmom / QO$   
=  $253.73 / 2532.86$   
=  $100.17 \text{ mm.} > Bplen / 6 \rightarrow \text{Uplift in Transverse direction}$

$f = Bplen / 2 - Edgedis$   
=  $460.0 / 2 - 50.8$   
=  $179.20 \text{ mm.}$

Modular Ratio Of Steel/Concrete (n1):  
=  $ES / EC$   
=  $203402.5 / 21526.32$   
=  $9.45$

$K1 = 3 \cdot (e - 0.5 \cdot Bplen)$   
=  $3 \cdot (100.17 - 0.5 \cdot 460.0)$   
=  $-389.48 \text{ mm.}$

$K2 = 6 \cdot n1 \cdot A_t / Bpwid \cdot (f + e)$   
=  $6 \cdot 9.45 \cdot 2.26 / 150.0 \cdot (179.2 + 100.17)$   
=  $23848.60 \text{ mm.}^2$

$K3 = -K2 \cdot (0.5 \cdot Bplen + f)$   
=  $-23848.6 \cdot (0.5 \cdot 460.0 + 179.2)$   
=  $-9758845.92 \text{ mm.}^3$

Iteratively Solving for the Effective Bearing Length:  
 $Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$   
 $Y^3 + -389.48 \cdot Y^2 + 23848.6 \cdot Y + -9758846.0 = 0$   
 $Y = 392.13 \text{ mm.}$

$Num = (Bplen / 2 - Y / 3 - e)$   
=  $(460.0 / 2 - 392.13 / 3 - 100.17)$   
=  $-0.88$

$Denom = (Bplen / 2 - Y / 3 + f)$   
=  $(460.0 / 2 - 392.13 / 3 + 179.2)$   
=  $278.49$

Total Bolt Tension Force [Tforce]:  
=  $-QO \cdot Num / Denom$   
=  $-2532.86 \cdot -0.88 / 278.49$   
=  $8.03 \text{ Kg}$

Bolt Area Required due to Transverse Load [Bltareart]:  
=  $Tforce / (Stba \cdot Nbt)$   
=  $8.03 / (137.3 \cdot 1.0)$   
=  $0.0057 \text{ cm}^2$

Required Area of a Single Bolt [Bltarear]:  
=  $\max[Bltarearl, Bltarears, Bltareart]$   
=  $\max[0.0, 0.1633, 0.0057]$   
=  $0.1633 \text{ cm}^2$

**Baseplate Thickness Calculation per D. Moss:**

Bearing Pressure (fc)

$$= 2(QO + Tforce) / (Y \cdot Bpwid)$$

$$= 2(2532.86 + 8.03) / (392.13 \cdot 150.0)$$

$$= 8.47 \text{ bars}$$

Distance from Baseplate Edge to the Web [ADIST]:

$$= (Bplen - WebIngh) / 2$$

$$= (460.0 - 409.2) / 2$$

$$= 25.4000 \text{ mm.}$$

Overturning Moment due To Bolt Tension [Mt]:

$$= Tforce \cdot Adist$$

$$= 8.03 \cdot 25.4$$

$$= 0.20 \text{ Kg-m.}$$

Equivalent Bearing Pressure (f1):

$$= fc \cdot (Y - Adist) / Y$$

$$= 8.47 \cdot (392.13 - 25.4) / 392.13$$

$$= 7.92 \text{ bars}$$

Overturning Moment due to Bearing Pressure [Mc]:

$$= (Adist^2 \cdot Bpwid / 6) \cdot (f1 + 2 \cdot fc)$$

$$= (25.4^2 \cdot 150.0 / 6) \cdot (7.92 + 2 \cdot 8.47)$$

$$= 4.09 \text{ Kg-m.}$$

Baseplate Required Thickness [Treq]:

$$= \sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))}$$

$$= \sqrt{(6 \cdot \max(0.2, 4.09) / (150.0 \cdot 207.0))}$$

$$= 2.7841 \text{ mm.}$$

#### MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 12.0, tr = 5.092, c = 3.175 mm., E\* = 1.0  
 Thickness Ratio = tr \* E\* / (tg - c) = 0.577, Temp. Reduction = 24 °C

|                                                  |        |
|--------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -23 °C |
| Min Metal Temp. at Required thickness (UCS 66.1) | -47 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C |

#### ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

##### Input and Calculated Values:

|                                               |                  |         |                    |
|-----------------------------------------------|------------------|---------|--------------------|
| Vessel Mean Radius                            | Rm               | 252.59  | mm.                |
| Shell Thickness used in this Case             | t                | 12.000  | mm.                |
| Stiffened Vessel Length per 4.15.6            | L                | 6221.80 | mm.                |
| Distance from Saddle to Vessel tangent        | a1 or a          | 1220.00 | mm.                |
| Saddle Width                                  | b1 or b          | 150.00  | mm.                |
| Saddle Bearing Angle                          | delta or theta   | 120.00  | degrees            |
| Wear Plate Width                              | b2 or b1         | 225.00  | mm.                |
| Wear Plate Bearing Angle                      | delta2 or theta1 | 132.00  | degrees            |
| Wear Plate Thickness                          | e2 or tr         | 12.0    | mm.                |
| Wear Plate Allowable Stress                   | fw or Sr         | 138.00  | N./mm <sup>2</sup> |
| Inside or Mean Depth of Head                  | Hi or hm         | 130.09  | mm.                |
| Shell Allowable Stress used in Calculation    |                  | 138.00  | N./mm <sup>2</sup> |
| Head Allowable Stress used in Calculation     |                  | 138.00  | N./mm <sup>2</sup> |
| Circumferential Efficiency in Plane of Saddle |                  | 1.00    |                    |
| Circumferential Efficiency at Mid-Span        |                  | 1.00    |                    |
| Distance from Saddle Base to Centerline       | B                | 500.00  | mm.                |
| Coefficient of Friction                       | mu               | 0.45    |                    |

##### Results for Horizontal Vessel Analysis:

|                                |   |         |      |
|--------------------------------|---|---------|------|
| Saddle Force Q, Operating Case |   | 3238.65 | Kgf  |
| Pressure used in Analysis      | P | 25.535  | bars |



| Horizontal Vessel Analysis Results: | Actual<br>N./mm <sup>2</sup> | Allowable<br>N./mm <sup>2</sup> |
|-------------------------------------|------------------------------|---------------------------------|
| Long. Stress at Top of Midspan      | 31.21                        | 138.00                          |
| Long. Stress at Bottom of Midspan   | 41.88                        | 138.00                          |
| Long. Stress at Top of Saddles      | 80.70                        | 138.00                          |
| Long. Stress at Bottom of Saddles   | 12.07                        | 138.00                          |
| Tangential Shear in Shell           | 9.86                         | 110.40                          |
| Circ. Stress at Horn of Saddle      | 7.51                         | 172.50                          |
| Circ. Compressive Stress in Shell   | 0.52                         | 138.00                          |

#### Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr( Ft/Num \text{ of Saddles} + Z \text{ Force Load} ) \cdot B / E$$

$$= 3.0( 1014.9/2 + 0 ) \cdot 500.0/460.0$$

$$= 1654.7 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max( Fl,wind; \text{Sum of X Forces} ) \cdot B / Ls$$

$$= \max( 66.51, 0 ) \cdot 500.0/3656.0$$

$$= 9.1 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max( Fl,eq; \text{Sum of X Forces} ) \cdot B / Ls$$

$$= \max( 359.24, 0 ) \cdot 500.0/3656.0$$

$$= 49.1 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= Ftr( Ft/Num \text{ of Saddles} + Z \text{ Force Load} ) \cdot B / E$$

$$= 3.0( 359/2 + 0 ) \cdot 500.0/460.0$$

$$= 585.7 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max( Fwl, Fwt, Fsl, Fst )$$

$$= 1584 + \max( 9, 1655, 49, 586 )$$

$$= 3238.6 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}( \text{Platform Area} + ( \pi/4( OD \cdot \text{WindDiaMult} )^2 ) )$$

$$= 1.1 \cdot 1984.306( 0.0 + \pi/4( 0.514 \cdot 1.2 )^2 )$$

$$= 652.203 \text{ N}$$

#### Summary of Loads at the Base of this Saddle:

|                                         |         |     |
|-----------------------------------------|---------|-----|
| Vertical Load (including saddle weight) | 3288.37 | Kgf |
| Transverse Shear Load                   | 507.45  | Kgf |
| Longitudinal Shear Load                 | 359.24  | Kgf |

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 460.000 \text{ mm.}$$

#### The Computed K values from Table 4.15.1:

|              |              |              |              |
|--------------|--------------|--------------|--------------|
| K1 = 0.1066  | K2 = 1.1707  | K3 = 0.8799  | K4 = 0.4011  |
| K5 = 0.7603  | K6 = 0.0529  | K7 = 0.0529  | K8 = 0.3405  |
| K9 = 0.2711  | K10 = 0.0581 | K1* = 0.1923 | K6p = 0.0434 |
| K7p = 0.0434 |              |              |              |

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [ 1 - ( 1 - a/L + (Rm^2 - hm^2)/(2a \cdot L) ) / ( 1 + (4hm)/3L ) ]$$

$$= -3239 \cdot 1220.0 [ 1 - ( 1 - 1220.0/6221.8 + (252.587^2 - 130.087^2) /$$



$$\begin{aligned} & (2 \cdot 1220.0 \cdot 6221.8) / (1 + (4 \cdot 130.09) / (3 \cdot 6221.8)) \\ & = -849.1 \text{ Kg-m.} \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned} & = Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a/L) \\ & = 3239 \cdot 6222 / 4 \cdot ((1 + 2(253^2 - 130^2) / (6222^2)) / (1 + (4 \cdot 130) / (3 \cdot 6222)) - 4 \cdot 1220 / 6222) \\ & = 961.7 \text{ Kg-m.} \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [ $\sigma_1$ ]:

$$\begin{aligned} & = P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t) \\ & = 25.535 \cdot 252.587 / (2 \cdot 8.825) - 961.7 / (\pi \cdot 252.6^2 \cdot 8.825) \\ & = 31.21 \text{ N./mm}^2 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [ $\sigma_2$ ]:

$$\begin{aligned} & = P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\ & = 25.535 \cdot 252.587 / (2 \cdot 8.825) + 961.7 / (\pi \cdot 252.6^2 \cdot 8.825) \\ & = 41.88 \text{ N./mm}^2 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [ $\sigma^*3$ ]:

$$\begin{aligned} & = P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t) \\ & = 25.535 \cdot 252.587 / (2 \cdot 8.825) - 849.1 / (0.1066 \cdot \pi \cdot 252.6^2 \cdot 8.825) \\ & = 80.70 \text{ N./mm}^2 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [ $\sigma^*4$ ]:

$$\begin{aligned} & = P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\ & = 25.535 \cdot 252.587 / (2 \cdot 8.825) + 849.1 / (0.1923 \cdot \pi \cdot 252.6^2 \cdot 8.825) \\ & = 12.07 \text{ N./mm}^2 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned} & = Q(L - 2a) / (L + (4 \cdot hm / 3)) \\ & = 3239(6221.8 - 2 \cdot 1220.0) / (6221.8 + (4 \cdot 130.09 / 3)) \\ & = 1915.2 \text{ Kgf} \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [ $\tau_2$ ]:

$$\begin{aligned} & = K2 \cdot T / (Rm \cdot t) \\ & = 1.1707 \cdot 1915.16 / (252.5875 \cdot 8.825) \\ & = 9.86 \text{ N./mm}^2 \end{aligned}$$

Decay Length (4.15.20) [x1,x2]:

$$\begin{aligned} & = 0.78 \cdot \sqrt{(Rm \cdot t)} \\ & = 0.78 \cdot \sqrt{(252.587 \cdot 8.825)} \\ & = 36.826 \text{ mm.} \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned} & = \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\ & = \min(150.0 + 1.56 \cdot \sqrt{(252.587 \cdot 8.825)}, 2 \cdot 1220.0) \\ & = 223.65 \text{ mm.} \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$\begin{aligned} & = \min(Sr/S, 1) \\ & = \min(137.997/137.997, 1) \\ & = 1.0000 \end{aligned}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [ $\sigma_6, r$ ]:

$$\begin{aligned} & = -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\ & = -0.7603 \cdot 3239 \cdot 0.1 / (223.653(8.825 + 1 \cdot 12.0)) \\ & = -0.52 \text{ N./mm}^2 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle,  $L \geq 8Rm$  (4.15.28) [ $\sigma_7, r$ ]:

$$\begin{aligned} & = -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2) \\ & = -3239 / (4(8.825 + 1 \cdot 12.0)223.653) - 3 \cdot 0.053 \cdot 3239 / (2(8.825 + 1 \cdot 12.0)^2) \\ & = -7.51 \text{ N./mm}^2 \end{aligned}$$

#### Results for Vessel Ribs, Web and Base:

|                     |       |          |     |
|---------------------|-------|----------|-----|
| Baseplate Length    | Bplen | 460.0000 | mm. |
| Baseplate Thickness | Bpthk | 16.0000  | mm. |
| Baseplate Width     | Bpwid | 150.0000 | mm. |

|                                      |        |         |                    |
|--------------------------------------|--------|---------|--------------------|
| Number of Ribs ( inc. outside ribs ) | Nribs  | 2       |                    |
| Rib Thickness                        | Ribtk  | 12.0000 | mm.                |
| Web Thickness                        | Webtk  | 12.0000 | mm.                |
| Web Location                         | Webloc | Side    |                    |
| Saddle Design Temperature            |        | 21.1    | °C                 |
| Saddle Yield Stress                  | Sy     | 234.8   | N./mm <sup>2</sup> |
| Height of Web at Center              | Hw,c   | 306.0   | mm.                |
| Friction Coefficient                 | mu     | 0.450   |                    |

#### Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:  
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$   
 $= ((1.56 \cdot \sqrt{(248.175 \cdot 8.825)} + 225.0)^3) / 12$   
 $= 1.707 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:  
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$   
 $= 1.707 + 26.299 (93.88 - 4.412)^2$   
 $= 2106.781 \text{ cm}^4$

|           | B<br>mm. | D<br>mm. | Y<br>mm. | A<br>cm <sup>2</sup> | AY<br>mm. <sup>3</sup> | I + AD <sup>(2)</sup><br>cm <sup>4</sup> |
|-----------|----------|----------|----------|----------------------|------------------------|------------------------------------------|
| Shell     | 298.0    | 8.8      | 4.4      | 26.3                 | 11604.5                | 0.211E+04                                |
| Wearplate | 225.0    | 12.0     | 14.8     | 27.0                 | 40027.5                | 0.169E+04                                |
| Web       | 12.0     | 215.0    | 128.3    | 25.8                 | 331078.6               | 0.130E+04                                |
| BasePlate | 150.0    | 16.0     | 243.8    | 24.0                 | 585180.1               | 0.540E+04                                |
| Totals    | ...      | ...      | ...      | 103.1                | 967890.6               | 0.105E+05                                |

Distance to Centroid [C1]:  
 $= AY / A$   
 $= 967891 / 103.099$   
 $= 93.880 \text{ mm.}$

Angle [β]:  
 $= 180 - \text{Saddle Angle} / 2$   
 $= 180 - 120.0 / 2$   
 $= 120.0$

Saddle Splitting Coefficient [K1]:  
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$   
 $= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$   
 $= 0.2035$

Saddle Splitting Force [Fh]:  
 $= K1 \cdot Q$   
 $= 0.204 \cdot 3238.647$   
 $= 659.1349 \text{ Kg}$

Tension Stress, St = ( Fh/As ) = 0.8417 N./mm<sup>2</sup>  
 Allowed Stress, Sa = 0.6 \* Yield Str = 140.8543 N./mm<sup>2</sup>

Saddle Splitting Dimension [d]:  
 $= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$   
 $= 500.0 - 248.17 \cdot \sin(120.0 / 2) / 1.0472$   
 $= 294.761 \text{ mm.}$

Bending Moment, M = Fh \* d = 194.2911 Kg-m.

Bending Stress, Sb = ( M \* C1 / I ) = 1.7038 N./mm<sup>2</sup>  
 Allowed Stress, Sa = 2/3 \* Yield Str = 156.5048 N./mm<sup>2</sup>

#### Minimum Thickness of Baseplate per Moss:

$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$   
 $= \sqrt{3(3239 + 50)150.0 / (2 \cdot 460.0 \cdot 156.505)}$   
 $= 10.039 \text{ mm.}$

#### Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [ Web Length ]:  
 $= 2 \cdot \cos( 90 - \text{Saddle Angle}/2 )( \text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk} )$   
 $= 2 \cdot \cos( 90 - 120.0/2 )( 245.0 + 12.0 + 12.0 )$   
 $= 465.922 \text{ mm.}$

Distance between Ribs [e]:  
 $= \text{Web Length} / ( \text{Nribs} - 1 )$   
 $= 465.9216 / ( 2 - 1 )$   
 $= 465.922 \text{ mm.}$

Baseplate Pressure Area [Ap]:  
 $= e \cdot \text{Bpwid} / 2$   
 $= 465.9216 \cdot 150.0 / 2$   
 $= 349.441 \text{ cm}^2$

Bearing Pressure [Bp]:  
 $= ( Q + \text{SaddleWt} ) / ( \text{BasePlateLength} \cdot \text{BasePlateWidth} )$   
 $= 3288.37 / ( 460.0 \cdot 150.0 )$   
 $= 4.766 \text{ Kg/cm}^2$

Axial Load [P]:  
 $= \text{Ap} \cdot \text{Bp}$   
 $= 349.4 \cdot 4.77$   
 $= 1665.351 \text{ Kg}$

Area of the Rib and Web [Ar]:  
 $= \text{Rib Area} + \text{Web Area}$   
 $= 16.56 + 27.955$   
 $= 44.515 \text{ cm}^2$

Compressive Stress [Sc]:  
 $= P / \text{Ar}$   
 $= 1665.4 / 44.5153$   
 $= 3.669 \text{ N./mm}^2$

#### Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

|        | B     | D     | Y    | A    | AY       | Io   |
|--------|-------|-------|------|------|----------|------|
| Rib    | 12.0  | 138.0 | 81.0 | 16.6 | 134136.0 | 630. |
| Web    | 233.0 | 12.0  | 6.0  | 28.0 | 16773.2  | 221. |
| Totals | ...   | ...   | ...  | 44.5 | 150909.2 | 851. |

Rib dimension [D]:  
 $= \text{Saddle Width} - \text{Web Thickness}$   
 $= 150.0 - 12.0$   
 $= 138.000 \text{ mm.}$

Distance to Centroid from Datum [ytot]:  
 $= \text{AY} / \text{A}$   
 $= 150909.172 / 44.515$   
 $= 33.901 \text{ mm.}$

Distance to Centroid [C1]:  
 $= \max( \text{ytot}, \text{Saddle Width} - \text{ytot} )$   
 $= \max( 33.901, 150.0 - 33.901 )$   
 $= 116.099 \text{ mm.}$

Radius of Gyration [r]:  
 $= \sqrt{ ( \text{Total Inertia} / \text{Total Area} )}$   
 $= \sqrt{ ( 851.1 / 44.515 )}$   
 $= 43.727 \text{ mm.}$

Length of Outer Rib, step 1, [Lv]:  
 $= \text{Saddle Height} - \cos( \text{saddle bearing angle} / 2 )( \text{radius} + \text{shlthk} + \text{wpdthk} ) - \text{bpthk}$   
 $= 500.0 - \cos( 120.0/2 )( 245.0 + 12.0 + 12.0 ) - 16.0$   
 $= 349.500 \text{ mm.}$



Length of Outer Rib, step 2, [Lw]:  
 $= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$   
 $= 2 \cdot \sin(120.0/2) (245.0 + 12.0 + 12.0)$   
 $= 465.922 \text{ mm.}$

Length of Outer Rib, step 3, [Bd]:  
 $= \text{Baseplate length} - \text{clearance}$   
 $= 460.0 - 50.8$   
 $= 409.200 \text{ mm.}$

Length of Outer Rib, step 4, [Dd]:  
 $= (Lw - Bd) / 2$   
 $= (465.922 - 409.2) / 2$   
 $= 28.361 \text{ mm.}$

Length of Outer Rib [L]:  
 $= \sqrt{Lv^2 + Dd^2}$   
 $= \sqrt{349.5^2 + 28.361^2}$   
 $= 350.649 \text{ mm.}$

Intermediate Term [Cc]:  
 $= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$   
 $= \sqrt{(2 \cdot \pi^2 \cdot 0.19648\text{E}+09 / 234.8)}$   
 $= 128.536$

Slenderness ratio [KL/r]:  
 $= KL/r$   
 $= 1 \cdot 350.649 / 43.727$   
 $= 8.019$

Bending Moment [Rm]:  
 $= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$   
 $= 359.2 / (2 \cdot 460.0) \cdot 465.922 \cdot 350.65 / 2$   
 $= 31.898 \text{ Kg-m.}$

Compressive Allowable,  $KL/r < Cc$  ( $8.0191 < 128.5358$ ) per AISC E2-1 [Sca]:  
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$   
 $= (1 - (8.02)^2 / (2 \cdot 128.54^2)) 235 /$   
 $(5/3 + 3 \cdot (8.02) / (8 \cdot 128.54) - (8.02^3) / (8 \cdot 128.54^3))$   
 $= 138.6 \text{ N./mm}^2$

**AISC Unity Check of Outside Ribs (must be  $\leq 1$ )**  
 $= Sc/Sca + (Rm \cdot C1 / I) / Sba$   
 $= 3.67 / 138.64 + (31.9 \cdot 116.099 / 8511374) / 156.5$   
 $= 0.054$

#### **Input Data for Base Plate Bolting Calculations:**

|                                            |         |                           |
|--------------------------------------------|---------|---------------------------|
| Total Number of Bolts per BasePlate        | Nbolts  | 2                         |
| Total Number of Bolts in Tension/Baseplate | Nbt     | 1                         |
| Bolt Design Temperature                    |         | 21.1 °C                   |
| Bolt Material Specification                |         | SA-193 B7                 |
| Bolt Allowable Stress                      | Stba    | 137.30 N./mm <sup>2</sup> |
| Bolt Corrosion Allowance                   | Bca     | 1.5 mm.                   |
| Distance from Bolts to Edge                | Edgedis | 50.8 mm.                  |
| Nominal Bolt Diameter                      | Bnd     | 24.0000 mm.               |
| Thread Series                              | Series  | TEMA Metric               |
| BasePlate Allowable Stress                 | S       | 138.00 N./mm <sup>2</sup> |
| Area Available in a Single Bolt            | BlArea  | 2.2586 cm <sup>2</sup>    |
| Saddle Load Q0 (Weight)                    | Q0      | 1633.6 Kg                 |
| Saddle Load QL (Wind/Seismic contribution) | QL      | 49.1 Kg                   |
| Maximum Transverse Force                   | Ft      | 507.5 Kg                  |
| Maximum Longitudinal Force                 | Fl      | 359.2 Kg                  |
| Saddle Bolted to Steel Foundation          | No      |                           |

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:  
 $= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$   
 $= 359.2 / (2.259 \cdot 2)$   
 $= 7.8 \text{ N./mm}^2$ . Must be less than 107.9 N./mm<sup>2</sup>.



Shear Stress in a Single Bolt, Transverse Direction [Tb,t]:  
= Ft / ( Bolt Area • Number of Bolts )  
= 507.5 / ( 2.259 • 2 )  
= 11.0 N./mm<sup>2</sup>. Must be less than 107.9 N./mm<sup>2</sup>.

**Bolt Area Calculation per Dennis R. Moss**

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:  
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:  
= FI / ( BoltShearAllowable • Nbolts )  
= 359.24 / (107.87 • 2.0)  
= 0.1633 cm<sup>2</sup>

**Bolt Area due to Transverse Load:**

Moment on Baseplate Due to Transverse Load [Rmom]:  
= B • Ft + Sum of X Moments  
= 500.0 • 507.45 + 0.0  
= 253.73 Kg-m.

Eccentricity (e):  
= Rmom / QO  
= 253.73 / 1633.63  
= 155.31 mm. > Bplen/6 --> Uplift in Transverse direction

f = Bplen / 2 - Edgedis  
= 460.0 / 2 - 50.8  
= 179.20 mm.

Modular Ratio Of Steel/Concrete (n1):  
= ES / EC  
= 203402.5 / 21526.32  
= 9.45

K1 = 3 (e - 0.5 • Bplen)  
= 3 (155.31 - 0.5 • 460.0)  
= -224.06 mm.

K2 = 6 • n1 • At / Bpwid • (f + e)  
= 6 • 9.45 • 2.26 / 150.0 • (179.2 + 155.31)  
= 28555.65 mm.<sup>2</sup>

K3 = -K2 • (0.5 • Bplen + f)  
= -28555.65 • (0.5 • 460.0 + 179.2)  
= -11684972.28 mm.<sup>3</sup>

Iteratively Solving for the Effective Bearing Length:  
 $Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$   
 $Y^3 + -224.06 \cdot Y^2 + 28555.65 \cdot Y + -0.1E+08 = 0$   
Y = 274.85 mm.

Num = (Bplen / 2 - Y / 3 - e)  
= (460.0 / 2 - 274.85 / 3 - 155.31)  
= -16.93

Denom = (Bplen / 2 - Y / 3 + f)  
= (460.0 / 2 - 274.85 / 3 + 179.2)  
= 317.58

Total Bolt Tension Force [Tforce]:  
= - QO • Num / Denom  
= - 1633.63 • -16.93 / 317.58  
= 87.08 Kg

Bolt Area Required due to Transverse Load [Bltareart]:  
= Tforce / (Stba • Nbt)  
= 87.08 / (137.3 • 1.0)  
= 0.0622 cm<sup>2</sup>

Required Area of a Single Bolt [Bltarear]:  
= max[Bltarearl, Bltarears, Bltareart]  
= max[0.0, 0.1633, 0.0622]  
= 0.1633 cm<sup>2</sup>

**Baseplate Thickness Calculation per D. Moss:**

Bearing Pressure (fc)  
=  $2(QO + Tforce) / (Y \cdot Bpwid)$   
=  $2(1633.63 + 87.08) / (274.85 \cdot 150.0)$   
= 8.19 bars

Distance from Baseplate Edge to the Web [ADIST]:  
=  $(Bplen - WebIngh) / 2$   
=  $(460.0 - 409.2) / 2$   
= 25.4000 mm.

Overturning Moment due To Bolt Tension [Mt]:  
=  $Tforce \cdot Adist$   
=  $87.08 \cdot 25.4$   
= 2.21 Kg-m.

Equivalent Bearing Pressure (f1):  
=  $fc \cdot (Y - Adist) / Y$   
=  $8.19 \cdot (274.85 - 25.4) / 274.85$   
= 7.43 bars

Overturning Moment due to Bearing Pressure [Mc]:  
=  $(Adist^2 \cdot Bpwid / 6) \cdot (f1 + 2 \cdot fc)$   
=  $(25.4^2 \cdot 150.0 / 6) \cdot (7.43 + 2 \cdot 8.19)$   
= 3.91 Kg-m.

Baseplate Required Thickness [Treq]:  
=  $\sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))}$   
=  $\sqrt{(6 \cdot \max(2.21, 3.91) / (150.0 \cdot 207.0))}$   
= 2.7237 mm.

**MDMT Calculations for: RIGHT SADDLE, Saddle to Wearplate Junction, Curve: B**

Govrn. thk, tg = 12.0, tr = 5.092, c = 3.175 mm., E\* = 1.0  
Thickness Ratio =  $tr \cdot E* / (tg - c) = 0.577$ , Temp. Reduction = 24 °C

|                                                  |        |
|--------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -23 °C |
| Min Metal Temp. at Required thickness (UCS 66.1) | -47 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C |

*Review notes about nozzle loadings and supports in the Warnings report.*

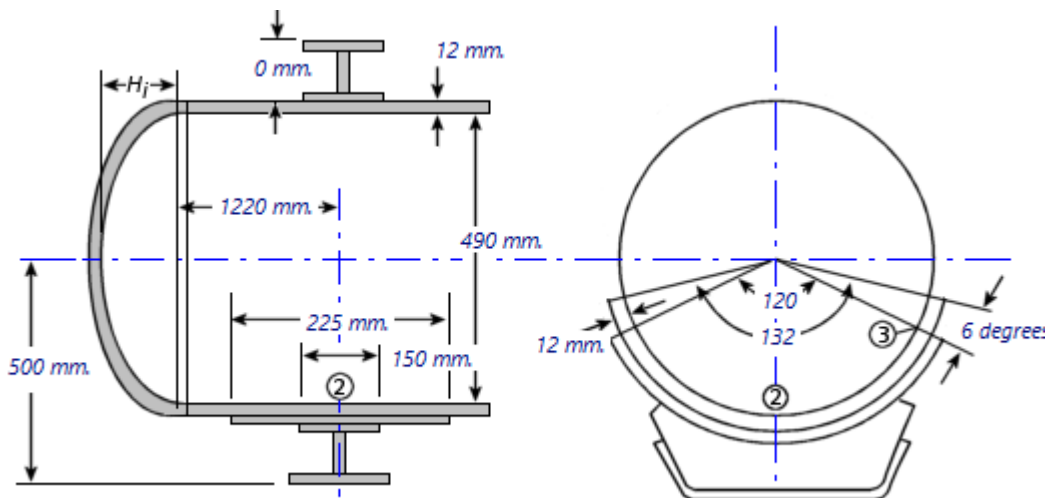
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### ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

#### Input and Calculated Values:

|                                               |                  |         |                    |
|-----------------------------------------------|------------------|---------|--------------------|
| Vessel Mean Radius                            | Rm               | 252.59  | mm.                |
| Shell Thickness used in this Case             | t                | 12.000  | mm.                |
| Stiffened Vessel Length per 4.15.6            | L                | 6221.80 | mm.                |
| Distance from Saddle to Vessel tangent        | a1 or a          | 1220.00 | mm.                |
| Saddle Width                                  | b1 or b          | 150.00  | mm.                |
| Saddle Bearing Angle                          | delta or theta   | 120.00  | degrees            |
| Wear Plate Width                              | b2 or b1         | 225.00  | mm.                |
| Wear Plate Bearing Angle                      | delta2 or theta1 | 132.00  | degrees            |
| Wear Plate Thickness                          | e2 or tr         | 12.0    | mm.                |
| Wear Plate Allowable Stress                   | fw or Sr         | 138.00  | N./mm <sup>2</sup> |
| Shell Allowable Stress used in Calculation    |                  | 247.00  | N./mm <sup>2</sup> |
| Head Allowable Stress used in Calculation     |                  | 247.00  | N./mm <sup>2</sup> |
| Circumferential Efficiency in Plane of Saddle |                  | 1.00    |                    |
| Circumferential Efficiency at Mid-Span        |                  | 1.00    |                    |
| Distance from Saddle Base to Centerline       | B                | 500.00  | mm.                |
| Coefficient of Friction                       | mu               | 0.45    |                    |



#### Results for Horizontal Vessel Analysis:

|                                           |   |         |      |
|-------------------------------------------|---|---------|------|
| Saddle Force Q, Test Case, no Ext. Forces |   | 3083.79 | Kgf  |
| Pressure used in Analysis                 | P | 36.385  | bars |

| Horizontal Vessel Analysis Results: | Actual<br>N./mm <sup>2</sup> | Allowable<br>N./mm <sup>2</sup> |
|-------------------------------------|------------------------------|---------------------------------|
| Long. Stress at Top of Midspan      | 46.25                        | 247.00                          |
| Long. Stress at Bottom of Midspan   | 57.90                        | 247.00                          |
| Long. Stress at Top of Saddles      | 89.62                        | 247.00                          |
| Long. Stress at Bottom of Saddles   | 31.27                        | 247.00                          |
| Tangential Shear in Shell           | 9.65                         | 138.58                          |
| Circ. Stress at Horn of Saddle      | 12.12                        | 370.49                          |
| Circ. Compressive Stress in Shell   | 0.66                         | 247.00                          |

#### Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:



$$\begin{aligned} &= \text{Ftr}(\text{Ft}/\text{Num of Saddles} + \text{Z Force Load}) \cdot B / E \\ &= 3.0(334.9/2 + 0) \cdot 500.0/460.0 \\ &= 546.1 \text{ Kgf} \end{aligned}$$

$$\begin{aligned} &\text{Longitudinal Saddle Reaction Force [Fwl]:} \\ &= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / L_s \\ &= \max(21.95, 0) \cdot 500.0/3656.0 \\ &= 3.0 \text{ Kgf} \end{aligned}$$

$$\begin{aligned} &\text{Load Combination Results for Q + Wind or Seismic [Q]:} \\ &= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst}) \\ &= 2538 + \max(3, 546, 0, 0) \\ &= 3083.8 \text{ Kgf} \end{aligned}$$

$$\begin{aligned} &\text{Longitudinal Wind Force [FI]:} \\ &= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2)) \\ &= 1.1 \cdot 1984.306(0.0 + \pi/4(0.514 \cdot 1.2)^2) \\ &= 652.203 \text{ N} \end{aligned}$$

**Summary of Loads at the Base of this Saddle:**

|                                         |         |     |
|-----------------------------------------|---------|-----|
| Vertical Load (including saddle weight) | 3133.51 | Kgf |
| Transverse Shear Load                   | 167.46  | Kgf |
| Longitudinal Shear Load                 | 21.95   | Kgf |

Note: Wear Plate is Welded to the Shell,  $k = 0.1$

$$\begin{aligned} &\text{Saddle Dimension [E]:} \\ &= \text{Baseplate Length} \\ &= 460.000 \text{ mm.} \end{aligned}$$

**The Computed K values from Table 4.15.1:**

|              |              |              |              |
|--------------|--------------|--------------|--------------|
| K1 = 0.1066  | K2 = 1.1707  | K3 = 0.8799  | K4 = 0.4011  |
| K5 = 0.7603  | K6 = 0.0529  | K7 = 0.0529  | K8 = 0.3405  |
| K9 = 0.2711  | K10 = 0.0581 | K1* = 0.1923 | K6p = 0.0434 |
| K7p = 0.0434 |              |              |              |

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to  $R_m/2$ .

$$\begin{aligned} &\text{Moment per Equation 4.15.1 [M1]:} \\ &= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)] \\ &= -3084 \cdot 1220.0 [1 - (1 - 1220.0/6221.8 + (252.587^2 - 0.0^2)/(2 \cdot 1220.0 \cdot 6221.8))/(1 + (4 \cdot 0.0)/(3 \cdot 6221.8))] \\ &= -721.9 \text{ Kg-m.} \end{aligned}$$

$$\begin{aligned} &\text{Moment per Equation 4.15.2 [M2]:} \\ &= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L) \\ &= 3084 \cdot 6222/4((1 + 2(253^2 - 0^2)/(6222^2))/(1 + (4 \cdot 0)/(3 \cdot 6222)) - 4 \cdot 1220/6222) \\ &= 1050.3 \text{ Kg-m.} \end{aligned}$$

$$\begin{aligned} &\text{Longitudinal Stress at Top of Shell (4.15.4) [\sigma_1]:} \\ &= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 \cdot t) \\ &= 36.385 \cdot 252.587/(2 \cdot 8.825) - 1050.3/(\pi \cdot 252.6^2 \cdot 8.825) \\ &= 46.25 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} &\text{Longitudinal Stress at Bottom of Shell (4.15.5) [\sigma_2]:} \\ &= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t) \\ &= 36.385 \cdot 252.587/(2 \cdot 8.825) + 1050.3/(\pi \cdot 252.6^2 \cdot 8.825) \\ &= 57.90 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} &\text{Longitudinal Stress at Top of Shell at Support (4.15.8) [\sigma*3]:} \\ &= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ &= 36.385 \cdot 252.587/(2 \cdot 8.825) - 721.9/(0.1066 \cdot \pi \cdot 252.6^2 \cdot 8.825) \\ &= 89.62 \text{ N./mm}^2 \end{aligned}$$

$$\begin{aligned} &\text{Longitudinal Stress at Bottom of Shell at Support (4.15.9) [\sigma*4]:} \\ &= P \cdot R_m/(2t) + M_1/(K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ &= 36.385 \cdot 252.587/(2 \cdot 8.825) + 721.9/(0.1923 \cdot \pi \cdot 252.6^2 \cdot 8.825) \end{aligned}$$

= 31.27 N./mm<sup>2</sup>

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a)/(L+(4 \cdot hm/3))$$

$$= 3084(6221.8 - 2 \cdot 1220.0)/(6221.8 + (4 \cdot 0.0/3))$$

$$= 1874.4 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [ $\tau_2$ ]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 1874.42 / (252.5875 \cdot 8.825)$$

$$= 9.65 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(252.587 \cdot 8.825)}$$

$$= 36.826 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{(252.587 \cdot 8.825)}, 2 \cdot 1220.0)$$

$$= 223.65 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(S_r/S, 1)$$

$$= \min(137.997/246.996, 1)$$

$$= 0.5587$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [ $\sigma_6, r$ ]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 3084 \cdot 0.1 / (223.653(8.825 + 0.559 \cdot 12.0))$$

$$= -0.66 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle,  $L \geq 8R_m$  (4.15.28) [ $\sigma_7, r$ ]:

$$= -Q/(4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -3084/(4(8.825 + 0.559 \cdot 12.0)223.653) -$$

$$3 \cdot 0.053 \cdot 3084 / (2(8.825 + 0.559 \cdot 12.0)^2)$$

$$= -12.12 \text{ N./mm}^2$$

#### Results for Vessel Ribs, Web and Base:

|                                      |        |          |                    |
|--------------------------------------|--------|----------|--------------------|
| Baseplate Length                     | Bplen  | 460.0000 | mm.                |
| Baseplate Thickness                  | Bpthk  | 16.0000  | mm.                |
| Baseplate Width                      | Bpwid  | 150.0000 | mm.                |
| Number of Ribs ( inc. outside ribs ) | Nribs  | 2        |                    |
| Rib Thickness                        | Ribtk  | 12.0000  | mm.                |
| Web Thickness                        | Webtk  | 12.0000  | mm.                |
| Web Location                         | Webloc | Side     |                    |
| Saddle Yield Stress                  | Sy     | 260.0    | N./mm <sup>2</sup> |
| Height of Web at Center              | Hw,c   | 306.0    | mm.                |
| Friction Coefficient                 | mu     | 0.450    |                    |

#### Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(248.175 \cdot 8.825)} + 225.0)8.825^3)/12$$

$$= 1.707 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 1.707 + 26.299(93.88 - 4.412)^2$$

$$= 2106.781 \text{ cm}^4$$

|           | B<br>mm. | D<br>mm. | Y<br>mm. | A<br>cm <sup>2</sup> | AY<br>mm. <sup>3</sup> | I + AD <sup>(2)</sup><br>cm <sup>4</sup> |
|-----------|----------|----------|----------|----------------------|------------------------|------------------------------------------|
| Shell     | 298.0    | 8.8      | 4.4      | 26.3                 | 11604.5                | 0.211E+04                                |
| Wearplate | 225.0    | 12.0     | 14.8     | 27.0                 | 40027.5                | 0.169E+04                                |
| Web       | 12.0     | 215.0    | 128.3    | 25.8                 | 331078.6               | 0.130E+04                                |
| BasePlate | 150.0    | 16.0     | 243.8    | 24.0                 | 585180.1               | 0.540E+04                                |
| Totals    | ...      | ...      | ...      | 103.1                | 967890.6               | 0.105E+05                                |



Distance to Centroid [C1]:

$$\begin{aligned} &= AY / A \\ &= 967891/103.099 \\ &= 93.880 \text{ mm.} \end{aligned}$$

Angle [β]:

$$\begin{aligned} &= 180 - \text{Saddle Angle}/2 \\ &= 180 - 120.0/2 \\ &= 120.0 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0)) \\ &= 0.2035 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.204 \cdot 3083.791 \\ &= 627.6182 \text{ Kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh/As) = 0.8014 \text{ N./mm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\ &= 500.0 - 248.17 \cdot \sin(120.0/2) / 1.0472 \\ &= 294.761 \text{ mm.} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 185.0010 \text{ Kg-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 1.6223 \text{ N./mm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2 \end{aligned}$$

**Minimum Thickness of Baseplate per Moss:**

$$\begin{aligned} &= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\ &= \sqrt{3(3084 + 50)150.0 / (2 \cdot 460.0 \cdot 173.33)} \\ &= 9.312 \text{ mm.} \end{aligned}$$

**Calculation of Axial Load, Intermediate Values and Compressive Stress:**

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120.0/2) (245.0 + 12.0 + 12.0) \\ &= 465.922 \text{ mm.} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 465.9216 / (2 - 1) \\ &= 465.922 \text{ mm.} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 465.9216 \cdot 150.0/2 \\ &= 349.441 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 3133.513 / (460.0 \cdot 150.0) \\ &= 4.541 \text{ Kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 349.4 \cdot 4.54 \\ &= 1586.926 \text{ Kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 16.56 + 27.955 \\ &= 44.515 \text{ cm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 1586.9/44.5153$$

$$= 3.496 \text{ N./mm}^2$$

#### Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

|        | B     | D     | Y    | A    | AY       | Io   |
|--------|-------|-------|------|------|----------|------|
| Rib    | 12.0  | 138.0 | 81.0 | 16.6 | 134136.0 | 630. |
| Web    | 233.0 | 12.0  | 6.0  | 28.0 | 16773.2  | 221. |
| Totals | ...   | ...   | ...  | 44.5 | 150909.2 | 851. |

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 150.0 - 12.0$$

$$= 138.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 150909.172/44.515$$

$$= 33.901 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max( y_{tot}, \text{Saddle Width} - y_{tot} )$$

$$= \max( 33.901, 150.0 - 33.901 )$$

$$= 116.099 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(851.1/44.515)}$$

$$= 43.727 \text{ mm.}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos( \text{saddle bearing angle} / 2 )( \text{radius} + \text{shlthk} + \text{wpdthk} ) - \text{bpthk}$$

$$= 500.0 - \cos( 120.0/2 )( 245.0 + 12.0 + 12.0 ) - 16.0$$

$$= 349.500 \text{ mm.}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin( \text{saddle bearing angle} / 2 )( \text{radius} + \text{shlthk} + \text{wpdthk} )$$

$$= 2 \cdot \sin( 120.0/2 )( 245.0 + 12.0 + 12.0 )$$

$$= 465.922 \text{ mm.}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 460.0 - 50.8$$

$$= 409.200 \text{ mm.}$$

Length of Outer Rib, step 4, [Dd]:

$$= ( Lw - Bd )/2$$

$$= ( 465.922 - 409.2 )/2$$

$$= 28.361 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \sqrt{(\text{Lv}^2 + \text{Dd}^2)}$$

$$= \sqrt{(349.5^2 + 28.361^2)}$$

$$= 350.649 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19648\text{E}+09/260.0)}$$

$$= 122.138$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 350.649/43.727$$

$$= 8.019$$



Bending Moment [Rm]:

$$= FI / ( 2 \cdot Bpln ) \cdot e \cdot L / 2$$

$$= 21.9 / ( 2 \cdot 460.0 ) \cdot 465.922 \cdot 350.65 / 2$$

$$= 1.949 \text{ Kg-m.}$$

Compressive Allowable,  $KL/r < Cc$  (  $8.0191 < 122.1379$  ) per AISC E2-1 [Sca]:

$$= ( 1 - (KL/r)^2 / (2 \cdot Cc^2) ) Fy / ( 5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3) )$$

$$= ( 1 - ( 8.02 )^2 / ( 2 \cdot 122.14^2 ) ) 260 /$$

$$( 5/3 + 3 \cdot ( 8.02 ) / ( 8 \cdot 122.14 ) - ( 8.02^3 ) / ( 8 \cdot 122.14^3 ) )$$

$$= 153.4 \text{ N./mm}^2$$

**AISC Unity Check of Outside Ribs ( must be  $\leq 1$  )**

$$= Sc/Sca + ( Rm \cdot C1 / I ) / Sba$$

$$= 3.5 / 153.4 + ( 1.95 \cdot 116.099 / 8511374 ) / 173.33$$

$$= 0.024$$

#### **Input Data for Base Plate Bolting Calculations:**

|                                            |         |                           |
|--------------------------------------------|---------|---------------------------|
| Total Number of Bolts per BasePlate        | Nbolts  | 2                         |
| Total Number of Bolts in Tension/Baseplate | Nbt     | 1                         |
| Bolt Material Specification                |         | SA-193 B7                 |
| Bolt Allowable Stress                      | Stba    | 172.00 N./mm <sup>2</sup> |
| Bolt Corrosion Allowance                   | Bca     | 1.5 mm.                   |
| Distance from Bolts to Edge                | Edgedis | 50.8 mm.                  |
| Nominal Bolt Diameter                      | Bnd     | 24.0000 mm.               |
| Thread Series                              | Series  | TEMA Metric               |
| BasePlate Allowable Stress                 | S       | 138.00 N./mm <sup>2</sup> |
| Area Available in a Single Bolt            | BlArea  | 2.2586 cm <sup>2</sup>    |
| Saddle Load Q0 (Weight)                    | Q0      | 2587.4 Kg                 |
| Saddle Load QL (Wind/Seismic contribution) | QL      | 3.0 Kg                    |
| Maximum Transverse Force                   | Ft      | 167.5 Kg                  |
| Maximum Longitudinal Force                 | F1      | 21.9 Kg                   |
| Saddle Bolted to Steel Foundation          |         | No                        |

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= FI / ( Bolt Area \cdot Number of Bolts )$$

$$= 21.9 / ( 2.259 \cdot 2 )$$

$$= 0.5 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / ( Bolt Area \cdot Number of Bolts )$$

$$= 167.5 / ( 2.259 \cdot 2 )$$

$$= 3.6 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

#### **Bolt Area Calculation per Dennis R. Moss**

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (Q0 > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= FI / ( BoltShearAllowable \cdot Nbolts )$$

$$= 21.95 / ( 103.2 \cdot 2.0 )$$

$$= 0.0104 \text{ cm}^2$$

#### **Bolt Area due to Transverse Load:**

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 500.0 \cdot 167.46 + 0.0$$

$$= 83.73 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 83.73 / 2587.45$$

$$= 32.36 \text{ mm.} < Bpln / 6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:  
 = max[Bltarearl, Bltarears, Bltareart]  
 = max[0.0, 0.0104, 0.0]  
 = 0.0104 cm<sup>2</sup>

## ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

### Input and Calculated Values:

|                                               |                  |         |                    |
|-----------------------------------------------|------------------|---------|--------------------|
| Vessel Mean Radius                            | Rm               | 252.59  | mm.                |
| Shell Thickness used in this Case             | t                | 12.000  | mm.                |
| Stiffened Vessel Length per 4.15.6            | L                | 6221.80 | mm.                |
| Distance from Saddle to Vessel tangent        | a1 or a          | 1220.00 | mm.                |
| Saddle Width                                  | b1 or b          | 150.00  | mm.                |
| Saddle Bearing Angle                          | delta or theta   | 120.00  | degrees            |
| Wear Plate Width                              | b2 or b1         | 225.00  | mm.                |
| Wear Plate Bearing Angle                      | delta2 or theta1 | 132.00  | degrees            |
| Wear Plate Thickness                          | e2 or tr         | 12.0    | mm.                |
| Wear Plate Allowable Stress                   | fw or Sr         | 138.00  | N./mm <sup>2</sup> |
| Inside or Mean Depth of Head                  | Hi or hm         | 130.09  | mm.                |
| Shell Allowable Stress used in Calculation    |                  | 247.00  | N./mm <sup>2</sup> |
| Head Allowable Stress used in Calculation     |                  | 247.00  | N./mm <sup>2</sup> |
| Circumferential Efficiency in Plane of Saddle |                  | 1.00    |                    |
| Circumferential Efficiency at Mid-Span        |                  | 1.00    |                    |
| Distance from Saddle Base to Centerline       | B                | 500.00  | mm.                |
| Coefficient of Friction                       | mu               | 0.45    |                    |

### Results for Horizontal Vessel Analysis:

|                                           |   |         |      |
|-------------------------------------------|---|---------|------|
| Saddle Force Q, Test Case, no Ext. Forces |   | 2137.13 | Kgf  |
| Pressure used in Analysis                 | P | 36.385  | bars |

| Horizontal Vessel Analysis Results: | Actual<br>N./mm <sup>2</sup> | Allowable<br>N./mm <sup>2</sup> |
|-------------------------------------|------------------------------|---------------------------------|
| Long. Stress at Top of Midspan      | 48.56                        | 247.00                          |
| Long. Stress at Bottom of Midspan   | 55.59                        | 247.00                          |
| Long. Stress at Top of Saddles      | 81.21                        | 247.00                          |
| Long. Stress at Bottom of Saddles   | 35.92                        | 247.00                          |
| Tangential Shear in Shell           | 6.51                         | 138.58                          |
| Circ. Stress at Horn of Saddle      | 8.40                         | 370.49                          |
| Circ. Compressive Stress in Shell   | 0.46                         | 247.00                          |

### Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:  
 = Ftr( Ft/Num of Saddles + Z Force Load ) • B / E  
 = 3.0( 334.9/2 + 0 ) • 500.0/460.0  
 = 546.1 Kgf

Longitudinal Saddle Reaction Force [Fwl]:  
 = max( Fl,wind; Sum of X Forces ) • B / Ls  
 = max( 21.95, 0 ) • 500.0/3656.0  
 = 3.0 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:  
 = Saddle Load + max( Fwl, Fwt, Fsl, Fst )  
 = 1591 + max( 3, 546, 0, 0 )  
 = 2137.1 Kgf

Longitudinal Wind Force [Fl]:  
 = WindScalar • WindPress( Platform Area + (  $\pi/4$ ( OD • WindDiaMult )<sup>2</sup> ) )



$$= 1.1 \cdot 1984.306(0.0 + \pi/4(0.514 \cdot 1.2)^2)$$

$$= 652.203 \text{ N}$$

**Summary of Loads at the Base of this Saddle:**

|                                         |         |     |
|-----------------------------------------|---------|-----|
| Vertical Load (including saddle weight) | 2186.85 | Kgf |
| Transverse Shear Load                   | 167.46  | Kgf |
| Longitudinal Shear Load                 | 21.95   | Kgf |

**Note: Wear Plate is Welded to the Shell, k = 0.1**

Saddle Dimension [E]:

= Baseplate Length  
 = 460.000 mm.

**The Computed K values from Table 4.15.1:**

|              |              |              |              |
|--------------|--------------|--------------|--------------|
| K1 = 0.1066  | K2 = 1.1707  | K3 = 0.8799  | K4 = 0.4011  |
| K5 = 0.7603  | K6 = 0.0529  | K7 = 0.0529  | K8 = 0.3405  |
| K9 = 0.2711  | K10 = 0.0581 | K1* = 0.1923 | K6p = 0.0434 |
| K7p = 0.0434 |              |              |              |

*The suffix 'p' denotes the values for a wear plate if it exists.*

**Note: Dimension a is greater than or equal to Rm/2.**

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -2137 \cdot 1220.0 [1 - (1 - 1220.0/6221.8 + (252.587^2 - 130.087^2)/(2 \cdot 1220.0 \cdot 6221.8))/(1 + (4 \cdot 130.09)/(3 \cdot 6221.8))]$$

$$= -560.3 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 2137 \cdot 6222/4((1 + 2(253^2 - 130^2)/(6222^2))/(1 + (4 \cdot 130)/(3 \cdot 6222)) - 4 \cdot 1220/6222)$$

$$= 634.6 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [ $\sigma_1$ ]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$

$$= 36.385 \cdot 252.587/(2 \cdot 8.825) - 634.6/(\pi \cdot 252.6^2 \cdot 8.825)$$

$$= 48.56 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [ $\sigma_2$ ]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 36.385 \cdot 252.587/(2 \cdot 8.825) + 634.6/(\pi \cdot 252.6^2 \cdot 8.825)$$

$$= 55.59 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [ $\sigma_{*3}$ ]:

$$= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t)$$

$$= 36.385 \cdot 252.587/(2 \cdot 8.825) - 560.3/(0.1066 \cdot \pi \cdot 252.6^2 \cdot 8.825)$$

$$= 81.21 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [ $\sigma_{*4}$ ]:

$$= P \cdot Rm/(2t) + M1/(K1* \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 36.385 \cdot 252.587/(2 \cdot 8.825) + 560.3/(0.1923 \cdot \pi \cdot 252.6^2 \cdot 8.825)$$

$$= 35.92 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot hm/3))$$

$$= 2137(6221.8 - 2 \cdot 1220.0)/(6221.8 + (4 \cdot 130.09/3))$$

$$= 1263.8 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [ $\tau_2$ ]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 1263.78 / (252.5875 \cdot 8.825)$$

$$= 6.51 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1, x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(252.587 \cdot 8.825)}$$

$$= 36.826 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:  
 $= \min( b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a )$   
 $= \min( 150.0 + 1.56 \cdot \sqrt{ 252.587 \cdot 8.825 }, 2 \cdot 1220.0 )$   
 $= 223.65 \text{ mm.}$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:  
 $= \min( S_r/S, 1 )$   
 $= \min( 137.997/246.996, 1 )$   
 $= 0.5587$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [ $\sigma_6, r$ ]:  
 $= -K5 \cdot Q \cdot k / ( B1( t + \eta \cdot tr ) )$   
 $= -0.7603 \cdot 2137 \cdot 0.1 / ( 223.653( 8.825 + 0.559 \cdot 12.0 ) )$   
 $= -0.46 \text{ N./mm}^2$

Circ. Comp. Stress at Horn of Saddle,  $L \geq 8R_m$  (4.15.28) [ $\sigma_7, r$ ]:  
 $= -Q / ( 4( t + \eta \cdot tr ) b1 ) - 3 \cdot K7 \cdot Q / ( 2( t + \eta \cdot tr )^2 )$   
 $= -2137 / ( 4( 8.825 + 0.559 \cdot 12.0 ) 223.653 ) -$   
 $3 \cdot 0.053 \cdot 2137 / ( 2( 8.825 + 0.559 \cdot 12.0 )^2 )$   
 $= -8.40 \text{ N./mm}^2$

#### Results for Vessel Ribs, Web and Base:

|                                      |        |          |                    |
|--------------------------------------|--------|----------|--------------------|
| Baseplate Length                     | Bplen  | 460.0000 | mm.                |
| Baseplate Thickness                  | Bpthk  | 16.0000  | mm.                |
| Baseplate Width                      | Bpwid  | 150.0000 | mm.                |
| Number of Ribs ( inc. outside ribs ) | Nribs  | 2        |                    |
| Rib Thickness                        | Ribtk  | 12.0000  | mm.                |
| Web Thickness                        | Webtk  | 12.0000  | mm.                |
| Web Location                         | Webloc | Side     |                    |
| Saddle Yield Stress                  | Sy     | 260.0    | N./mm <sup>2</sup> |
| Height of Web at Center              | Hw,c   | 306.0    | mm.                |
| Friction Coefficient                 | mu     | 0.450    |                    |

#### Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:  
 $= (( 1.56 \cdot \sqrt{ R \cdot t } + \text{WearPlateWidth} ) t^3 ) / 12$   
 $= (( 1.56 \cdot \sqrt{ 248.175 \cdot 8.825 } + 225.0 ) 8.825^3 ) / 12$   
 $= 1.707 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:  
 $= \text{ShellInertia} + A_{\text{Shell}}( C1 - Y )^2$   
 $= 1.707 + 26.299( 93.88 - 4.412 )^2$   
 $= 2106.781 \text{ cm}^4$

|           | B<br>mm. | D<br>mm. | Y<br>mm. | A<br>cm <sup>2</sup> | AY<br>mm. <sup>3</sup> | I + AD <sup>(2)</sup><br>cm <sup>4</sup> |
|-----------|----------|----------|----------|----------------------|------------------------|------------------------------------------|
| Shell     | 298.0    | 8.8      | 4.4      | 26.3                 | 11604.5                | 0.211E+04                                |
| Wearplate | 225.0    | 12.0     | 14.8     | 27.0                 | 40027.5                | 0.169E+04                                |
| Web       | 12.0     | 215.0    | 128.3    | 25.8                 | 331078.6               | 0.130E+04                                |
| BasePlate | 150.0    | 16.0     | 243.8    | 24.0                 | 585180.1               | 0.540E+04                                |
| Totals    | ...      | ...      | ...      | 103.1                | 967890.6               | 0.105E+05                                |

Distance to Centroid [C1]:  
 $= AY / A$   
 $= 967891/103.099$   
 $= 93.880 \text{ mm.}$

Angle [β]:  
 $= 180 - \text{Saddle Angle}/2$   
 $= 180 - 120.0/2$   
 $= 120.0$

Saddle Splitting Coefficient [K1]:  
 $= ( 1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2 ) / ( \pi - \beta + \sin(\beta) \cos(\beta) )$   
 $= ( 1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2 ) / ( \pi - 2.094 + \sin(120.0) \cos(120.0) )$   
 $= 0.2035$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 2137.128$$

$$= 434.9519 \text{ Kgf}$$

$$\text{Tension Stress, } St = ( Fh/As ) = 0.5554 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 500.0 - 248.17 \cdot \sin(120.0/2) / 1.0472$$

$$= 294.761 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 128.2094 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = ( M \cdot C1 / I ) = 1.1243 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

#### Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(2137 + 50) \cdot 150.0 / (2 \cdot 460.0 \cdot 173.33)}$$

$$= 7.779 \text{ mm.}$$

#### Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) \cdot (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) \cdot (245.0 + 12.0 + 12.0)$$

$$= 465.922 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 465.9216 / (2 - 1)$$

$$= 465.922 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 465.9216 \cdot 150.0/2$$

$$= 349.441 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 2186.851 / (460.0 \cdot 150.0)$$

$$= 3.169 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 349.4 \cdot 3.17$$

$$= 1107.501 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 16.56 + 27.955$$

$$= 44.515 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 1107.5/44.5153$$

$$= 2.440 \text{ N./mm}^2$$

#### Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

|        | B     | D     | Y    | A    | AY       | Io   |
|--------|-------|-------|------|------|----------|------|
| Rib    | 12.0  | 138.0 | 81.0 | 16.6 | 134136.0 | 630. |
| Web    | 233.0 | 12.0  | 6.0  | 28.0 | 16773.2  | 221. |
| Totals | ...   | ...   | ...  | 44.5 | 150909.2 | 851. |

Rib dimension [D]:

$$\begin{aligned} &= \text{Saddle Width} - \text{Web Thickness} \\ &= 150.0 - 12.0 \\ &= 138.000 \text{ mm.} \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned} &= AY / A \\ &= 150909.172 / 44.515 \\ &= 33.901 \text{ mm.} \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned} &= \max( ytot, \text{Saddle Width} - ytot ) \\ &= \max( 33.901, 150.0 - 33.901 ) \\ &= 116.099 \text{ mm.} \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned} &= \sqrt{ (\text{Total Inertia} / \text{Total Area}) } \\ &= \sqrt{ (851.1 / 44.515) } \\ &= 43.727 \text{ mm.} \end{aligned}$$

Length of Outer Rib, step 1, [Lv]:

$$\begin{aligned} &= \text{Saddle Height} - \cos( \text{saddle bearing angle} / 2 ) ( \text{radius} + \text{shlthk} + \text{wpdthk} ) - \text{bpthk} \\ &= 500.0 - \cos( 120.0 / 2 ) ( 245.0 + 12.0 + 12.0 ) - 16.0 \\ &= 349.500 \text{ mm.} \end{aligned}$$

Length of Outer Rib, step 2, [Lw]:

$$\begin{aligned} &= 2 \cdot \sin( \text{saddle bearing angle} / 2 ) ( \text{radius} + \text{shlthk} + \text{wpdthk} ) \\ &= 2 \cdot \sin( 120.0 / 2 ) ( 245.0 + 12.0 + 12.0 ) \\ &= 465.922 \text{ mm.} \end{aligned}$$

Length of Outer Rib, step 3, [Bd]:

$$\begin{aligned} &= \text{Baseplate length} - \text{clearance} \\ &= 460.0 - 50.8 \\ &= 409.200 \text{ mm.} \end{aligned}$$

Length of Outer Rib, step 4, [Dd]:

$$\begin{aligned} &= ( Lw - Bd ) / 2 \\ &= ( 465.922 - 409.2 ) / 2 \\ &= 28.361 \text{ mm.} \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned} &= \sqrt{ (Lv^2 + Dd^2) } \\ &= \sqrt{ (349.5^2 + 28.361^2) } \\ &= 350.649 \text{ mm.} \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned} &= \sqrt{ ( 2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress} ) } \\ &= \sqrt{ ( 2 \cdot \pi^2 \cdot 0.19648\text{E}+09 / 260.0 ) } \\ &= 122.138 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned} &= KL / r \\ &= 1 \cdot 350.649 / 43.727 \\ &= 8.019 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned} &= FI / ( 2 \cdot \text{Bplen} ) \cdot e \cdot L / 2 \\ &= 21.9 / ( 2 \cdot 460.0 ) \cdot 465.922 \cdot 350.65 / 2 \\ &= 1.949 \text{ Kg-m.} \end{aligned}$$

Compressive Allowable,  $KL/r < Cc$  (  $8.0191 < 122.1379$  ) per AISC E2-1 [Sca]:

$$\begin{aligned} &= ( 1 - (KL/r)^2 / (2 \cdot Cc^2) ) Fy / ( 5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3) ) \\ &= ( 1 - ( 8.02 )^2 / ( 2 \cdot 122.14^2 ) ) 260 / \\ &\quad ( 5/3 + 3 \cdot ( 8.02 ) / ( 8 \cdot 122.14 ) - ( 8.02^3 ) / ( 8 \cdot 122.14^3 ) ) \\ &= 153.4 \text{ N./mm}^2 \end{aligned}$$

**AISC Unity Check of Outside Ribs ( must be  $\leq 1$  )**

$$\begin{aligned} &= Sc / Sca + ( Rm \cdot C1 / I ) / Sba \\ &= 2.44 / 153.4 + ( 1.95 \cdot 116.099 / 8511374 ) / 173.33 \end{aligned}$$



= 0.017

#### **Input Data for Base Plate Bolting Calculations:**

|                                            |         |                           |
|--------------------------------------------|---------|---------------------------|
| Total Number of Bolts per BasePlate        | Nbolts  | 2                         |
| Total Number of Bolts in Tension/Baseplate | Nbt     | 1                         |
| Bolt Material Specification                |         | SA-193 B7                 |
| Bolt Allowable Stress                      | Stba    | 172.00 N./mm <sup>2</sup> |
| Bolt Corrosion Allowance                   | Bca     | 1.5 mm.                   |
| Distance from Bolts to Edge                | Edgedis | 50.8 mm.                  |
| Nominal Bolt Diameter                      | Bnd     | 24.0000 mm.               |
| Thread Series                              | Series  | TEMA Metric               |
| BasePlate Allowable Stress                 | S       | 138.00 N./mm <sup>2</sup> |
| Area Available in a Single Bolt            | BltArea | 2.2586 cm <sup>2</sup>    |
| Saddle Load Q0 (Weight)                    | Q0      | 1640.8 Kg                 |
| Saddle Load QL (Wind/Seismic contribution) | QL      | 3.0 Kg                    |
| Maximum Transverse Force                   | Ft      | 167.5 Kg                  |
| Maximum Longitudinal Force                 | Fl      | 21.9 Kg                   |
| Saddle Bolted to Steel Foundation          |         | No                        |

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= Fl / ( \text{Bolt Area} \cdot \text{Number of Bolts} )$$

$$= 21.9 / ( 2.259 \cdot 2 )$$

$$= 0.5 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / ( \text{Bolt Area} \cdot \text{Number of Bolts} )$$

$$= 167.5 / ( 2.259 \cdot 2 )$$

$$= 3.6 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

#### **Bolt Area Calculation per Dennis R. Moss**

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / ( \text{BoltShearAllowable} \cdot \text{Nbolts} )$$

$$= 21.95 / ( 103.2 \cdot 2.0 )$$

$$= 0.0104 \text{ cm}^2$$

#### **Bolt Area due to Transverse Load:**

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 500.0 \cdot 167.46 + 0.0$$

$$= 83.73 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 83.73 / 1640.79$$

$$= 51.03 \text{ mm.} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0.0104, 0.0]$$

$$= 0.0104 \text{ cm}^2$$

*Review notes about nozzle loadings and supports in the Warnings report.*

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**Input, Nozzle Desc: TS IN NOZ T1**

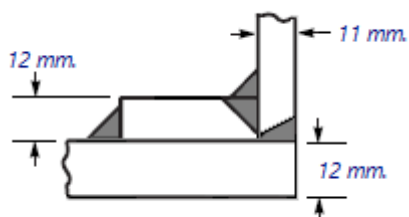
**From: 20**

|                                               |      |           |                    |
|-----------------------------------------------|------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P    | 5.564     | bars               |
| Temperature for Internal Pressure             | Temp | 77        | °C                 |
| Parent Material                               |      | SA-516 70 |                    |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D    | 490.00    | mm.                |
| Shell Finished (Minimum) Thickness            | t    | 12.0000   | mm.                |
| Shell Internal Corrosion Allowance            | c    | 3.1750    | mm.                |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |      | 250.00    | mm.                |
| User Entered Minimum Design Metal Temperature |      | -28.89    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                             |      |                  |                    |
|---------------------------------------------|------|------------------|--------------------|
| Material                                    |      | SA-106 B         |                    |
| Material UNS Number                         |      | K03006           |                    |
| Material Specification/Type                 |      | Smls. pipe       |                    |
| Allowable Stress at Temperature             | Sn   | 118.00           | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                 | Sna  | 118.00           | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)           |      | Inside           |                    |
| Layout Angle                                |      | -90.00           | deg                |
| Diameter                                    |      | 6.0000           | in.                |
| Size and Thickness Basis                    |      | Nominal          |                    |
| Nominal Thickness                           |      | 80               |                    |
| Flange Material                             |      | SA-105           |                    |
| Flange Type                                 |      | Weld Neck Flange |                    |
| Corrosion Allowance                         | can  | 3.1750           | mm.                |
| Joint Efficiency of Shell Seam at Nozzle    | E1   | 1.00             |                    |
| Joint Efficiency of Nozzle Neck             | En   | 1.00             |                    |
| Outside Projection                          | ho   | 150.0000         | mm.                |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000          | mm.                |
| Groove weld depth between Nozzle and Vessel | Wgnv | 0.0000           | mm.                |
| Pad Material                                |      | SA-516 70        |                    |
| Pad Allowable Stress at Temperature         | Sp   | 138.00           | N./mm <sup>2</sup> |
| Pad Allowable Stress At Ambient             | Spa  | 138.00           | N./mm <sup>2</sup> |
| Diameter of Pad along vessel surface        | Dp   | 268.2750         | mm.                |
| Thickness of Pad                            | te   | 12.0000          | mm.                |
| Weld leg size between Pad and Shell         | Wp   | 10.0000          | mm.                |
| Groove weld depth between Pad and Nozzle    | Wgpn | 12.0000          | mm.                |
| Reinforcing Pad Width                       |      | 50.0000          | mm.                |
| Flange Class                                |      | 300              |                    |
| Flange Grade                                |      | GR 1.1           |                    |

The Pressure Design option was Design Pressure + static head.



# Reinforcement CALCULATION, Description: TS IN NOZ T1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 5.761 in.  
 Actual Thickness Used in Calculation 0.432 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 5.56 \cdot 248.175 / (138 \cdot 1.0 - 0.6 \cdot 5.56)$   
 $= 1.0031$  mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 5.56 \cdot 76.34 / (118 \cdot 1.0 - 0.6 \cdot 5.56)$   
 $= 0.3610$  mm.

## UG-40, Limits of Reinforcement: [Internal Pressure]

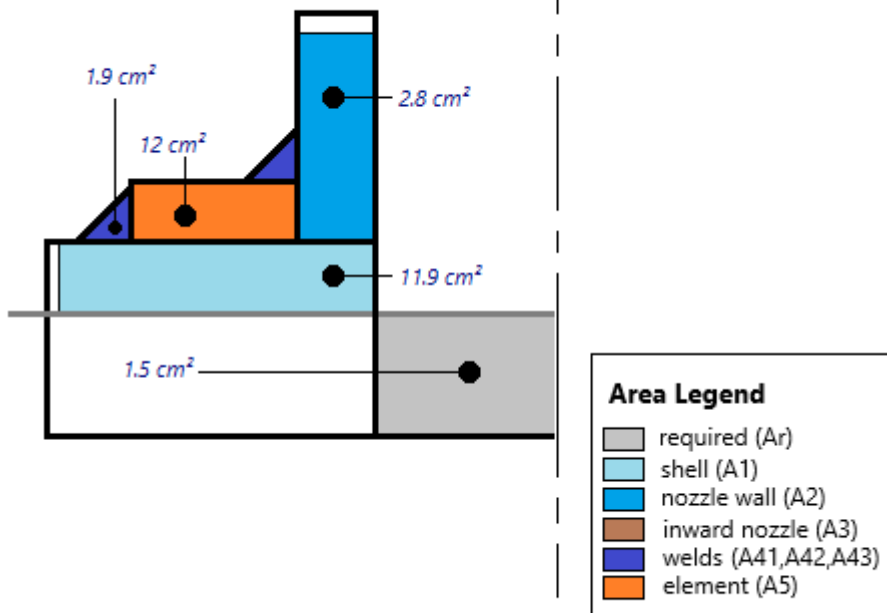
|                                                        |    |          |     |
|--------------------------------------------------------|----|----------|-----|
| Parallel to Vessel Wall (Diameter Limit)               | D1 | 305.3588 | mm. |
| Parallel to Vessel Wall, opening length                | d  | 152.6794 | mm. |
| Normal to Vessel Wall (Thickness Limit), pad side Tlwp |    | 22.0625  | mm. |

Weld Strength Reduction Factor [fr1]:  
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 118.0 / 138.0)$   
 $= 0.855$

Weld Strength Reduction Factor [fr4]:  
 $= \min(1, S_p / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(0.855, 1.0)$   
 $= 0.855$



**Figure UG 37.1 Nozzle Areas**

**Results of Nozzle Reinforcement Area Calculations: (cm²)**

| AREA AVAILABLE, A1 to A5 |             | Design | External | Mapnc |
|--------------------------|-------------|--------|----------|-------|
| Area Required            | Ar          | 1.532  | NA       | NA    |
| Area in Shell            | A1          | 11.942 | NA       | NA    |
| Area in Nozzle Wall      | A2          | 2.806  | NA       | NA    |
| Area in Inward Nozzle    | A3          | 0.000  | NA       | NA    |
| Area in Welds            | A41+A42+A43 | 1.855  | NA       | NA    |
| Area in Element          | A5          | 12.000 | NA       | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 28.603 | NA       | NA    |

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.  
 The area available with the given pad is Sufficient.

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$  UG-37(c)  
 $= (152.6794 \cdot 1.0031 \cdot 1.0 + 2 \cdot 7.7978 \cdot 1.0031 \cdot 1.0 \cdot (1 - 1.0))$   
 $= 1.532 \text{ cm}^2$

**Reinforcement Areas per Figure UG-37.1**

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 152.679(1.0 \cdot 8.825 - 1.0 \cdot 1.003) - 2 \cdot 7.798$   
 $(1.0 \cdot 8.825 - 1.0 \cdot 1.0031) \cdot (1 - 1.0)$   
 $= 11.942 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:  
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$   
 $= (2 \cdot 22.06) \cdot (7.8 - 0.36) \cdot 0.8551$   
 $= 2.806 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:  
 $= Wo^2 \cdot fr3 + (Wi\text{-}can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$   
 $= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 10.0^2 \cdot 1.0$



= 1.855 cm<sup>2</sup>

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))\*(min(tp,Tlwp,te)) \* fr4  
= ( 268.275 - 168.275 ) \* 12.0 \* 1.0  
= 12.000 cm<sup>2</sup>

**UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]**

Wall Thickness for Internal/External pressures      ta = 3.5360 mm.  
Wall Thickness per UG16(b),      tr16b = 5.6750 mm.  
Wall Thickness, shell/head, internal pressure      trb1 = 4.1781 mm.  
Wall Thickness      tb1 = max(trb1, tr16b) = 5.6750 mm.  
Wall Thickness      tb2 = max(trb2, tr16b) = 5.6750 mm.  
Wall Thickness per table UG-45      tb3 = 9.3950 mm.

Determine Nozzle Thickness candidate [tb]:

= min[ tb3, max( tb1,tb2) ]  
= min[ 9.395, max( 5.675, 5.675 ) ]  
= 5.6750 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max( ta, tb )  
= max( 3.536, 5.675 )  
= 5.6750 mm.

Available Nozzle Neck Thickness = 0.875 \* 10.973 = 9.601 mm. -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):**

|            |   |       |           |   |                          |        |
|------------|---|-------|-----------|---|--------------------------|--------|
| Sustained  | : | 31.4, | Allowable | : | 118.0 N./mm <sup>2</sup> | Passed |
| Expansion  | : | 0.0,  | Allowable | : | 263.6 N./mm <sup>2</sup> | Passed |
| Occasional | : | 2.6,  | Allowable | : | 156.9 N./mm <sup>2</sup> | Passed |
| Shear      | : | 20.3, | Allowable | : | 82.6 N./mm <sup>2</sup>  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**

**Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)**

Govrn. thk, tg = 9.601, tr = 0.361, c = 3.175 mm., E\* = 1.0  
Thickness Ratio = tr \* E\*/(tg - c) = 0.056, Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve A   | -8 °C   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

**Nozzle Neck to Pad Weld for the Nozzle, Curve: B**

Govrn. thk, tg = 9.601, tr = 0.361, c = 3.175 mm., E\* = 1.0  
Thickness Ratio = tr \* E\*/(tg - c) = 0.056, Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -29 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |

**Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B**

Govrn. thk, tg = 9.601, tr = 0.361, c = 3.175 mm., E\* = 1.0  
Thickness Ratio = tr \* E\*/(tg - c) = 0.056, Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -29 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |

**Shell to Pad Weld Junction at Pad OD, Curve: B**

Govrn. thk, tg = 12.0, tr = 1.003, c = 3.175 mm., E\* = 1.0  
Thickness Ratio = tr \* E\*/(tg - c) = 0.114, Temp. Reduction = 78 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

#### Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.601, tr = 0.361, c = 3.175 mm., E\* = 1.0  
 Thickness Ratio =  $tr \cdot E^* / (tg - c) = 0.056$ , Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C  
 Gov. MDMT of the Reinforcement Pad : -104 °C  
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

#### Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
 Design Pressure/Ambient Rating =  $5.56/51.00 = 0.109$

#### Weld Size Calculations, Description: TS IN NOZ T1

Intermediate Calc. for nozzle/shell Welds Tmin 7.7978 mm.  
 Intermediate Calc. for pad/shell Welds TminPad 8.8250 mm.

#### Results Per UW-16.1:

|             | Required Thickness              | Actual Thickness             |
|-------------|---------------------------------|------------------------------|
| Nozzle Weld | $5.4585 = 0.7 \cdot t_{min}$    | $7.0700 = 0.7 \cdot W_o$ mm. |
| Pad Weld    | $4.4125 = 0.5 \cdot t_{minPad}$ | $7.0700 = 0.7 \cdot W_p$ mm. |

#### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
 $= (A_r - A_1) \cdot S_v$   
 $= (1.5316 - 11.9424) \cdot 138$   
 $= 0.00 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A_2 + A_5 + A_4 - (W_i - C_{an} / .707)^2 \cdot f_{r2}) \cdot S_v$   
 $= (2.8059 + 12.0 + 1.8551 - 0.0 \cdot 0.86) \cdot 138$   
 $= 23444.62 \text{ Kgf}$

#### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$   
 $= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 118$   
 $= 15584. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:  
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$   
 $= (3.1416/2.0) \cdot 268.275 \cdot 10.0 \cdot 0.49 \cdot 138$   
 $= 29056. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:  
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$   
 $= (3.1416/2) \cdot 168.275 \cdot 12.0 \cdot 0.74 \cdot 138$   
 $= 33029. \text{ Kgf}$

#### Strength of Failure Paths:

Failure Path [Path 1-1]:  
 $= (S_{PEW} + S_{NGW}) = (29056 + 0) = 29056 \text{ Kgf}$



### Summary of Failure Path Calculations:

Path 1-1 = 29056 Kgf, must exceed W = 0 Kgf or W1 = 23444 Kgf

### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 48 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 14.9003 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 176.9003 mm.

### Input Echo, WRC297 Item 1, Description: TS IN NOZ T1 :

|                                      |            |                    |
|--------------------------------------|------------|--------------------|
| Diameter Basis for Cylindrical Shell | ID         |                    |
| Shell Internal Corrosion Allowance   | 3.1750     | mm.                |
| Shell External Corrosion Allowance   | 0.0000     | mm.                |
| Shell Diameter                       | 490.000    | mm.                |
| Shell Thickness                      | 12.0000    | mm.                |
| Shell Stress Concentration Factor    | 1.000      |                    |
| Vessel Material                      | SA-516 70  |                    |
| Vessel Cold S.I. Allowable           | Smc 138.00 | N./mm <sup>2</sup> |
| Vessel Hot S.I. Allowable            | Smh 138.00 | N./mm <sup>2</sup> |

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

|                                    |             |                    |
|------------------------------------|-------------|--------------------|
| Diameter Basis for Nozzle          | ID          |                    |
| Nozzle Corrosion Allowance         | 3.1750      | mm.                |
| Nozzle Diameter                    | 146.329     | mm.                |
| Nozzle Thickness                   | 10.9728     | mm.                |
| Nozzle Stress Concentration Factor | 1.000       |                    |
| Nozzle Material                    | SA-106 B    |                    |
| Nozzle Cold S.I. Allowable         | SNmc 118.00 | N./mm <sup>2</sup> |
| Nozzle Hot S.I. Allowable          | SNmh 118.00 | N./mm <sup>2</sup> |

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

|                              |         |     |
|------------------------------|---------|-----|
| Diameter of Reinforcing Pad  | 268.275 | mm. |
| Thickness of Reinforcing Pad | 12.0000 | mm. |

### Note:

External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUS tained loads.

|                          |    |         |       |
|--------------------------|----|---------|-------|
| Design Internal Pressure | Dp | 5.56    | bars  |
| Radial Load              | P  | -407.91 | Kgf   |
| Circumferential Shear    | Vc | 611.87  | Kgf   |
| Longitudinal Shear       | Vl | 611.87  | Kgf   |
| Circumferential Moment   | Mc | 275.24  | Kg-m. |
| Longitudinal Moment      | Ml | 366.98  | Kg-m. |
| Torsional Moment         | Mt | 458.73  | Kg-m. |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

### Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(258.587(12.0 - 3.175))}$$

$$= 325.918 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 517.175 mm.

Nozzle Outside Diameter (d) = 168.275 mm.  
 Vessel Thickness used (T) = 20.825 mm.  
 Nozzle Thickness used (t) = 7.798 mm.  
 T / t = 2.671  
 d / t = 21.580

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.621$$

Nr/P = 0.079  
 Mr/P = 0.083  
 M0/P = 0.071  
 N0/P = 0.185  
 MrD/Mc = 0.178  
 NrDL/Mc = 0.128  
 M0d/Mc = 0.170  
 N0DL/Mc = 0.152  
 MrD/Ml = 0.127  
 NrDL/Ml = 0.091  
 M0D/Ml = 0.122  
 N0DL/Ml = 0.281

**Vessel Stresses (N./mm²):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | 1                    | 1                   | 1                       | 1                      |
| Outplane Bending (P )                                            | 3                    | -3                  | 3                       | -3                     |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (ML)                                           | -13                  | -13                 | 13                      | 13                     |
| Outplane Bending (ML)                                            | -36                  | 36                  | 36                      | -36                    |
| Normal Pressure Stress                                           | 6                    | 6                   | 6                       | 6                      |
| Outplane Stress Summary                                          | -37                  | 26                  | 62                      | -17                    |

**Vessel Stresses (N./mm²):**

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (P )                                               | 4                    | -4                  | 4                       | -4                     |
| Inplane Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Membrane (ML)                                              | -4                   | -4                  | 4                       | 4                      |
| Inplane Bending (ML)                                               | -37                  | 37                  | 37                      | -37                    |
| Inplane Pressure Stress                                            | 3                    | 3                   | 3                       | 3                      |
| Inplane Stress Summary                                             | -33                  | 32                  | 50                      | -33                    |

**Vessel Stresses (N./mm²):**

| LONGITUDINAL PLANE<br>(Shear stress normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|----------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Shear (Vc)                                                  | 1                    | 1                   | -1                      | -1                     |
| Outplane Shear (Vl)                                                  | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Mt)                                                  | 4                    | 4                   | 4                       | 4                      |
| Shear Stress Summary                                                 | 5                    | 5                   | 3                       | 3                      |

**Vessel Stresses (N./mm²):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Two * Max Shear Stress                     | 42                   | 36                  | 63                      | 34                     |

**Vessel Stresses (N./mm²):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Membrane (P )                                                 | 1                     | 1                    | 1                      | 1                     |
| Outplane Bending (P )                                                  | 3                     | -3                   | 3                      | -3                    |
| Outplane Membrane (Mc)                                                 | -5                    | -5                   | 5                      | 5                     |
| Outplane Bending (Mc)                                                  | -37                   | 37                   | 37                     | -37                   |
| Outplane Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Outplane Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Normal Pressure Stress                                                 | 3                     | 3                    | 3                      | 3                     |
| Outplane Stress Summary                                                | -34                   | 32                   | 52                     | -31                   |

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses parallel to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Inplane Membrane (P )                                                    | 0                     | 0                    | 0                      | 0                     |
| Inplane Bending (P )                                                     | 4                     | -4                   | 4                      | -4                    |
| Inplane Membrane (Mc)                                                    | -4                    | -4                   | 4                      | 4                     |
| Inplane Bending (Mc)                                                     | -39                   | 39                   | 39                     | -39                   |
| Inplane Membrane (ML)                                                    | 0                     | 0                    | 0                      | 0                     |
| Inplane Bending (ML)                                                     | 0                     | 0                    | 0                      | 0                     |
| Inplane Pressure Stress                                                  | 6                     | 6                    | 6                      | 6                     |
| Inplane Stress Summary                                                   | -32                   | 37                   | 55                     | -31                   |

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|----------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Shear (Vc)                                                        | 0                     | 0                    | 0                      | 0                     |
| Outplane Shear (Vl)                                                        | -1                    | -1                   | 1                      | 1                     |
| Torsional Shear (Mt)                                                       | 4                     | 4                    | 4                      | 4                     |
| Shear Stress Summary                                                       | 3                     | 3                    | 5                      | 5                     |

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|-----------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Two * Max Shear Stress                        | 37                    | 39                   | 60                     | 37                    |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses in the<br>hoop direction) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|-----------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Hoop Membrane (P )                                        | 1                    | 1                   | 1                       | 1                      |
| Hoop Bending (P )                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (Mc)                                        | 0                    | 0                   | 0                       | 0                      |
| Hoop Bending (Mc)                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (ML)                                        | -13                  | -13                 | 13                      | 13                     |
| Hoop Bending (ML)                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Pressure Stress                                      | 6                    | 6                   | 6                       | 6                      |
| Hoop Stress Summary                                       | -5                   | -5                  | 21                      | 22                     |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Axial Membrane (P )                                              | 1                    | 1                   | 1                       | 1                      |
| Axial Bending (P )                                               | 17                   | -17                 | 17                      | -17                    |
| Axial Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |



|                       |      |     |     |      |
|-----------------------|------|-----|-----|------|
| Axial Bending (Mc)    | 0    | 0   | 0   | 0    |
| Axial Membrane (ML)   | -23  | -23 | 23  | 23   |
| Axial Bending (ML)    | -171 | 171 | 171 | -171 |
| Axial Pressure Stress | 3    | 3   | 3   | 3    |

|                      |      |     |     |      |
|----------------------|------|-----|-----|------|
| Axial Stress Summary | -174 | 134 | 216 | -160 |
|----------------------|------|-----|-----|------|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                    |         |        |         |        |
|--------------------|---------|--------|---------|--------|
| LONGITUDINAL PLANE | Au      | Al     | Bu      | Bl     |
| (Shear stress)     | Outside | Inside | Outside | Inside |

|                      |    |    |    |    |
|----------------------|----|----|----|----|
| Shear due to (Vc)    | 2  | 2  | -2 | -2 |
| Shear due to (Vl)    | 0  | 0  | 0  | 0  |
| Shear due to Torsion | 12 | 12 | 12 | 12 |

|                      |    |    |    |    |
|----------------------|----|----|----|----|
| Shear Stress Summary | 15 | 15 | 10 | 10 |
|----------------------|----|----|----|----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                     |         |        |         |        |
|---------------------|---------|--------|---------|--------|
| LONGITUDINAL PLANE  | Au      | Al     | Bu      | Bl     |
| (Stress Intensities | Outside | Inside | Outside | Inside |

|                        |     |     |     |     |
|------------------------|-----|-----|-----|-----|
| Two * Max Shear Stress | 175 | 143 | 217 | 184 |
|------------------------|-----|-----|-----|-----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                       |         |        |         |        |
|-----------------------|---------|--------|---------|--------|
| CIRCUMFERENTIAL PLANE | Cu      | Cl     | Du      | Dl     |
| (Stresses in the      | Left    | Left   | Right   | Right  |
| hoop direction)       | Outside | Inside | Outside | Inside |

|                      |    |    |   |   |
|----------------------|----|----|---|---|
| Hoop Membrane (P )   | 1  | 1  | 1 | 1 |
| Hoop Bending (P )    | 0  | 0  | 0 | 0 |
| Hoop Membrane (Mc)   | -5 | -5 | 5 | 5 |
| Hoop Bending (Mc)    | 0  | 0  | 0 | 0 |
| Hoop Membrane (ML)   | 0  | 0  | 0 | 0 |
| Hoop Bending (ML)    | 0  | 0  | 0 | 0 |
| Hoop Pressure Stress | 6  | 6  | 6 | 6 |

|                     |   |   |    |    |
|---------------------|---|---|----|----|
| Hoop Stress Summary | 2 | 2 | 13 | 14 |
|---------------------|---|---|----|----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                       |         |        |         |        |
|-----------------------|---------|--------|---------|--------|
| CIRCUMFERENTIAL PLANE | Cu      | Cl     | Du      | Dl     |
| (Stresses Normal to   | Left    | Left   | Right   | Right  |
| pipe cross-section)   | Outside | Inside | Outside | Inside |

|                       |      |     |     |      |
|-----------------------|------|-----|-----|------|
| Axial Membrane (P )   | 1    | 1   | 1   | 1    |
| Axial Bending (P )    | 17   | -17 | 17  | -17  |
| Axial Membrane (Mc)   | -17  | -17 | 17  | 17   |
| Axial Bending (Mc)    | -179 | 179 | 179 | -179 |
| Axial Membrane (ML)   | 0    | 0   | 0   | 0    |
| Axial Bending (ML)    | 0    | 0   | 0   | 0    |
| Axial Pressure Stress | 3    | 3   | 3   | 3    |

|                      |      |     |     |      |
|----------------------|------|-----|-----|------|
| Axial Stress Summary | -176 | 148 | 218 | -174 |
|----------------------|------|-----|-----|------|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                       |         |        |         |        |
|-----------------------|---------|--------|---------|--------|
| CIRCUMFERENTIAL PLANE | Cu      | Cl     | Du      | Dl     |
| (Shear stress)        | Outside | Inside | Outside | Inside |

|                      |    |    |    |    |
|----------------------|----|----|----|----|
| Shear due to (Vc)    | 0  | 0  | 0  | 0  |
| Shear due to (Vl)    | -2 | -2 | 2  | 2  |
| Shear due to Torsion | 12 | 12 | 12 | 12 |

|                      |    |    |    |    |
|----------------------|----|----|----|----|
| Shear Stress Summary | 10 | 10 | 15 | 15 |
|----------------------|----|----|----|----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                       |         |        |         |        |
|-----------------------|---------|--------|---------|--------|
| CIRCUMFERENTIAL PLANE | Cu      | Cl     | Du      | Dl     |
| (Stress Intensities   | Outside | Inside | Outside | Inside |

|                        |     |     |     |     |
|------------------------|-----|-----|-----|-----|
| Two * Max Shear Stress | 179 | 149 | 219 | 191 |
|------------------------|-----|-----|-----|-----|

### Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{252.587(12.0 - 3.175)}$$

$$= 324.078 \text{ mm.}$$

WRC 297 Curve Access Parameters:

|                         |       |             |
|-------------------------|-------|-------------|
| Vessel Mean Diameter    | (D) = | 505.175 mm. |
| Nozzle Outside Diameter | (d) = | 268.275 mm. |
| Vessel Thickness used   | (T) = | 8.825 mm.   |
| Nozzle Thickness used   | (t) = | 57.798 mm.  |
| T / t                   | =     | 0.153       |
| d / t                   | =     | 10.000      |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.018$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

|         |   |       |
|---------|---|-------|
| Nr/P    | = | 0.050 |
| Mr/P    | = | 0.082 |
| M0/P    | = | 0.025 |
| N0/P    | = | 0.044 |
| MrD/Mc  | = | 0.253 |
| NrDL/Mc | = | 0.096 |
| M0d/Mc  | = | 0.076 |
| N0DL/Mc | = | 0.073 |
| MrD/M1  | = | 0.064 |
| NrDL/M1 | = | 0.053 |
| M0D/M1  | = | 0.019 |
| N0DL/M1 | = | 0.047 |

### Vessel Stresses (N./mm²):

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | 2                    | 2                   | 2                       | 2                      |
| Outplane Bending (P )                                            | 7                    | -7                  | 7                       | -7                     |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (ML)                                           | -8                   | -8                  | 8                       | 8                      |
| Outplane Bending (ML)                                            | -20                  | 20                  | 20                      | -20                    |
| Normal Pressure Stress                                           | 15                   | 15                  | 15                      | 15                     |
| Outplane Stress Summary                                          | -2                   | 22                  | 53                      | -1                     |

### Vessel Stresses (N./mm²):

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | 2                    | 2                   | 2                       | 2                      |
| Inplane Bending (P )                                               | 25                   | -25                 | 25                      | -25                    |
| Inplane Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Membrane (ML)                                              | -9                   | -9                  | 9                       | 9                      |
| Inplane Bending (ML)                                               | -66                  | 66                  | 66                      | -66                    |
| Inplane Pressure Stress                                            | 7                    | 7                   | 7                       | 7                      |
| Inplane Stress Summary                                             | -39                  | 41                  | 111                     | -72                    |

### Vessel Stresses (N./mm²):

| LONGITUDINAL PLANE<br>(Shear stress normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|----------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Shear (Vc)                                                  | 1                    | 1                   | -1                      | -1                     |
| Outplane Shear (Vl)                                                  | 0                    | 0                   | 0                       | 0                      |



|                     |   |   |   |   |
|---------------------|---|---|---|---|
| Outplane Shear (Mt) | 4 | 4 | 4 | 4 |
|---------------------|---|---|---|---|

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 6 | 6 | 2 | 2 |
|----------------------|---|---|---|---|

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Two * Max Shear Stress                     | 40                   | 43                  | 111                     | 72                     |

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Membrane (P )                                                 | 2                     | 2                    | 2                      | 2                     |
| Outplane Bending (P )                                                  | 7                     | -7                   | 7                      | -7                    |
| Outplane Membrane (Mc)                                                 | -9                    | -9                   | 9                      | 9                     |
| Outplane Bending (Mc)                                                  | -58                   | 58                   | 58                     | -58                   |
| Outplane Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Outplane Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Normal Pressure Stress                                                 | 7                     | 7                    | 7                      | 7                     |
| Outplane Stress Summary                                                | -50                   | 51                   | 85                     | -47                   |

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses parallel to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Inplane Membrane (P )                                                    | 2                     | 2                    | 2                      | 2                     |
| Inplane Bending (P )                                                     | 25                    | -25                  | 25                     | -25                   |
| Inplane Membrane (Mc)                                                    | -12                   | -12                  | 12                     | 12                    |
| Inplane Bending (Mc)                                                     | -196                  | 196                  | 196                    | -196                  |
| Inplane Membrane (ML)                                                    | 0                     | 0                    | 0                      | 0                     |
| Inplane Bending (ML)                                                     | 0                     | 0                    | 0                      | 0                     |
| Inplane Pressure Stress                                                  | 15                    | 15                   | 15                     | 15                    |
| Inplane Stress Summary                                                   | -165                  | 177                  | 252                    | -191                  |

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|----------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Shear (Vc)                                                        | 0                     | 0                    | 0                      | 0                     |
| Outplane Shear (Vl)                                                        | -1                    | -1                   | 1                      | 1                     |
| Torsional Shear (Mt)                                                       | 4                     | 4                    | 4                      | 4                     |
| Shear Stress Summary                                                       | 2                     | 2                    | 6                      | 6                     |

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|-----------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Two * Max Shear Stress                        | 165                   | 177                  | 252                    | 191                   |

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm<sup>2</sup>):**

| Type of<br>Stress Int.<br>Location | Stress Values at |     |    |    |    |    |    |    |
|------------------------------------|------------------|-----|----|----|----|----|----|----|
|                                    | Au               | Al  | Bu | Bl | Cu | Cl | Du | Dl |
| Circ. Pm (SUS)                     | 6                | 6   | 6  | 6  | 6  | 6  | 6  | 6  |
| Circ. Pl (SUS)                     | -12              | -12 | 14 | 14 | -4 | -4 | 4  | 4  |



|                 |      |      |      |      |      |      |      |      |
|-----------------|------|------|------|------|------|------|------|------|
| Circ. Q (SUS)   | -33  | 33   | 39   | -39  | -35  | 35   | 43   | -43  |
| Long. Pm (SUS)  | 3    | 3    | 3    | 3    | 3    | 3    | 3    | 3    |
| Long. Pl (SUS)  | -4   | -4   | 4    | 4    | -4   | -4   | 6    | 6    |
| Long. Q (SUS)   | -33  | 33   | 41   | -41  | -34  | 34   | 40   | -40  |
| Shear Pm (SUS)  | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear Pl (SUS)  | 1    | 1    | -1   | -1   | -1   | -1   | 1    | 1    |
| Shear Q (SUS)   | 4    | 4    | 4    | 4    | 4    | 4    | 4    | 4    |
| Pm (SUS)        | 6.0  | 6.0  | 6.0  | 6.0  | 6.0  | 6.0  | 6.0  | 6.0  |
| Pm+Pl (SUS)     | 6.2  | 6.2  | 20.1 | 20.1 | 3.6  | 3.6  | 10.6 | 10.6 |
| Pm+Pl+Q (Total) | 42.1 | 35.1 | 59.8 | 34.6 | 37.2 | 38.6 | 56.4 | 37.1 |

#### Vessel Stress Summation Comparison (N./mm<sup>2</sup>):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 6.00      | 138.00         | Passed |
| Pm+Pl (SUS)         | 20.08     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 59.76     | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Vessel Stress Summation at Reinforcing Pad Edge (N./mm<sup>2</sup>):

| Type of Stress Int. Location | Stress Values at |      |       |      |       |       |       |       |
|------------------------------|------------------|------|-------|------|-------|-------|-------|-------|
|                              | Au               | Al   | Bu    | Bl   | Cu    | Cl    | Du    | Dl    |
| Circ. Pm (SUS)               | 15               | 15   | 15    | 15   | 15    | 15    | 15    | 15    |
| Circ. Pl (SUS)               | -6               | -6   | 10    | 10   | -10   | -10   | 14    | 14    |
| Circ. Q (SUS)                | -13              | 13   | 27    | -27  | -171  | 171   | 221   | -221  |
| Long. Pm (SUS)               | 7                | 7    | 7     | 7    | 7     | 7     | 7     | 7     |
| Long. Pl (SUS)               | -7               | -7   | 11    | 11   | -7    | -7    | 11    | 11    |
| Long. Q (SUS)                | -41              | 41   | 91    | -91  | -51   | 51    | 65    | -65   |
| Shear Pm (SUS)               | 0                | 0    | 0     | 0    | 0     | 0     | 0     | 0     |
| Shear Pl (SUS)               | 1                | 1    | -1    | -1   | -1    | -1    | 1     | 1     |
| Shear Q (SUS)                | 4                | 4    | 4     | 4    | 4     | 4     | 4     | 4     |
| Pm (SUS)                     | 15.0             | 15.0 | 15.0  | 15.0 | 15.0  | 15.0  | 15.0  | 15.0  |
| Pm+Pl (SUS)                  | 9.2              | 9.2  | 25.1  | 25.1 | 5.4   | 5.4   | 29.1  | 29.1  |
| Pm+Pl+Q (Total)              | 41.7             | 42.2 | 109.2 | 73.1 | 166.1 | 176.1 | 250.1 | 192.2 |

#### Vessel Stress Summation Comparison (N./mm<sup>2</sup>):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 15.00     | 138.00         | Passed |
| Pm+Pl (SUS)         | 29.09     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 250.15    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm<sup>2</sup>):

| Type of | Stress Values at |
|---------|------------------|
|---------|------------------|



| Stress Int. Location | Au    | Al    | Bu    | Bl    | Cu    | Cl    | Du    | Dl    |
|----------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| Circ. Pm (SUS)       | 6     | 6     | 6     | 6     | 6     | 6     | 6     | 6     |
| Circ. Pl (SUS)       | -12   | -12   | 14    | 14    | -4    | -4    | 6     | 6     |
| Circ. Q (SUS)        | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Long. Pm (SUS)       | 3     | 3     | 3     | 3     | 3     | 3     | 3     | 3     |
| Long. Pl (SUS)       | -22   | -22   | 24    | 24    | -16   | -16   | 18    | 18    |
| Long. Q (SUS)        | -154  | 154   | 188   | -188  | -162  | 162   | 196   | -196  |
| Shear Pm (SUS)       | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Pl (SUS)       | 2     | 2     | -2    | -2    | -2    | -2    | 2     | 2     |
| Shear Q (SUS)        | 12    | 12    | 12    | 12    | 12    | 12    | 12    | 12    |
| Pm (SUS)             | 6.0   | 6.0   | 6.0   | 6.0   | 6.0   | 6.0   | 6.0   | 6.0   |
| Pm+Pl (SUS)          | 19.3  | 19.3  | 27.5  | 27.5  | 15.5  | 15.5  | 21.4  | 21.4  |
| Pm+Pl+Q (Total)      | 174.2 | 143.8 | 215.5 | 182.1 | 178.1 | 149.7 | 218.0 | 189.1 |

**Nozzle Stress Summation Comparison (N./mm<sup>2</sup>):**

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 6.00      | 118.00         | Passed |
| Pm+Pl (SUS)         | 27.53     | 177.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 217.95    | 353.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

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**Input, Nozzle Desc: TS OUT NOZ T2**

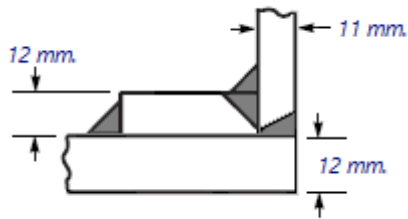
**From: 20**

|                                               |      |           |                    |
|-----------------------------------------------|------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P    | 5.516     | bars               |
| Temperature for Internal Pressure             | Temp | 77        | °C                 |
| Parent Material                               |      | SA-516 70 |                    |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D    | 490.00    | mm.                |
| Shell Finished (Minimum) Thickness            | t    | 12.0000   | mm.                |
| Shell Internal Corrosion Allowance            | c    | 3.1750    | mm.                |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |      | 250.00    | mm.                |
| User Entered Minimum Design Metal Temperature |      | -28.89    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                             |      |                  |                    |
|---------------------------------------------|------|------------------|--------------------|
| Material                                    |      | SA-106 B         |                    |
| Material UNS Number                         |      | K03006           |                    |
| Material Specification/Type                 |      | Smls. pipe       |                    |
| Allowable Stress at Temperature             | Sn   | 118.00           | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                 | Sna  | 118.00           | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)           |      | Inside           |                    |
| Layout Angle                                |      | 90.00            | deg                |
| Diameter                                    |      | 6.0000           | in.                |
| Size and Thickness Basis                    |      | Nominal          |                    |
| Nominal Thickness                           |      | 80               |                    |
| Flange Material                             |      | SA-105           |                    |
| Flange Type                                 |      | Weld Neck Flange |                    |
| Corrosion Allowance                         | can  | 3.1750           | mm.                |
| Joint Efficiency of Shell Seam at Nozzle    | E1   | 1.00             |                    |
| Joint Efficiency of Nozzle Neck             | En   | 1.00             |                    |
| Outside Projection                          | ho   | 150.0000         | mm.                |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000          | mm.                |
| Groove weld depth between Nozzle and Vessel | Wgnv | 0.0000           | mm.                |
| Pad Material                                |      | SA-516 70        |                    |
| Pad Allowable Stress at Temperature         | Sp   | 138.00           | N./mm <sup>2</sup> |
| Pad Allowable Stress At Ambient             | Spa  | 138.00           | N./mm <sup>2</sup> |
| Diameter of Pad along vessel surface        | Dp   | 268.2750         | mm.                |
| Thickness of Pad                            | te   | 12.0000          | mm.                |
| Weld leg size between Pad and Shell         | Wp   | 10.0000          | mm.                |
| Groove weld depth between Pad and Nozzle    | Wgpn | 12.0000          | mm.                |
| Reinforcing Pad Width                       |      | 50.0000          | mm.                |
| Flange Class                                |      | 300              |                    |
| Flange Grade                                |      | GR 1.1           |                    |

The Pressure Design option was Design Pressure + static head.



### Reinforcement CALCULATION, Description: TS OUT NOZ T2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |           |
|--------------------------------------------|-----------|
| Actual Inside Diameter Used in Calculation | 5.761 in. |
| Actual Thickness Used in Calculation       | 0.432 in. |

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 5.52 \cdot 248.175 / (138 \cdot 1.0 - 0.6 \cdot 5.52)$   
 $= 0.9944$  mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 5.52 \cdot 76.34 / (118 \cdot 1.0 - 0.6 \cdot 5.52)$   
 $= 0.3579$  mm.

### UG-40, Limits of Reinforcement: [Internal Pressure]

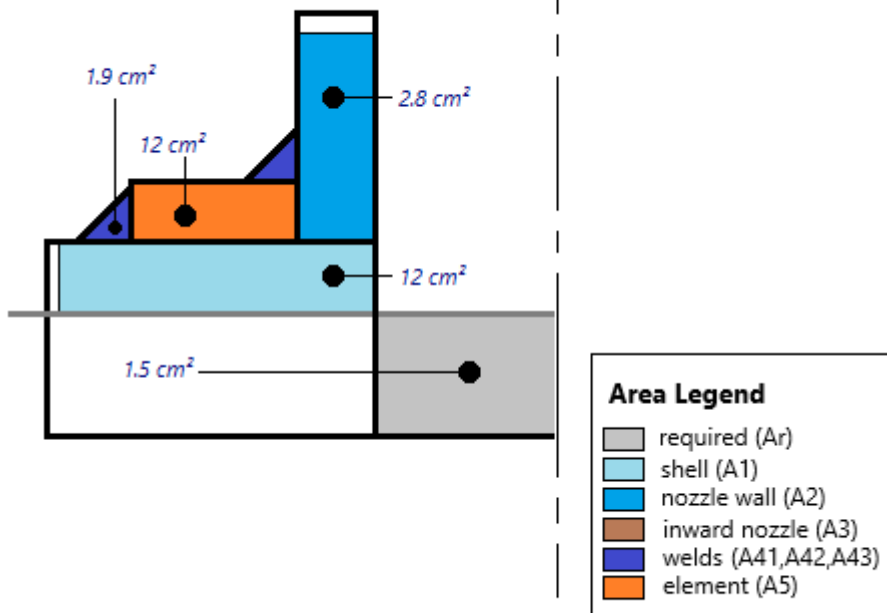
|                                                        |    |              |
|--------------------------------------------------------|----|--------------|
| Parallel to Vessel Wall (Diameter Limit)               | D1 | 305.3588 mm. |
| Parallel to Vessel Wall, opening length                | d  | 152.6794 mm. |
| Normal to Vessel Wall (Thickness Limit), pad side Tlwp |    | 22.0625 mm.  |

Weld Strength Reduction Factor [fr1]:  
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 118.0 / 138.0)$   
 $= 0.855$

Weld Strength Reduction Factor [fr4]:  
 $= \min(1, S_p / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(0.855, 1.0)$   
 $= 0.855$



**Figure UG 37.1 Nozzle Areas**

**Results of Nozzle Reinforcement Area Calculations: (cm²)**

| AREA AVAILABLE, A1 to A5 |             | Design | External | Mapnc |
|--------------------------|-------------|--------|----------|-------|
| Area Required            | Ar          | 1.518  | NA       | NA    |
| Area in Shell            | A1          | 11.956 | NA       | NA    |
| Area in Nozzle Wall      | A2          | 2.807  | NA       | NA    |
| Area in Inward Nozzle    | A3          | 0.000  | NA       | NA    |
| Area in Welds            | A41+A42+A43 | 1.855  | NA       | NA    |
| Area in Element          | A5          | 12.000 | NA       | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 28.618 | NA       | NA    |

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.  
 The area available with the given pad is Sufficient.

Area Required [A]:  

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (152.6794 \cdot 0.9944 \cdot 1.0 + 2 \cdot 7.7978 \cdot 0.9944 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 1.518 \text{ cm}^2$$

**Reinforcement Areas per Figure UG-37.1**

Area Available in Shell [A1]:  

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 152.679 (1.0 \cdot 8.825 - 1.0 \cdot 0.994) - 2 \cdot 7.798$$

$$(1.0 \cdot 8.825 - 1.0 \cdot 0.994) \cdot (1 - 1.0)$$

$$= 11.956 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:  

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 22.06) \cdot (7.8 - 0.36) \cdot 0.8551$$

$$= 2.807 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:  

$$= Wo^2 \cdot fr3 + (Wi\text{-}can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$

$$= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 10.0^2 \cdot 1.0$$



= 1.855 cm<sup>2</sup>

Area Available in Element [A5]:

$$\begin{aligned} &= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lwp}, t_e)) \cdot f_{r4} \\ &= (268.275 - 168.275) \cdot 12.0 \cdot 1.0 \\ &= 12.000 \text{ cm}^2 \end{aligned}$$

**UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]**

|                                                |                                     |
|------------------------------------------------|-------------------------------------|
| Wall Thickness for Internal/External pressures | ta = 3.5329 mm.                     |
| Wall Thickness per UG16(b),                    | tr16b = 5.6750 mm.                  |
| Wall Thickness, shell/head, internal pressure  | trb1 = 4.1694 mm.                   |
| Wall Thickness                                 | tb1 = max(trb1, tr16b) = 5.6750 mm. |
| Wall Thickness                                 | tb2 = max(trb2, tr16b) = 5.6750 mm. |
| Wall Thickness per table UG-45                 | tb3 = 9.3950 mm.                    |

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[ tb3, \max( tb1, tb2 ) ] \\ &= \min[ 9.395, \max( 5.675, 5.675 ) ] \\ &= 5.6750 \text{ mm.} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max( ta, tb ) \\ &= \max( 3.5329, 5.675 ) \\ &= 5.6750 \text{ mm.} \end{aligned}$$

Available Nozzle Neck Thickness = 0.875 \* 10.973 = 9.601 mm. -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):**

|            |         |           |                            |        |
|------------|---------|-----------|----------------------------|--------|
| Sustained  | : 31.4, | Allowable | : 118.0 N./mm <sup>2</sup> | Passed |
| Expansion  | : 0.0,  | Allowable | : 263.6 N./mm <sup>2</sup> | Passed |
| Occasional | : 2.6,  | Allowable | : 156.9 N./mm <sup>2</sup> | Passed |
| Shear      | : 20.3, | Allowable | : 82.6 N./mm <sup>2</sup>  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**

**Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)**

Govrn. thk, tg = 9.601, tr = 0.358, c = 3.175 mm., E\* = 1.0  
Thickness Ratio = tr \* E\*/(tg - c) = 0.056, Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve A   | -8 °C   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

**Nozzle Neck to Pad Weld for the Nozzle, Curve: B**

Govrn. thk, tg = 9.601, tr = 0.358, c = 3.175 mm., E\* = 1.0  
Thickness Ratio = tr \* E\*/(tg - c) = 0.056, Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -29 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |

**Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B**

Govrn. thk, tg = 9.601, tr = 0.358, c = 3.175 mm., E\* = 1.0  
Thickness Ratio = tr \* E\*/(tg - c) = 0.056, Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -29 °C  |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |

**Shell to Pad Weld Junction at Pad OD, Curve: B**

Govrn. thk, tg = 12.0, tr = 0.994, c = 3.175 mm., E\* = 1.0  
Thickness Ratio = tr \* E\*/(tg - c) = 0.113, Temp. Reduction = 78 °C



Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C  
Min Metal Temp. w/o impact per UG-20(f) -29 °C

#### Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.601, tr = 0.358, c = 3.175 mm., E\* = 1.0  
Thickness Ratio = tr \* E\*/(tg - c) = 0.056, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C  
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C  
Gov. MDMT of the Reinforcement Pad : -104 °C  
Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

#### Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating = 5.52/51.00 = 0.108

#### Weld Size Calculations, Description: TS OUT NOZ T2

Intermediate Calc. for nozzle/shell Welds Tmin 7.7978 mm.  
Intermediate Calc. for pad/shell Welds TminPad 8.8250 mm.

#### Results Per UW-16.1:

|             | Required Thickness     | Actual Thickness      |
|-------------|------------------------|-----------------------|
| Nozzle Weld | 5.4585 = 0.7 * tmin.   | 7.0700 = 0.7 * Wo mm. |
| Pad Weld    | 4.4125 = 0.5 * TminPad | 7.0700 = 0.7 * Wp mm. |

#### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
= ( Ar - A1 ) \* Sv  
= ( 1.5183 - 11.9557 ) \* 138  
= 0.00 Kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
= (A2+A5+A4-(Wi-Can/.707)<sup>2</sup>\*fr2)\*Sv  
= ( 2.8071 + 12.0 + 1.8551 - 0.0 \* 0.86 ) \* 138  
= 23446.27 Kgf

#### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
= (π/2) \* Dlo \* Wo \* 0.49 \* Snw  
= ( 3.1416/2.0 ) \* 168.275 \* 10.0 \* 0.49 \* 118  
= 15584. Kgf

Shear, Pad Element Weld [Spew]:  
= (π/2) \* DP \* WP \* 0.49 \* SEW  
= ( 3.1416/2.0 ) \* 268.275 \* 10.0 \* 0.49 \* 138  
= 29056. Kgf

Tension, Pad Groove Weld [Tpgw]:  
= ( π/2 ) \* Dlo \* Wgpn \* 0.74 \* Seg  
= ( 3.1416/2 ) \* 168.275 \* 12.0 \* 0.74 \* 138  
= 33029. Kgf

#### Strength of Failure Paths:

Failure Path [Path 1-1]:  
= ( SPEW + SNGW ) = ( 29056 + 0 ) = 29056 Kgf



### Summary of Failure Path Calculations:

Path 1-1 = 29056 Kgf, must exceed W = 0 Kgf or W1 = 23446 Kgf

### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 48 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 14.9003 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 176.9003 mm.

### Input Echo, WRC297 Item 1, Description: TS OUT NOZ T2 :

|                                      |            |                    |
|--------------------------------------|------------|--------------------|
| Diameter Basis for Cylindrical Shell | ID         |                    |
| Shell Internal Corrosion Allowance   | 3.1750     | mm.                |
| Shell External Corrosion Allowance   | 0.0000     | mm.                |
| Shell Diameter                       | 490.000    | mm.                |
| Shell Thickness                      | 12.0000    | mm.                |
| Shell Stress Concentration Factor    | 1.000      |                    |
| Vessel Material                      | SA-516 70  |                    |
| Vessel Cold S.I. Allowable           | Smc 138.00 | N./mm <sup>2</sup> |
| Vessel Hot S.I. Allowable            | Smh 138.00 | N./mm <sup>2</sup> |

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

|                                    |             |                    |
|------------------------------------|-------------|--------------------|
| Diameter Basis for Nozzle          | ID          |                    |
| Nozzle Corrosion Allowance         | 3.1750      | mm.                |
| Nozzle Diameter                    | 146.329     | mm.                |
| Nozzle Thickness                   | 10.9728     | mm.                |
| Nozzle Stress Concentration Factor | 1.000       |                    |
| Nozzle Material                    | SA-106 B    |                    |
| Nozzle Cold S.I. Allowable         | SNmc 118.00 | N./mm <sup>2</sup> |
| Nozzle Hot S.I. Allowable          | SNmh 118.00 | N./mm <sup>2</sup> |

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

|                              |         |     |
|------------------------------|---------|-----|
| Diameter of Reinforcing Pad  | 268.275 | mm. |
| Thickness of Reinforcing Pad | 12.0000 | mm. |

### Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

|                          |    |         |       |
|--------------------------|----|---------|-------|
| Design Internal Pressure | Dp | 5.52    | bars  |
| Radial Load              | P  | -407.91 | Kgf   |
| Circumferential Shear    | Vc | 611.87  | Kgf   |
| Longitudinal Shear       | Vl | 611.87  | Kgf   |
| Circumferential Moment   | Mc | 275.24  | Kg-m. |
| Longitudinal Moment      | Ml | 366.98  | Kg-m. |
| Torsional Moment         | Mt | 458.73  | Kg-m. |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

### Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))} \\
 &= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(258.587(12.0 - 3.175))} \\
 &= 325.918 \text{ mm.}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 517.175 mm.



Nozzle Outside Diameter (d) = 168.275 mm.  
 Vessel Thickness used (T) = 20.825 mm.  
 Nozzle Thickness used (t) = 7.798 mm.  
 T / t = 2.671  
 d / t = 21.580

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.621$$

Nr/P = 0.079  
 Mr/P = 0.083  
 M0/P = 0.071  
 N0/P = 0.185  
 MrD/Mc = 0.178  
 NrDL/Mc = 0.128  
 M0d/Mc = 0.170  
 N0DL/Mc = 0.152  
 MrD/Ml = 0.127  
 NrDL/Ml = 0.091  
 M0D/Ml = 0.122  
 N0DL/Ml = 0.281

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | 1                    | 1                   | 1                       | 1                      |
| Outplane Bending (P )                                            | 3                    | -3                  | 3                       | -3                     |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (ML)                                           | -13                  | -13                 | 13                      | 13                     |
| Outplane Bending (ML)                                            | -36                  | 36                  | 36                      | -36                    |
| Normal Pressure Stress                                           | 6                    | 6                   | 6                       | 6                      |
| Outplane Stress Summary                                          | -38                  | 26                  | 61                      | -17                    |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (P )                                               | 4                    | -4                  | 4                       | -4                     |
| Inplane Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Membrane (ML)                                              | -4                   | -4                  | 4                       | 4                      |
| Inplane Bending (ML)                                               | -37                  | 37                  | 37                      | -37                    |
| Inplane Pressure Stress                                            | 3                    | 3                   | 3                       | 3                      |
| Inplane Stress Summary                                             | -33                  | 32                  | 50                      | -33                    |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Shear stress normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|----------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Shear (Vc)                                                  | 1                    | 1                   | -1                      | -1                     |
| Outplane Shear (Vl)                                                  | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Mt)                                                  | 4                    | 4                   | 4                       | 4                      |
| Shear Stress Summary                                                 | 5                    | 5                   | 3                       | 3                      |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Two * Max Shear Stress                     | 42                   | 36                  | 63                      | 34                     |

**Vessel Stresses (N./mm<sup>2</sup>):**



| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Membrane (P )                                                 | 1                     | 1                    | 1                      | 1                     |
| Outplane Bending (P )                                                  | 3                     | -3                   | 3                      | -3                    |
| Outplane Membrane (Mc)                                                 | -5                    | -5                   | 5                      | 5                     |
| Outplane Bending (Mc)                                                  | -37                   | 37                   | 37                     | -37                   |
| Outplane Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Outplane Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Normal Pressure Stress                                                 | 3                     | 3                    | 3                      | 3                     |
| Outplane Stress Summary                                                | -34                   | 32                   | 52                     | -31                   |

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses parallel to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Inplane Membrane (P )                                                    | 0                     | 0                    | 0                      | 0                     |
| Inplane Bending (P )                                                     | 4                     | -4                   | 4                      | -4                    |
| Inplane Membrane (Mc)                                                    | -4                    | -4                   | 4                      | 4                     |
| Inplane Bending (Mc)                                                     | -39                   | 39                   | 39                     | -39                   |
| Inplane Membrane (ML)                                                    | 0                     | 0                    | 0                      | 0                     |
| Inplane Bending (ML)                                                     | 0                     | 0                    | 0                      | 0                     |
| Inplane Pressure Stress                                                  | 6                     | 6                    | 6                      | 6                     |
| Inplane Stress Summary                                                   | -32                   | 37                   | 55                     | -31                   |

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|----------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Shear (Vc)                                                        | 0                     | 0                    | 0                      | 0                     |
| Outplane Shear (Vl)                                                        | -1                    | -1                   | 1                      | 1                     |
| Torsional Shear (Mt)                                                       | 4                     | 4                    | 4                      | 4                     |
| Shear Stress Summary                                                       | 3                     | 3                    | 5                      | 5                     |

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|-----------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Two * Max Shear Stress                        | 37                    | 39                   | 60                     | 37                    |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses in the<br>hoop direction) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|-----------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Hoop Membrane (P )                                        | 1                    | 1                   | 1                       | 1                      |
| Hoop Bending (P )                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (Mc)                                        | 0                    | 0                   | 0                       | 0                      |
| Hoop Bending (Mc)                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Membrane (ML)                                        | -13                  | -13                 | 13                      | 13                     |
| Hoop Bending (ML)                                         | 0                    | 0                   | 0                       | 0                      |
| Hoop Pressure Stress                                      | 6                    | 6                   | 6                       | 6                      |
| Hoop Stress Summary                                       | -5                   | -5                  | 21                      | 22                     |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Axial Membrane (P )                                              | 1                    | 1                   | 1                       | 1                      |
| Axial Bending (P )                                               | 17                   | -17                 | 17                      | -17                    |
| Axial Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |



|                       |      |     |     |      |
|-----------------------|------|-----|-----|------|
| Axial Bending (Mc)    | 0    | 0   | 0   | 0    |
| Axial Membrane (ML)   | -23  | -23 | 23  | 23   |
| Axial Bending (ML)    | -171 | 171 | 171 | -171 |
| Axial Pressure Stress | 3    | 3   | 3   | 3    |

|                      |      |     |     |      |
|----------------------|------|-----|-----|------|
| Axial Stress Summary | -174 | 134 | 216 | -160 |
|----------------------|------|-----|-----|------|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                    |         |        |         |        |
|--------------------|---------|--------|---------|--------|
| LONGITUDINAL PLANE | Au      | Al     | Bu      | Bl     |
| (Shear stress)     | Outside | Inside | Outside | Inside |

|                      |    |    |    |    |
|----------------------|----|----|----|----|
| Shear due to (Vc)    | 2  | 2  | -2 | -2 |
| Shear due to (Vl)    | 0  | 0  | 0  | 0  |
| Shear due to Torsion | 12 | 12 | 12 | 12 |

|                      |    |    |    |    |
|----------------------|----|----|----|----|
| Shear Stress Summary | 15 | 15 | 10 | 10 |
|----------------------|----|----|----|----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                     |         |        |         |        |
|---------------------|---------|--------|---------|--------|
| LONGITUDINAL PLANE  | Au      | Al     | Bu      | Bl     |
| (Stress Intensities | Outside | Inside | Outside | Inside |

|                        |     |     |     |     |
|------------------------|-----|-----|-----|-----|
| Two * Max Shear Stress | 175 | 143 | 217 | 184 |
|------------------------|-----|-----|-----|-----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                       |         |        |         |        |
|-----------------------|---------|--------|---------|--------|
| CIRCUMFERENTIAL PLANE | Cu      | Cl     | Du      | Dl     |
| (Stresses in the      | Left    | Left   | Right   | Right  |
| hoop direction)       | Outside | Inside | Outside | Inside |

|                      |    |    |   |   |
|----------------------|----|----|---|---|
| Hoop Membrane (P )   | 1  | 1  | 1 | 1 |
| Hoop Bending (P )    | 0  | 0  | 0 | 0 |
| Hoop Membrane (Mc)   | -5 | -5 | 5 | 5 |
| Hoop Bending (Mc)    | 0  | 0  | 0 | 0 |
| Hoop Membrane (ML)   | 0  | 0  | 0 | 0 |
| Hoop Bending (ML)    | 0  | 0  | 0 | 0 |
| Hoop Pressure Stress | 6  | 6  | 6 | 6 |

|                     |   |   |    |    |
|---------------------|---|---|----|----|
| Hoop Stress Summary | 2 | 2 | 13 | 14 |
|---------------------|---|---|----|----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                       |         |        |         |        |
|-----------------------|---------|--------|---------|--------|
| CIRCUMFERENTIAL PLANE | Cu      | Cl     | Du      | Dl     |
| (Stresses Normal to   | Left    | Left   | Right   | Right  |
| pipe cross-section)   | Outside | Inside | Outside | Inside |

|                       |      |     |     |      |
|-----------------------|------|-----|-----|------|
| Axial Membrane (P )   | 1    | 1   | 1   | 1    |
| Axial Bending (P )    | 17   | -17 | 17  | -17  |
| Axial Membrane (Mc)   | -17  | -17 | 17  | 17   |
| Axial Bending (Mc)    | -179 | 179 | 179 | -179 |
| Axial Membrane (ML)   | 0    | 0   | 0   | 0    |
| Axial Bending (ML)    | 0    | 0   | 0   | 0    |
| Axial Pressure Stress | 3    | 3   | 3   | 3    |

|                      |      |     |     |      |
|----------------------|------|-----|-----|------|
| Axial Stress Summary | -176 | 148 | 218 | -174 |
|----------------------|------|-----|-----|------|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                       |         |        |         |        |
|-----------------------|---------|--------|---------|--------|
| CIRCUMFERENTIAL PLANE | Cu      | Cl     | Du      | Dl     |
| (Shear stress)        | Outside | Inside | Outside | Inside |

|                      |    |    |    |    |
|----------------------|----|----|----|----|
| Shear due to (Vc)    | 0  | 0  | 0  | 0  |
| Shear due to (Vl)    | -2 | -2 | 2  | 2  |
| Shear due to Torsion | 12 | 12 | 12 | 12 |

|                      |    |    |    |    |
|----------------------|----|----|----|----|
| Shear Stress Summary | 10 | 10 | 15 | 15 |
|----------------------|----|----|----|----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                       |         |        |         |        |
|-----------------------|---------|--------|---------|--------|
| CIRCUMFERENTIAL PLANE | Cu      | Cl     | Du      | Dl     |
| (Stress Intensities   | Outside | Inside | Outside | Inside |

|                        |     |     |     |     |
|------------------------|-----|-----|-----|-----|
| Two * Max Shear Stress | 179 | 149 | 219 | 191 |
|------------------------|-----|-----|-----|-----|

### Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{252.587(12.0 - 3.175)}$$

$$= 324.078 \text{ mm.}$$

WRC 297 Curve Access Parameters:

|                         |       |             |
|-------------------------|-------|-------------|
| Vessel Mean Diameter    | (D) = | 505.175 mm. |
| Nozzle Outside Diameter | (d) = | 268.275 mm. |
| Vessel Thickness used   | (T) = | 8.825 mm.   |
| Nozzle Thickness used   | (t) = | 57.798 mm.  |
| T / t                   | =     | 0.153       |
| d / t                   | =     | 10.000      |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.018$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

|         |   |       |
|---------|---|-------|
| Nr/P    | = | 0.050 |
| Mr/P    | = | 0.082 |
| M0/P    | = | 0.025 |
| N0/P    | = | 0.044 |
| MrD/Mc  | = | 0.253 |
| NrDL/Mc | = | 0.096 |
| M0d/Mc  | = | 0.076 |
| N0DL/Mc | = | 0.073 |
| MrD/M1  | = | 0.064 |
| NrDL/M1 | = | 0.053 |
| M0D/M1  | = | 0.019 |
| N0DL/M1 | = | 0.047 |

### Vessel Stresses (N./mm²):

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | 2                    | 2                   | 2                       | 2                      |
| Outplane Bending (P )                                            | 7                    | -7                  | 7                       | -7                     |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (ML)                                           | -8                   | -8                  | 8                       | 8                      |
| Outplane Bending (ML)                                            | -20                  | 20                  | 20                      | -20                    |
| Normal Pressure Stress                                           | 15                   | 15                  | 15                      | 15                     |
| Outplane Stress Summary                                          | -3                   | 22                  | 53                      | -1                     |

### Vessel Stresses (N./mm²):

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | 2                    | 2                   | 2                       | 2                      |
| Inplane Bending (P )                                               | 25                   | -25                 | 25                      | -25                    |
| Inplane Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Membrane (ML)                                              | -9                   | -9                  | 9                       | 9                      |
| Inplane Bending (ML)                                               | -66                  | 66                  | 66                      | -66                    |
| Inplane Pressure Stress                                            | 7                    | 7                   | 7                       | 7                      |
| Inplane Stress Summary                                             | -39                  | 41                  | 111                     | -72                    |

### Vessel Stresses (N./mm²):

| LONGITUDINAL PLANE<br>(Shear stress normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|----------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Shear (Vc)                                                  | 1                    | 1                   | -1                      | -1                     |
| Outplane Shear (Vl)                                                  | 0                    | 0                   | 0                       | 0                      |

| Outplane Shear (Mt) | 4 | 4 | 4 | 4 |
|---------------------|---|---|---|---|
|---------------------|---|---|---|---|

| Shear Stress Summary | 6 | 6 | 2 | 2 |
|----------------------|---|---|---|---|
|----------------------|---|---|---|---|

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Two * Max Shear Stress                     | 40                   | 43                  | 111                     | 72                     |

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Membrane (P )                                                 | 2                     | 2                    | 2                      | 2                     |
| Outplane Bending (P )                                                  | 7                     | -7                   | 7                      | -7                    |
| Outplane Membrane (Mc)                                                 | -9                    | -9                   | 9                      | 9                     |
| Outplane Bending (Mc)                                                  | -58                   | 58                   | 58                     | -58                   |
| Outplane Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Outplane Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Normal Pressure Stress                                                 | 7                     | 7                    | 7                      | 7                     |
| Outplane Stress Summary                                                | -50                   | 51                   | 85                     | -47                   |

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses parallel to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Inplane Membrane (P )                                                    | 2                     | 2                    | 2                      | 2                     |
| Inplane Bending (P )                                                     | 25                    | -25                  | 25                     | -25                   |
| Inplane Membrane (Mc)                                                    | -12                   | -12                  | 12                     | 12                    |
| Inplane Bending (Mc)                                                     | -196                  | 196                  | 196                    | -196                  |
| Inplane Membrane (ML)                                                    | 0                     | 0                    | 0                      | 0                     |
| Inplane Bending (ML)                                                     | 0                     | 0                    | 0                      | 0                     |
| Inplane Pressure Stress                                                  | 15                    | 15                   | 15                     | 15                    |
| Inplane Stress Summary                                                   | -165                  | 177                  | 252                    | -191                  |

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|----------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Shear (Vc)                                                        | 0                     | 0                    | 0                      | 0                     |
| Outplane Shear (Vl)                                                        | -1                    | -1                   | 1                      | 1                     |
| Torsional Shear (Mt)                                                       | 4                     | 4                    | 4                      | 4                     |
| Shear Stress Summary                                                       | 2                     | 2                    | 6                      | 6                     |

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|-----------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Two * Max Shear Stress                        | 165                   | 177                  | 252                    | 191                   |

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm<sup>2</sup>):**

| Type of<br>Stress Int.<br>Location | Stress Values at |     |    |    |    |    |    |    |
|------------------------------------|------------------|-----|----|----|----|----|----|----|
|                                    | Au               | Al  | Bu | Bl | Cu | Cl | Du | Dl |
| Circ. Pm (SUS)                     | 6                | 6   | 6  | 6  | 6  | 6  | 6  | 6  |
| Circ. Pl (SUS)                     | -12              | -12 | 14 | 14 | -4 | -4 | 4  | 4  |

|                 |      |      |      |      |      |      |      |      |
|-----------------|------|------|------|------|------|------|------|------|
| Circ. Q (SUS)   | -33  | 33   | 39   | -39  | -35  | 35   | 43   | -43  |
| Long. Pm (SUS)  | 3    | 3    | 3    | 3    | 3    | 3    | 3    | 3    |
| Long. Pl (SUS)  | -4   | -4   | 4    | 4    | -4   | -4   | 6    | 6    |
| Long. Q (SUS)   | -33  | 33   | 41   | -41  | -34  | 34   | 40   | -40  |
| Shear Pm (SUS)  | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear Pl (SUS)  | 1    | 1    | -1   | -1   | -1   | -1   | 1    | 1    |
| Shear Q (SUS)   | 4    | 4    | 4    | 4    | 4    | 4    | 4    | 4    |
| Pm (SUS)        | 6.0  | 6.0  | 6.0  | 6.0  | 6.0  | 6.0  | 6.0  | 6.0  |
| Pm+Pl (SUS)     | 6.2  | 6.2  | 20.1 | 20.1 | 3.6  | 3.6  | 10.6 | 10.6 |
| Pm+Pl+Q (Total) | 42.1 | 35.1 | 59.8 | 34.6 | 37.2 | 38.6 | 56.4 | 37.1 |

#### Vessel Stress Summation Comparison (N./mm<sup>2</sup>):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 6.00      | 138.00         | Passed |
| Pm+Pl (SUS)         | 20.08     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 59.76     | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Vessel Stress Summation at Reinforcing Pad Edge (N./mm<sup>2</sup>):

| Type of Stress Int. Location | Stress Values at |      |       |      |       |       |       |       |
|------------------------------|------------------|------|-------|------|-------|-------|-------|-------|
|                              | Au               | Al   | Bu    | Bl   | Cu    | Cl    | Du    | Dl    |
| Circ. Pm (SUS)               | 15               | 15   | 15    | 15   | 15    | 15    | 15    | 15    |
| Circ. Pl (SUS)               | -6               | -6   | 10    | 10   | -10   | -10   | 14    | 14    |
| Circ. Q (SUS)                | -13              | 13   | 27    | -27  | -171  | 171   | 221   | -221  |
| Long. Pm (SUS)               | 7                | 7    | 7     | 7    | 7     | 7     | 7     | 7     |
| Long. Pl (SUS)               | -7               | -7   | 11    | 11   | -7    | -7    | 11    | 11    |
| Long. Q (SUS)                | -41              | 41   | 91    | -91  | -51   | 51    | 65    | -65   |
| Shear Pm (SUS)               | 0                | 0    | 0     | 0    | 0     | 0     | 0     | 0     |
| Shear Pl (SUS)               | 1                | 1    | -1    | -1   | -1    | -1    | 1     | 1     |
| Shear Q (SUS)                | 4                | 4    | 4     | 4    | 4     | 4     | 4     | 4     |
| Pm (SUS)                     | 15.0             | 15.0 | 15.0  | 15.0 | 15.0  | 15.0  | 15.0  | 15.0  |
| Pm+Pl (SUS)                  | 9.2              | 9.2  | 25.1  | 25.1 | 5.4   | 5.4   | 29.1  | 29.1  |
| Pm+Pl+Q (Total)              | 41.7             | 42.2 | 109.2 | 73.1 | 166.1 | 176.1 | 250.1 | 192.2 |

#### Vessel Stress Summation Comparison (N./mm<sup>2</sup>):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 15.00     | 138.00         | Passed |
| Pm+Pl (SUS)         | 29.09     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 250.15    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm<sup>2</sup>):

| Type of | Stress Values at |
|---------|------------------|
|---------|------------------|



| Stress Int. Location | Au    | Al    | Bu    | Bl    | Cu    | Cl    | Du    | Dl    |
|----------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| Circ. Pm (SUS)       | 6     | 6     | 6     | 6     | 6     | 6     | 6     | 6     |
| Circ. Pl (SUS)       | -12   | -12   | 14    | 14    | -4    | -4    | 6     | 6     |
| Circ. Q (SUS)        | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Long. Pm (SUS)       | 3     | 3     | 3     | 3     | 3     | 3     | 3     | 3     |
| Long. Pl (SUS)       | -22   | -22   | 24    | 24    | -16   | -16   | 18    | 18    |
| Long. Q (SUS)        | -154  | 154   | 188   | -188  | -162  | 162   | 196   | -196  |
| Shear Pm (SUS)       | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Pl (SUS)       | 2     | 2     | -2    | -2    | -2    | -2    | 2     | 2     |
| Shear Q (SUS)        | 12    | 12    | 12    | 12    | 12    | 12    | 12    | 12    |
| Pm (SUS)             | 6.0   | 6.0   | 6.0   | 6.0   | 6.0   | 6.0   | 6.0   | 6.0   |
| Pm+Pl (SUS)          | 19.3  | 19.3  | 27.5  | 27.5  | 15.5  | 15.5  | 21.4  | 21.4  |
| Pm+Pl+Q (Total)      | 174.2 | 143.8 | 215.5 | 182.1 | 178.1 | 149.7 | 218.0 | 189.1 |

#### Nozzle Stress Summation Comparison (N./mm<sup>2</sup>):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 6.00      | 118.00         | Passed |
| Pm+Pl (SUS)         | 27.53     | 177.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 217.95    | 353.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

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**Input, Nozzle Desc: SH IN NOZ S1**

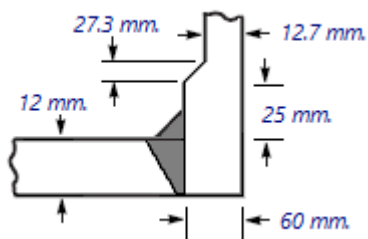
**From: 50**

|                                               |      |           |                    |
|-----------------------------------------------|------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P    | 25.511    | bars               |
| Temperature for Internal Pressure             | Temp | 127       | °C                 |
| Parent Material                               |      | SA-516 70 |                    |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D    | 490.00    | mm.                |
| Shell Finished (Minimum) Thickness            | t    | 12.0000   | mm.                |
| Shell Internal Corrosion Allowance            | c    | 3.1750    | mm.                |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |      | 6502.50   | mm.                |
| User Entered Minimum Design Metal Temperature |      | -28.89    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                                  |     |            |                    |
|--------------------------------------------------|-----|------------|--------------------|
| Material                                         |     | SA-106 B   |                    |
| Material UNS Number                              |     | K03006     |                    |
| Material Specification/Type                      |     | Smls. pipe |                    |
| Allowable Stress at Temperature                  | Sn  | 118.00     | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                      | Sna | 118.00     | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)                |     | Inside     |                    |
| Layout Angle                                     |     | 90.00      | deg                |
| Diameter                                         |     | 8.0000     | in.                |
| Size and Thickness Basis                         |     | Nominal    |                    |
| Nominal Thickness                                |     | 80         |                    |
| Flange Material                                  |     | SA-105     |                    |
| Hub Height of Integral Nozzle                    | h   | 25.0000    | mm.                |
| Height of Beveled Transition                     | L'  | 27.3087    | mm.                |
| Hub Thickness of Integral Nozzle ( tn or x+tp )  |     | 60.0000    | mm.                |
| Corrosion Allowance                              | can | 3.1750     | mm.                |
| Joint Efficiency of Shell Seam at Nozzle         | E1  | 1.00       |                    |
| Joint Efficiency of Nozzle Neck                  | En  | 1.00       |                    |
| Outside Projection                               | ho  | 150.0000   | mm.                |
| Weld leg size between Nozzle and Pad/Shell       | Wo  | 10.0000    | mm.                |
| Groove weld depth between Nozzle and Vessel Wgnv |     | 12.0000    | mm.                |
| Inside Projection                                | h   | 0.0000     | mm.                |
| Weld leg size, Inside Element to Shell           | Wi  | 0.0000     | mm.                |
| Flange Class                                     |     | 300        |                    |
| Flange Grade                                     |     | GR 1.1     |                    |

The Pressure Design option was Design Pressure + static head.



**Reinforcement CALCULATION, Description: SH IN NOZ S1**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 7.625 in.  
Actual Thickness Used in Calculation 0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $25.51 \cdot 248.175 / (138 \cdot 1.0 - 0.6 \cdot 25.51)$   
= 4.6396 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
=  $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $25.51 \cdot 100.01 / (118 \cdot 1.0 - 0.6 \cdot 25.51)$   
= 2.1908 mm.

#### Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:  
= ( Hub Thickness - Neck Thickness ) /  $\tan(30)$   
= 47.3/0.5773  
= 81.9260 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

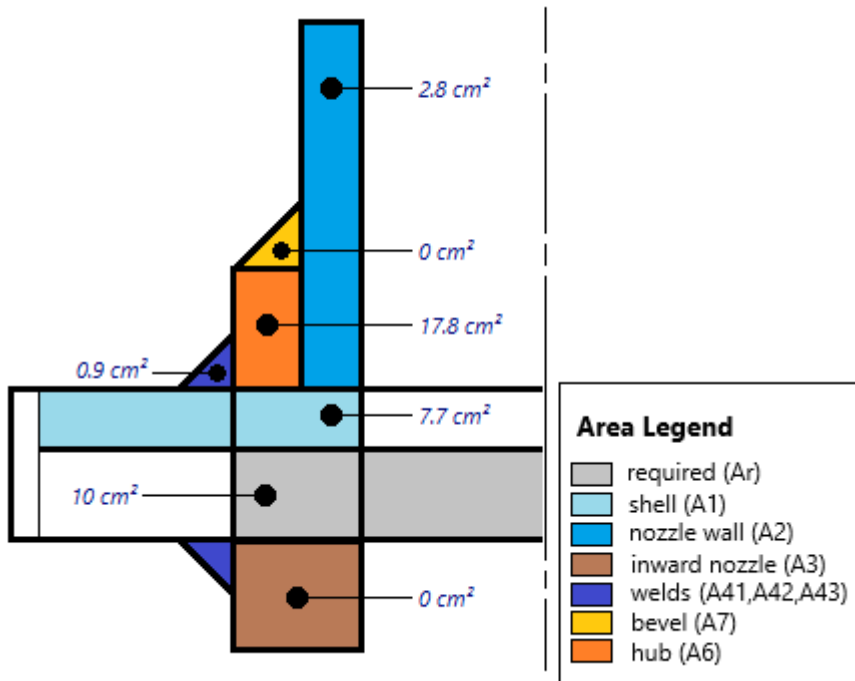
#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |      |          |     |
|-------------------------------------------------|------|----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | Dl   | 400.0500 | mm. |
| Parallel to Vessel Wall, opening length         | d    | 200.0250 | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 22.0625  | mm. |

Weld Strength Reduction Factor [fr1]:  
=  $\min(1, S_n/S_v)$   
=  $\min(1, 118.0/138.0)$   
= 0.855

Weld Strength Reduction Factor [fr2]:  
=  $\min(1, S_n/S_v)$   
=  $\min(1, 118.0/138.0)$   
= 0.855

Weld Strength Reduction Factor [fr3]:  
=  $\min(fr2, fr4)$   
=  $\min(0.855, 1.0)$   
= 0.855



#### Results of Nozzle Reinforcement Area Calculations: (cm²)

| AREA AVAILABLE, A1 to A5 |             | Design | External | Mapnc |
|--------------------------|-------------|--------|----------|-------|
| Area Required            | Ar          | 10.045 | NA       | NA    |
| Area in Shell            | A1          | 7.682  | NA       | NA    |
| Area in Nozzle Wall      | A2          | 2.767  | NA       | NA    |
| Area in Inward Nozzle    | A3          | 0.000  | NA       | NA    |
| Area in Welds            | A41+A42+A43 | 0.855  | NA       | NA    |
| Area in Element          | A5          | 0.000  | NA       | NA    |
| Area in Hub              | A6          | 17.846 | NA       | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 29.151 | NA       | NA    |

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$   
 $= (200.025 \cdot 4.6396 \cdot 1.0 + 2 \cdot 56.825 \cdot 4.6396 \cdot 1.0 \cdot (1 - 0.86))$   
 $= 10.045 \text{ cm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 200.025(1.0 \cdot 8.825 - 1.0 \cdot 4.64) - 2 \cdot 56.825$   
 $(1.0 \cdot 8.825 - 1.0 \cdot 4.6396) \cdot (1 - 0.855)$   
 $= 7.682 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:  
 $= (2 \cdot tlnp)(tn - trn)fr2$   
 $= (2 \cdot 22.06)(9.52 - 2.19)0.8551$   
 $= 2.767 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:  
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$   
 $= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$   
 $= 0.855 \text{ cm}^2$

Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lnp}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(22.1, 150.0, 25.0)) \cdot (60.0 - 12.7) \cdot 0.855 \\ &= 17.846 \text{ cm}^2 \end{aligned}$$

**UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]**

Wall Thickness for Internal/External pressures  $t_a = 5.3658 \text{ mm.}$   
Wall Thickness per UG16(b),  $tr_{16b} = 4.6750 \text{ mm.}$   
Wall Thickness, shell/head, internal pressure  $trb_1 = 7.8146 \text{ mm.}$   
Wall Thickness  $tb_1 = \max(trb_1, tr_{16b}) = 7.8146 \text{ mm.}$   
Wall Thickness  $tb_2 = \max(trb_2, tr_{16b}) = 4.6750 \text{ mm.}$   
Wall Thickness per table UG-45  $tb_3 = 10.3350 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[ tb_3, \max( tb_1, tb_2 ) ] \\ &= \min[ 10.335, \max( 7.8146, 4.675 ) ] \\ &= 7.8146 \text{ mm.} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max( t_a, t_b ) \\ &= \max( 5.3658, 7.8146 ) \\ &= 7.8146 \text{ mm.} \end{aligned}$$

Available Nozzle Neck Thickness =  $0.875 \cdot 12.700 = 11.113 \text{ mm.} \rightarrow \text{OK}$

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):**

|            |   |       |           |   |                          |        |
|------------|---|-------|-----------|---|--------------------------|--------|
| Sustained  | : | 37.3, | Allowable | : | 118.0 N./mm <sup>2</sup> | Passed |
| Expansion  | : | 0.0,  | Allowable | : | 257.7 N./mm <sup>2</sup> | Passed |
| Occasional | : | 12.8, | Allowable | : | 156.9 N./mm <sup>2</sup> | Passed |
| Shear      | : | 17.1, | Allowable | : | 82.6 N./mm <sup>2</sup>  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**

**Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)**

Govrn. thk,  $t_g = 11.113$ ,  $t_r = 2.191$ ,  $c = 3.175 \text{ mm.}$ ,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.276$ , Temp. Reduction =  $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A  $-5 \text{ }^\circ\text{C}$   
Min Metal Temp. at Required thickness (UCS 66.1)  $-104 \text{ }^\circ\text{C}$   
Min Metal Temp. w/o impact per UG-20(f)  $-29 \text{ }^\circ\text{C}$

**Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B**

Govrn. thk,  $t_g = 12.0$ ,  $t_r = 4.64$ ,  $c = 3.175 \text{ mm.}$ ,  $E^* = 1.0$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.526$ , Temp. Reduction =  $29 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B  $-23 \text{ }^\circ\text{C}$   
Min Metal Temp. at Required thickness (UCS 66.1)  $-48 \text{ }^\circ\text{C}$   
Min Metal Temp. w/o impact per UG-20(f)  $-29 \text{ }^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly :  $-48 \text{ }^\circ\text{C}$

**Flange MDMT including Temperature reduction per UCS-66.1:**

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)  $-18 \text{ }^\circ\text{C}$   
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)  $-48 \text{ }^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
Design Pressure/Ambient Rating =  $25.51/51.00 = 0.500$

Weld Size Calculations, Description: SH IN NOZ S1

Intermediate Calc. for nozzle/shell Welds  $T_{min} \quad 8.8250 \text{ mm.}$



### Results Per UW-16.1:

|             | Required Thickness    | Actual Thickness      |
|-------------|-----------------------|-----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm. |

### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
= max( 0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)  
= max( 0, (10.0446 - 7.6824 + 2 • 56.825 • 0.8551 •  
(1.0 • 8.825 - 4.6396 ) )138)  
= 9047.34 Kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
= (A2+A4-(Wi-Can/.707)<sup>2</sup>•fr2)•Sv  
= ( 20.6135 + 0.8551 - 0.0 • 0.86 ) • 138  
= 30209.69 Kgf

Weld Load [W2]:  
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv  
= ( 20.6135 + 0.0 + 0.8551 + ( 8.576 ) ) • 138  
= 42277.53 Kgf

Weld Load [W3]:  
= (A2+A3+A4+A5+(2•tn•t•fr1))•S  
= ( 20.6135 + 0.0 + 0.8551 + 0.0 + ( 8.576 ) ) • 138  
= 42277.53 Kgf

### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
= (π/2) • Dlo • Wo • 0.49 • Snw  
= ( 3.1416/2.0 ) • 313.675 • 10.0 • 0.49 • 118  
= 29050. Kgf

Shear, Nozzle Wall [Snw]:  
= (π • ( Dlr + Dlo )/4 ) • ( Thk - Can ) • 0.7 • Sn  
= (3.1416 • 128.425) • ( 60.0 - 3.175 ) • 0.7 • 118  
= 193100. Kgf

Tension, Shell Groove Weld [Tngw]:  
= (π/2) • Dlo • ( T - cas ) • 0.74 • Sng  
= ( 3.1416/2.0 ) • 313.675 • ( 12.0 - 3.175 ) • 0.74 • 138  
= 45278. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

### Strength of Failure Paths:

Failure Path [Path 1-1]:  
= ( SONW + SNW ) = ( 29050 + 193100 ) = 222150 Kgf

Failure Path [Path 2-2]:  
= ( Sonw + Tpgw + Tngw + Sinw )  
= ( 29050 + 0 + 45278 + 0 ) = 74328 Kgf

Failure Path [Path 3-3]:  
= ( Sonw + Tngw + Sinw )  
= ( 29050 + 45278 + 0 ) = 74328 Kgf

### Summary of Failure Path Calculations:

|                                                 |                   |
|-------------------------------------------------|-------------------|
| Path 1-1 = 222149 Kgf, must exceed W = 9047 Kgf | or W1 = 30209 Kgf |
| Path 2-2 = 74327 Kgf, must exceed W = 9047 Kgf  | or W2 = 42277 Kgf |
| Path 3-3 = 74327 Kgf, must exceed W = 9047 Kgf  | or W3 = 42277 Kgf |

### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 48 bars



Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 56.7794 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 218.7794 mm.

**Input Echo, WRC297 Item 1, Description: SH IN NOZ S1 :**

|                                      |            |                    |
|--------------------------------------|------------|--------------------|
| Diameter Basis for Cylindrical Shell | ID         |                    |
| Shell Internal Corrosion Allowance   | 3.1750     | mm.                |
| Shell External Corrosion Allowance   | 0.0000     | mm.                |
| Shell Diameter                       | 490.000    | mm.                |
| Shell Thickness                      | 12.0000    | mm.                |
| Shell Stress Concentration Factor    | 1.000      |                    |
| Vessel Material                      | SA-516 70  |                    |
| Vessel Cold S.I. Allowable           | Smc 138.00 | N./mm <sup>2</sup> |
| Vessel Hot S.I. Allowable            | Smh 138.00 | N./mm <sup>2</sup> |

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

|                                    |             |                    |
|------------------------------------|-------------|--------------------|
| Diameter Basis for Nozzle          | ID          |                    |
| Nozzle Corrosion Allowance         | 3.1750      | mm.                |
| Nozzle Diameter                    | 193.675     | mm.                |
| Nozzle Thickness                   | 60.0000     | mm.                |
| Nozzle Stress Concentration Factor | 1.000       |                    |
| Nozzle Material                    | SA-106 B    |                    |
| Nozzle Cold S.I. Allowable         | SNmc 118.00 | N./mm <sup>2</sup> |
| Nozzle Hot S.I. Allowable          | SNmh 118.00 | N./mm <sup>2</sup> |

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

**Note:**

*External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUSstained loads.*

|                          |    |         |       |
|--------------------------|----|---------|-------|
| Design Internal Pressure | Dp | 25.51   | bars  |
| Radial Load              | P  | -550.68 | Kgf   |
| Circumferential Shear    | Vc | 815.83  | Kgf   |
| Longitudinal Shear       | Vl | 815.83  | Kgf   |
| Circumferential Moment   | Mc | 489.31  | Kg-m. |
| Longitudinal Moment      | Ml | 652.41  | Kg-m. |
| Torsional Moment         | Mt | 815.51  | Kg-m. |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

*Warning - The ratio, Dn/Dv (.610) is > 0.5  
 check the limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (5.228) is < 10  
 check the limitations of WRC bulletin 297.*

**Stress Computations at the Edge of the Nozzle:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 313.675 + 2 \cdot 1.65 \cdot \sqrt{(252.587(12.0 - 3.175))}$$

$$= 469.479 \text{ mm.}$$

**WRC 297 Curve Access Parameters:**

|                         |       |         |     |
|-------------------------|-------|---------|-----|
| Vessel Mean Diameter    | (D) = | 505.175 | mm. |
| Nozzle Outside Diameter | (d) = | 313.675 | mm. |
| Vessel Thickness used   | (T) = | 8.825   | mm. |
| Nozzle Thickness used   | (t) = | 56.825  | mm. |
| T / t                   | =     | 0.155   |     |
| d / t                   | =     | 10.000  |     |



$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.698$$

Nr/P = 0.045  
 Mr/P = 0.071  
 M0/P = 0.022  
 N0/P = 0.041  
 MrD/Mc = 0.246  
 NrDL/Mc = 0.091  
 M0d/Mc = 0.074  
 N0DL/Mc = 0.071  
 MrD/Ml = 0.057  
 NrDL/Ml = 0.044  
 M0D/Ml = 0.018  
 N0DL/Ml = 0.038

#### Vessel Stresses (N./mm²):

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | 2                    | 2                   | 2                       | 2                      |
| Outplane Bending (P )                                            | 9                    | -9                  | 9                       | -9                     |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (ML)                                           | -10                  | -10                 | 10                      | 10                     |
| Outplane Bending (ML)                                            | -27                  | 27                  | 27                      | -27                    |
| Normal Pressure Stress                                           | 70                   | 73                  | 70                      | 73                     |
| Outplane Stress Summary                                          | 44                   | 84                  | 120                     | 49                     |

#### Vessel Stresses (N./mm²):

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | 3                    | 3                   | 3                       | 3                      |
| Inplane Bending (P )                                               | 29                   | -29                 | 29                      | -29                    |
| Inplane Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Membrane (ML)                                              | -11                  | -11                 | 11                      | 11                     |
| Inplane Bending (ML)                                               | -90                  | 90                  | 90                      | -90                    |
| Inplane Pressure Stress                                            | 35                   | 35                  | 35                      | 35                     |
| Inplane Stress Summary                                             | -33                  | 87                  | 169                     | -69                    |

#### Vessel Stresses (N./mm²):

| LONGITUDINAL PLANE<br>(Shear stress normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|----------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Shear (Vc)                                                  | 1                    | 1                   | -1                      | -1                     |
| Outplane Shear (Vl)                                                  | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Mt)                                                  | 5                    | 5                   | 5                       | 5                      |
| Shear Stress Summary                                                 | 7                    | 7                   | 4                       | 4                      |

#### Vessel Stresses (N./mm²):

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Two * Max Shear Stress                     | 79                   | 93                  | 169                     | 118                    |

#### Vessel Stresses (N./mm²):

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Membrane (P )                                                 | 2                     | 2                    | 2                      | 2                     |
| Outplane Bending (P )                                                  | 9                     | -9                   | 9                      | -9                    |

|                        |     |     |    |     |
|------------------------|-----|-----|----|-----|
| Outplane Membrane (Mc) | -13 | -13 | 13 | 13  |
| Outplane Bending (Mc)  | -86 | 86  | 86 | -86 |
| Outplane Membrane (ML) | 0   | 0   | 0  | 0   |
| Outplane Bending (ML)  | 0   | 0   | 0  | 0   |
| Normal Pressure Stress | 35  | 35  | 35 | 35  |

|                         |     |     |     |     |
|-------------------------|-----|-----|-----|-----|
| Outplane Stress Summary | -53 | 101 | 147 | -43 |
|-------------------------|-----|-----|-----|-----|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                              |              |             |               |              |
|----------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                        | Cu           | Cl          | Du            | Dl           |
| (Stresses parallel to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                         |      |     |     |      |
|-------------------------|------|-----|-----|------|
| Inplane Membrane (P )   | 3    | 3   | 3   | 3    |
| Inplane Bending (P )    | 29   | -29 | 29  | -29  |
| Inplane Membrane (Mc)   | -17  | -17 | 17  | 17   |
| Inplane Bending (Mc)    | -290 | 290 | 290 | -290 |
| Inplane Membrane (ML)   | 0    | 0   | 0   | 0    |
| Inplane Bending (ML)    | 0    | 0   | 0   | 0    |
| Inplane Pressure Stress | 70   | 73  | 70  | 73   |

|                        |      |     |     |      |
|------------------------|------|-----|-----|------|
| Inplane Stress Summary | -205 | 319 | 411 | -225 |
|------------------------|------|-----|-----|------|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                                |              |             |               |              |
|------------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                          | Cu           | Cl          | Du            | Dl           |
| (Shear stress normal to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                      |    |    |   |   |
|----------------------|----|----|---|---|
| Outplane Shear (Vc)  | 0  | 0  | 0 | 0 |
| Outplane Shear (Vl)  | -1 | -1 | 1 | 1 |
| Torsional Shear (Mt) | 5  | 5  | 5 | 5 |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 4 | 4 | 7 | 7 |
|----------------------|---|---|---|---|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                       |              |             |               |              |
|-----------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE | Cu           | Cl          | Du            | Dl           |
| (Stress Intensities)  | Left Outside | Left Inside | Right Outside | Right Inside |

|                        |     |     |     |     |
|------------------------|-----|-----|-----|-----|
| Two * Max Shear Stress | 205 | 319 | 411 | 226 |
|------------------------|-----|-----|-----|-----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                                  |             |            |                |               |
|----------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE               | Au          | Al         | Bu             | Bl            |
| (Stresses in the hoop direction) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                      |     |     |    |    |
|----------------------|-----|-----|----|----|
| Hoop Membrane (P )   | 2   | 2   | 2  | 2  |
| Hoop Bending (P )    | 0   | 0   | 0  | 0  |
| Hoop Membrane (Mc)   | 0   | 0   | 0  | 0  |
| Hoop Bending (Mc)    | 0   | 0   | 0  | 0  |
| Hoop Membrane (ML)   | -10 | -10 | 10 | 10 |
| Hoop Bending (ML)    | 0   | 0   | 0  | 0  |
| Hoop Pressure Stress | 4   | 7   | 4  | 7  |

|                     |    |   |    |    |
|---------------------|----|---|----|----|
| Hoop Stress Summary | -3 | 0 | 17 | 20 |
|---------------------|----|---|----|----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                                         |             |            |                |               |
|-----------------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE                      | Au          | Al         | Bu             | Bl            |
| (Stresses Normal to pipe cross-section) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                       |    |    |   |    |
|-----------------------|----|----|---|----|
| Axial Membrane (P )   | 0  | 0  | 0 | 0  |
| Axial Bending (P )    | 0  | 0  | 0 | 0  |
| Axial Membrane (Mc)   | 0  | 0  | 0 | 0  |
| Axial Bending (Mc)    | 0  | 0  | 0 | 0  |
| Axial Membrane (ML)   | -2 | -2 | 2 | 2  |
| Axial Bending (ML)    | -1 | 1  | 1 | -1 |
| Axial Pressure Stress | 2  | 2  | 2 | 2  |

|                      |    |   |   |   |
|----------------------|----|---|---|---|
| Axial Stress Summary | -1 | 0 | 6 | 2 |
|----------------------|----|---|---|---|

**Nozzle Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Shear stress) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                    | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                    | 0             | 0            | 0             | 0            |
| Shear due to Torsion                 | 0             | 0            | 0             | 0            |
| Shear Stress Summary                 | 1             | 1            | 0             | 0            |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                     | 3             | 2            | 17            | 20           |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses in the<br>hoop direction) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Hoop Membrane (P )                                           | 2                     | 2                    | 2                      | 2                     |
| Hoop Bending (P )                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (Mc)                                           | -13                   | -13                  | 13                     | 13                    |
| Hoop Bending (Mc)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (ML)                                           | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (ML)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Pressure Stress                                         | 4                     | 7                    | 4                      | 7                     |
| Hoop Stress Summary                                          | -6                    | -3                   | 20                     | 23                    |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Axial Membrane (P )                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (P )                                                  | 0                     | 0                    | 0                      | 0                     |
| Axial Membrane (Mc)                                                 | -1                    | -1                   | 1                      | 1                     |
| Axial Bending (Mc)                                                  | -5                    | 5                    | 5                      | -5                    |
| Axial Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Axial Pressure Stress                                               | 2                     | 2                    | 2                      | 2                     |
| Axial Stress Summary                                                | -4                    | 5                    | 10                     | -2                    |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                       | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                       | 0             | 0            | 0             | 0            |
| Shear due to Torsion                    | 0             | 0            | 0             | 0            |
| Shear Stress Summary                    | 0             | 0            | 1             | 1            |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                        | 7             | 9            | 21            | 26           |

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm<sup>2</sup>):**

| Type of<br>Stress Int.<br>Location | Stress Values at |    |    |    |    |    |    |    |
|------------------------------------|------------------|----|----|----|----|----|----|----|
|                                    | Au               | Al | Bu | Bl | Cu | Cl | Du | Dl |

|                 |      |      |       |       |       |       |       |       |
|-----------------|------|------|-------|-------|-------|-------|-------|-------|
| Circ. Pm (SUS)  | 70   | 73   | 70    | 73    | 70    | 73    | 70    | 73    |
| Circ. Pl (SUS)  | -8   | -8   | 12    | 12    | -14   | -14   | 20    | 20    |
| Circ. Q (SUS)   | -18  | 18   | 36    | -36   | -261  | 261   | 319   | -319  |
| Long. Pm (SUS)  | 35   | 35   | 35    | 35    | 35    | 35    | 35    | 35    |
| Long. Pl (SUS)  | -8   | -8   | 14    | 14    | -11   | -11   | 15    | 15    |
| Long. Q (SUS)   | -61  | 61   | 119   | -119  | -77   | 77    | 95    | -95   |
| Shear Pm (SUS)  | 0    | 0    | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Pl (SUS)  | 1    | 1    | -1    | -1    | -1    | -1    | 1     | 1     |
| Shear Q (SUS)   | 5    | 5    | 5     | 5     | 5     | 5     | 5     | 5     |
| Pm (SUS)        | 70.0 | 73.0 | 70.0  | 73.0  | 70.0  | 73.0  | 70.0  | 73.0  |
| Pm+Pl (SUS)     | 62.0 | 65.0 | 82.0  | 85.0  | 56.0  | 59.0  | 90.0  | 93.0  |
| Pm+Pl+Q (Total) | 78.9 | 92.0 | 168.3 | 119.3 | 205.1 | 320.1 | 409.1 | 226.2 |

#### Vessel Stress Summation Comparison (N./mm<sup>2</sup>):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 73.00     | 138.00         | Passed |
| Pm+Pl (SUS)         | 93.02     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 409.14    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm<sup>2</sup>):

| Type of Stress Int. Location | Stress Values at |     |      |      |     |      |      |      |
|------------------------------|------------------|-----|------|------|-----|------|------|------|
|                              | Au               | Al  | Bu   | Bl   | Cu  | Cl   | Du   | Dl   |
| Circ. Pm (SUS)               | 4                | 7   | 4    | 7    | 4   | 7    | 4    | 7    |
| Circ. Pl (SUS)               | -8               | -8  | 12   | 12   | -11 | -11  | 15   | 15   |
| Circ. Q (SUS)                | 0                | 0   | 0    | 0    | 0   | 0    | 0    | 0    |
| Long. Pm (SUS)               | 2                | 2   | 2    | 2    | 2   | 2    | 2    | 2    |
| Long. Pl (SUS)               | -2               | -2  | 2    | 2    | -1  | -1   | 1    | 1    |
| Long. Q (SUS)                | -1               | 1   | 1    | -1   | -5  | 5    | 5    | -5   |
| Shear Pm (SUS)               | 0                | 0   | 0    | 0    | 0   | 0    | 0    | 0    |
| Shear Pl (SUS)               | 0                | 0   | 0    | 0    | 0   | 0    | 0    | 0    |
| Shear Q (SUS)                | 0                | 0   | 0    | 0    | 0   | 0    | 0    | 0    |
| Pm (SUS)                     | 4.0              | 7.0 | 4.0  | 7.0  | 4.0 | 7.0  | 4.0  | 7.0  |
| Pm+Pl (SUS)                  | 4.0              | 1.0 | 16.0 | 19.0 | 8.0 | 5.0  | 19.0 | 22.0 |
| Pm+Pl+Q (Total)              | 4.0              | 2.0 | 16.0 | 19.0 | 7.0 | 10.0 | 19.0 | 24.0 |

#### Nozzle Stress Summation Comparison (N./mm<sup>2</sup>):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 7.00      | 118.00         | Passed |
| Pm+Pl (SUS)         | 22.00     | 177.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 24.00     | 353.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.



**Input, Nozzle Desc: SH OUT NOZ S2**

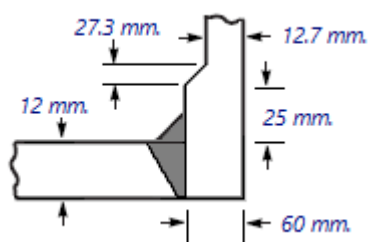
**From: 50**

|                                               |      |           |                    |
|-----------------------------------------------|------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P    | 25.559    | bars               |
| Temperature for Internal Pressure             | Temp | 127       | °C                 |
| Parent Material                               |      | SA-516 70 |                    |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D    | 490.00    | mm.                |
| Shell Finished (Minimum) Thickness            | t    | 12.0000   | mm.                |
| Shell Internal Corrosion Allowance            | c    | 3.1750    | mm.                |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |      | 906.50    | mm.                |
| User Entered Minimum Design Metal Temperature |      | -28.89    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                                  |     |            |                    |
|--------------------------------------------------|-----|------------|--------------------|
| Material                                         |     | SA-106 B   |                    |
| Material UNS Number                              |     | K03006     |                    |
| Material Specification/Type                      |     | Smls. pipe |                    |
| Allowable Stress at Temperature                  | Sn  | 118.00     | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                      | Sna | 118.00     | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)                |     | Inside     |                    |
| Layout Angle                                     |     | -90.00     | deg                |
| Diameter                                         |     | 8.0000     | in.                |
| Size and Thickness Basis                         |     | Nominal    |                    |
| Nominal Thickness                                |     | 80         |                    |
| Flange Material                                  |     | SA-105     |                    |
| Hub Height of Integral Nozzle                    | h   | 25.0000    | mm.                |
| Height of Beveled Transition                     | L'  | 27.3087    | mm.                |
| Hub Thickness of Integral Nozzle ( tn or x+tp )  |     | 60.0000    | mm.                |
| Corrosion Allowance                              | can | 3.1750     | mm.                |
| Joint Efficiency of Shell Seam at Nozzle         | E1  | 1.00       |                    |
| Joint Efficiency of Nozzle Neck                  | En  | 1.00       |                    |
| Outside Projection                               | ho  | 150.0000   | mm.                |
| Weld leg size between Nozzle and Pad/Shell       | Wo  | 10.0000    | mm.                |
| Groove weld depth between Nozzle and Vessel Wgnv |     | 12.0000    | mm.                |
| Inside Projection                                | h   | 0.0000     | mm.                |
| Weld leg size, Inside Element to Shell           | Wi  | 0.0000     | mm.                |
| Flange Class                                     |     | 300        |                    |
| Flange Grade                                     |     | GR 1.1     |                    |

The Pressure Design option was Design Pressure + static head.



**Reinforcement CALCULATION, Description: SH OUT NOZ S2**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 7.625 in.  
 Actual Thickness Used in Calculation 0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 25.56 \cdot 248.175 / (138 \cdot 1.0 - 0.6 \cdot 25.56)$   
 $= 4.6485$  mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 25.56 \cdot 100.01 / (118 \cdot 1.0 - 0.6 \cdot 25.56)$   
 $= 2.1950$  mm.

#### Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:  
 $= ( \text{Hub Thickness} - \text{Neck Thickness} ) / \tan(30)$   
 $= 47.3 / 0.5773$   
 $= 81.9260$  mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

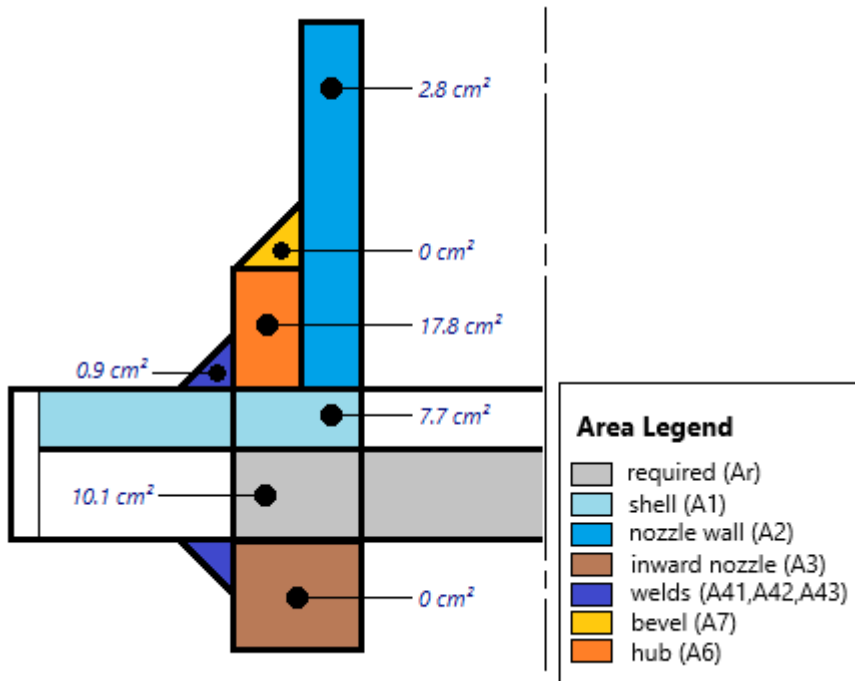
#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |      |          |     |
|-------------------------------------------------|------|----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 400.0500 | mm. |
| Parallel to Vessel Wall, opening length         | d    | 200.0250 | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 22.0625  | mm. |

Weld Strength Reduction Factor [fr1]:  
 $= \min( 1, S_n / S_v )$   
 $= \min( 1, 118.0 / 138.0 )$   
 $= 0.855$

Weld Strength Reduction Factor [fr2]:  
 $= \min( 1, S_n / S_v )$   
 $= \min( 1, 118.0 / 138.0 )$   
 $= 0.855$

Weld Strength Reduction Factor [fr3]:  
 $= \min( fr2, fr4 )$   
 $= \min( 0.855, 1.0 )$   
 $= 0.855$



#### Results of Nozzle Reinforcement Area Calculations: (cm²)

| AREA AVAILABLE, A1 to A5 |             | Design | External | Mapnc |
|--------------------------|-------------|--------|----------|-------|
| Area Required            | Ar          | 10.064 | NA       | NA    |
| Area in Shell            | A1          | 7.666  | NA       | NA    |
| Area in Nozzle Wall      | A2          | 2.766  | NA       | NA    |
| Area in Inward Nozzle    | A3          | 0.000  | NA       | NA    |
| Area in Welds            | A41+A42+A43 | 0.855  | NA       | NA    |
| Area in Element          | A5          | 0.000  | NA       | NA    |
| Area in Hub              | A6          | 17.846 | NA       | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 29.133 | NA       | NA    |

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:  

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (200.025 \cdot 4.6485 \cdot 1.0 + 2 \cdot 56.825 \cdot 4.6485 \cdot 1.0 \cdot (1 - 0.86))$$

$$= 10.064 \text{ cm}^2$$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 200.025(1.0 \cdot 8.825 - 1.0 \cdot 4.648) - 2 \cdot 56.825$$

$$(1.0 \cdot 8.825 - 1.0 \cdot 4.6485) \cdot (1 - 0.855)$$

$$= 7.666 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:  

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 22.06)(9.52 - 2.19)0.8551$$

$$= 2.766 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:  

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$$

$$= 0.855 \text{ cm}^2$$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(22.1, 150.0, 25.0)) \cdot (60.0 - 12.7) \cdot 0.855$$

$$= 17.846 \text{ cm}^2$$

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

|                                                |                                                   |
|------------------------------------------------|---------------------------------------------------|
| Wall Thickness for Internal/External pressures | $t_a = 5.3700 \text{ mm.}$                        |
| Wall Thickness per UG16(b),                    | $tr_{16b} = 4.6750 \text{ mm.}$                   |
| Wall Thickness, shell/head, internal pressure  | $trb1 = 7.8235 \text{ mm.}$                       |
| Wall Thickness                                 | $tb1 = \max(trb1, tr_{16b}) = 7.8235 \text{ mm.}$ |
| Wall Thickness                                 | $tb2 = \max(trb2, tr_{16b}) = 4.6750 \text{ mm.}$ |
| Wall Thickness per table UG-45                 | $tb3 = 10.3350 \text{ mm.}$                       |

Determine Nozzle Thickness candidate [tb]:

$$= \min[ tb3, \max( tb1, tb2 ) ]$$

$$= \min[ 10.335, \max( 7.8235, 4.675 ) ]$$

$$= 7.8235 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max( t_a, t_b )$$

$$= \max( 5.37, 7.8235 )$$

$$= 7.8235 \text{ mm.}$$

Available Nozzle Neck Thickness =  $0.875 \cdot 12.700 = 11.113 \text{ mm.} \rightarrow \text{OK}$

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):**

|            |         |           |                            |        |
|------------|---------|-----------|----------------------------|--------|
| Sustained  | : 37.3, | Allowable | : 118.0 N./mm <sup>2</sup> | Passed |
| Expansion  | : 0.0,  | Allowable | : 257.6 N./mm <sup>2</sup> | Passed |
| Occasional | : 12.8, | Allowable | : 156.9 N./mm <sup>2</sup> | Passed |
| Shear      | : 17.1, | Allowable | : 82.6 N./mm <sup>2</sup>  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, min( Curve:B, Curve:A)

Govrn. thk,  $t_g = 11.113$ ,  $t_r = 2.195$ ,  $c = 3.175 \text{ mm.}$ ,  $E^* = 1.0$   
 Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.277$ , Temp. Reduction =  $78 \text{ }^\circ\text{C}$

|                                                  |                       |
|--------------------------------------------------|-----------------------|
| Min Metal Temp. w/o impact per UCS-66, Curve A   | -5 $^\circ\text{C}$   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 $^\circ\text{C}$ |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 $^\circ\text{C}$  |

##### Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk,  $t_g = 12.0$ ,  $t_r = 4.648$ ,  $c = 3.175 \text{ mm.}$ ,  $E^* = 1.0$   
 Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.527$ , Temp. Reduction =  $29 \text{ }^\circ\text{C}$

|                                                  |                      |
|--------------------------------------------------|----------------------|
| Min Metal Temp. w/o impact per UCS-66, Curve B   | -23 $^\circ\text{C}$ |
| Min Metal Temp. at Required thickness (UCS 66.1) | -48 $^\circ\text{C}$ |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 $^\circ\text{C}$ |

Gov. MDMT of the nozzle to shell joint welded assembly : -48  $^\circ\text{C}$

#### Flange MDMT including Temperature reduction per UCS-66.1:

|                                                        |                      |
|--------------------------------------------------------|----------------------|
| Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) | -18 $^\circ\text{C}$ |
| Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)   | -48 $^\circ\text{C}$ |

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
 Design Pressure/Ambient Rating =  $25.56/51.00 = 0.501$

#### Weld Size Calculations, Description: SH OUT NOZ S2

Intermediate Calc. for nozzle/shell Welds  $T_{min} = 8.8250 \text{ mm.}$



### Results Per UW-16.1:

|             | Required Thickness    | Actual Thickness      |
|-------------|-----------------------|-----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm. |

### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
= max( 0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)  
= max( 0, (10.0637 - 7.6662 + 2 • 56.825 • 0.8551 •  
(1.0 • 8.825 - 4.6485 ) )138)  
= 9084.99 Kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
= (A2+A4-(Wi-Can/.707)<sup>2</sup>•fr2)•Sv  
= ( 20.6119 + 0.8551 - 0.0 • 0.86 ) • 138  
= 30207.47 Kgf

Weld Load [W2]:  
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv  
= ( 20.6119 + 0.0 + 0.8551 + ( 8.576 ) ) • 138  
= 42275.31 Kgf

Weld Load [W3]:  
= (A2+A3+A4+A5+(2•tn•t•fr1))•S  
= ( 20.6119 + 0.0 + 0.8551 + 0.0 + ( 8.576 ) ) • 138  
= 42275.31 Kgf

### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
= (π/2) • Dlo • Wo • 0.49 • Snw  
= ( 3.1416/2.0 ) • 313.675 • 10.0 • 0.49 • 118  
= 29050. Kgf

Shear, Nozzle Wall [Snw]:  
= (π • ( Dlr + Dlo )/4 ) • ( Thk - Can ) • 0.7 • Sn  
= (3.1416 • 128.425) • ( 60.0 - 3.175 ) • 0.7 • 118  
= 193100. Kgf

Tension, Shell Groove Weld [Tngw]:  
= (π/2) • Dlo • ( T - cas ) • 0.74 • Sng  
= ( 3.1416/2.0 ) • 313.675 • ( 12.0 - 3.175 ) • 0.74 • 138  
= 45278. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

### Strength of Failure Paths:

Failure Path [Path 1-1]:  
= ( SONW + SNW ) = ( 29050 + 193100 ) = 222150 Kgf

Failure Path [Path 2-2]:  
= ( Sonw + Tpgw + Tngw + Sinw )  
= ( 29050 + 0 + 45278 + 0 ) = 74328 Kgf

Failure Path [Path 3-3]:  
= ( Sonw + Tngw + Sinw )  
= ( 29050 + 45278 + 0 ) = 74328 Kgf

### Summary of Failure Path Calculations:

|                                                 |                   |
|-------------------------------------------------|-------------------|
| Path 1-1 = 222149 Kgf, must exceed W = 9084 Kgf | or W1 = 30207 Kgf |
| Path 2-2 = 74327 Kgf, must exceed W = 9084 Kgf  | or W2 = 42275 Kgf |
| Path 3-3 = 74327 Kgf, must exceed W = 9084 Kgf  | or W3 = 42275 Kgf |

### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 48 bars



Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 56.7794 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 218.7794 mm.

**Input Echo, WRC297 Item 1, Description: SH OUT NOZ S2 :**

|                                      |            |                    |
|--------------------------------------|------------|--------------------|
| Diameter Basis for Cylindrical Shell | ID         |                    |
| Shell Internal Corrosion Allowance   | 3.1750     | mm.                |
| Shell External Corrosion Allowance   | 0.0000     | mm.                |
| Shell Diameter                       | 490.000    | mm.                |
| Shell Thickness                      | 12.0000    | mm.                |
| Shell Stress Concentration Factor    | 1.000      |                    |
| Vessel Material                      | SA-516 70  |                    |
| Vessel Cold S.I. Allowable           | Smc 138.00 | N./mm <sup>2</sup> |
| Vessel Hot S.I. Allowable            | Smh 138.00 | N./mm <sup>2</sup> |

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

|                                    |             |                    |
|------------------------------------|-------------|--------------------|
| Diameter Basis for Nozzle          | ID          |                    |
| Nozzle Corrosion Allowance         | 3.1750      | mm.                |
| Nozzle Diameter                    | 193.675     | mm.                |
| Nozzle Thickness                   | 60.0000     | mm.                |
| Nozzle Stress Concentration Factor | 1.000       |                    |
| Nozzle Material                    | SA-106 B    |                    |
| Nozzle Cold S.I. Allowable         | SNmc 118.00 | N./mm <sup>2</sup> |
| Nozzle Hot S.I. Allowable          | SNmh 118.00 | N./mm <sup>2</sup> |

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

**Note:**

*External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUSstained loads.*

|                          |    |         |       |
|--------------------------|----|---------|-------|
| Design Internal Pressure | Dp | 25.56   | bars  |
| Radial Load              | P  | -550.68 | Kgf   |
| Circumferential Shear    | Vc | 815.83  | Kgf   |
| Longitudinal Shear       | Vl | 815.83  | Kgf   |
| Circumferential Moment   | Mc | 489.31  | Kg-m. |
| Longitudinal Moment      | Ml | 652.41  | Kg-m. |
| Torsional Moment         | Mt | 815.51  | Kg-m. |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

*Warning - The ratio, Dn/Dv (.610) is > 0.5  
 check the Limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (5.228) is < 10  
 check the Limitations of WRC bulletin 297.*

**Stress Computations at the Edge of the Nozzle:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 313.675 + 2 \cdot 1.65 \cdot \sqrt{(252.587(12.0 - 3.175))}$$

$$= 469.479 \text{ mm.}$$

**WRC 297 Curve Access Parameters:**

|                         |       |         |     |
|-------------------------|-------|---------|-----|
| Vessel Mean Diameter    | (D) = | 505.175 | mm. |
| Nozzle Outside Diameter | (d) = | 313.675 | mm. |
| Vessel Thickness used   | (T) = | 8.825   | mm. |
| Nozzle Thickness used   | (t) = | 56.825  | mm. |
| T / t                   | =     | 0.155   |     |
| d / t                   | =     | 10.000  |     |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.698$$

Nr/P = 0.045  
 Mr/P = 0.071  
 M0/P = 0.022  
 N0/P = 0.041  
 MrD/Mc = 0.246  
 NrDL/Mc = 0.091  
 M0d/Mc = 0.074  
 N0DL/Mc = 0.071  
 MrD/Ml = 0.057  
 NrDL/Ml = 0.044  
 M0D/Ml = 0.018  
 N0DL/Ml = 0.038

#### Vessel Stresses (N./mm²):

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | 2                    | 2                   | 2                       | 2                      |
| Outplane Bending (P )                                            | 9                    | -9                  | 9                       | -9                     |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (ML)                                           | -10                  | -10                 | 10                      | 10                     |
| Outplane Bending (ML)                                            | -27                  | 27                  | 27                      | -27                    |
| Normal Pressure Stress                                           | 70                   | 73                  | 70                      | 73                     |
| Outplane Stress Summary                                          | 44                   | 84                  | 120                     | 49                     |

#### Vessel Stresses (N./mm²):

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | 3                    | 3                   | 3                       | 3                      |
| Inplane Bending (P )                                               | 29                   | -29                 | 29                      | -29                    |
| Inplane Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Membrane (ML)                                              | -11                  | -11                 | 11                      | 11                     |
| Inplane Bending (ML)                                               | -90                  | 90                  | 90                      | -90                    |
| Inplane Pressure Stress                                            | 35                   | 35                  | 35                      | 35                     |
| Inplane Stress Summary                                             | -33                  | 87                  | 169                     | -69                    |

#### Vessel Stresses (N./mm²):

| LONGITUDINAL PLANE<br>(Shear stress normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|----------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Shear (Vc)                                                  | 1                    | 1                   | -1                      | -1                     |
| Outplane Shear (Vl)                                                  | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Mt)                                                  | 5                    | 5                   | 5                       | 5                      |
| Shear Stress Summary                                                 | 7                    | 7                   | 4                       | 4                      |

#### Vessel Stresses (N./mm²):

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Two * Max Shear Stress                     | 80                   | 93                  | 169                     | 118                    |

#### Vessel Stresses (N./mm²):

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Membrane (P )                                                 | 2                     | 2                    | 2                      | 2                     |
| Outplane Bending (P )                                                  | 9                     | -9                   | 9                      | -9                    |

|                        |     |     |    |     |
|------------------------|-----|-----|----|-----|
| Outplane Membrane (Mc) | -13 | -13 | 13 | 13  |
| Outplane Bending (Mc)  | -86 | 86  | 86 | -86 |
| Outplane Membrane (ML) | 0   | 0   | 0  | 0   |
| Outplane Bending (ML)  | 0   | 0   | 0  | 0   |
| Normal Pressure Stress | 35  | 35  | 35 | 35  |

|                         |     |     |     |     |
|-------------------------|-----|-----|-----|-----|
| Outplane Stress Summary | -53 | 101 | 147 | -43 |
|-------------------------|-----|-----|-----|-----|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                              |              |             |               |              |
|----------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                        | Cu           | Cl          | Du            | Dl           |
| (Stresses parallel to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                         |      |     |     |      |
|-------------------------|------|-----|-----|------|
| Inplane Membrane (P )   | 3    | 3   | 3   | 3    |
| Inplane Bending (P )    | 29   | -29 | 29  | -29  |
| Inplane Membrane (Mc)   | -17  | -17 | 17  | 17   |
| Inplane Bending (Mc)    | -290 | 290 | 290 | -290 |
| Inplane Membrane (ML)   | 0    | 0   | 0   | 0    |
| Inplane Bending (ML)    | 0    | 0   | 0   | 0    |
| Inplane Pressure Stress | 70   | 73  | 70  | 73   |

|                        |      |     |     |      |
|------------------------|------|-----|-----|------|
| Inplane Stress Summary | -205 | 319 | 411 | -225 |
|------------------------|------|-----|-----|------|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                                |              |             |               |              |
|------------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                          | Cu           | Cl          | Du            | Dl           |
| (Shear stress normal to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                      |    |    |   |   |
|----------------------|----|----|---|---|
| Outplane Shear (Vc)  | 0  | 0  | 0 | 0 |
| Outplane Shear (Vl)  | -1 | -1 | 1 | 1 |
| Torsional Shear (Mt) | 5  | 5  | 5 | 5 |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 4 | 4 | 7 | 7 |
|----------------------|---|---|---|---|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                       |              |             |               |              |
|-----------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE | Cu           | Cl          | Du            | Dl           |
| (Stress Intensities)  | Left Outside | Left Inside | Right Outside | Right Inside |

|                        |     |     |     |     |
|------------------------|-----|-----|-----|-----|
| Two * Max Shear Stress | 205 | 319 | 411 | 225 |
|------------------------|-----|-----|-----|-----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                                  |             |            |                |               |
|----------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE               | Au          | Al         | Bu             | Bl            |
| (Stresses in the hoop direction) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                      |     |     |    |    |
|----------------------|-----|-----|----|----|
| Hoop Membrane (P )   | 2   | 2   | 2  | 2  |
| Hoop Bending (P )    | 0   | 0   | 0  | 0  |
| Hoop Membrane (Mc)   | 0   | 0   | 0  | 0  |
| Hoop Bending (Mc)    | 0   | 0   | 0  | 0  |
| Hoop Membrane (ML)   | -10 | -10 | 10 | 10 |
| Hoop Bending (ML)    | 0   | 0   | 0  | 0  |
| Hoop Pressure Stress | 4   | 7   | 4  | 7  |

|                     |    |   |    |    |
|---------------------|----|---|----|----|
| Hoop Stress Summary | -3 | 0 | 17 | 20 |
|---------------------|----|---|----|----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                                         |             |            |                |               |
|-----------------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE                      | Au          | Al         | Bu             | Bl            |
| (Stresses Normal to pipe cross-section) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                       |    |    |   |    |
|-----------------------|----|----|---|----|
| Axial Membrane (P )   | 0  | 0  | 0 | 0  |
| Axial Bending (P )    | 0  | 0  | 0 | 0  |
| Axial Membrane (Mc)   | 0  | 0  | 0 | 0  |
| Axial Bending (Mc)    | 0  | 0  | 0 | 0  |
| Axial Membrane (ML)   | -2 | -2 | 2 | 2  |
| Axial Bending (ML)    | -1 | 1  | 1 | -1 |
| Axial Pressure Stress | 2  | 2  | 2 | 2  |

|                      |    |   |   |   |
|----------------------|----|---|---|---|
| Axial Stress Summary | -1 | 0 | 6 | 2 |
|----------------------|----|---|---|---|

**Nozzle Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Shear stress) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                    | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                    | 0             | 0            | 0             | 0            |
| Shear due to Torsion                 | 0             | 0            | 0             | 0            |
| Shear Stress Summary                 | 1             | 1            | 0             | 0            |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                     | 3             | 2            | 17            | 20           |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses in the<br>hoop direction) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Hoop Membrane (P )                                           | 2                     | 2                    | 2                      | 2                     |
| Hoop Bending (P )                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (Mc)                                           | -13                   | -13                  | 13                     | 13                    |
| Hoop Bending (Mc)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (ML)                                           | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (ML)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Pressure Stress                                         | 4                     | 7                    | 4                      | 7                     |
| Hoop Stress Summary                                          | -6                    | -3                   | 20                     | 23                    |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Axial Membrane (P )                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (P )                                                  | 0                     | 0                    | 0                      | 0                     |
| Axial Membrane (Mc)                                                 | -1                    | -1                   | 1                      | 1                     |
| Axial Bending (Mc)                                                  | -5                    | 5                    | 5                      | -5                    |
| Axial Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Axial Pressure Stress                                               | 2                     | 2                    | 2                      | 2                     |
| Axial Stress Summary                                                | -4                    | 5                    | 10                     | -2                    |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------|---------------|--------------|---------------|--------------|
| Shear due to (Vc)                       | 0             | 0            | 0             | 0            |
| Shear due to (Vl)                       | 0             | 0            | 0             | 0            |
| Shear due to Torsion                    | 0             | 0            | 0             | 0            |
| Shear Stress Summary                    | 0             | 0            | 1             | 1            |

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------------|---------------|--------------|---------------|--------------|
| Two * Max Shear Stress                        | 7             | 9            | 21            | 26           |

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**

**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm<sup>2</sup>):**

| Type of<br>Stress Int.<br>Location | Stress Values at |    |    |    |    |    |    |    |
|------------------------------------|------------------|----|----|----|----|----|----|----|
|                                    | Au               | Al | Bu | Bl | Cu | Cl | Du | Dl |

|                 |      |      |       |       |       |       |       |       |
|-----------------|------|------|-------|-------|-------|-------|-------|-------|
| Circ. Pm (SUS)  | 70   | 73   | 70    | 73    | 70    | 73    | 70    | 73    |
| Circ. Pl (SUS)  | -8   | -8   | 12    | 12    | -14   | -14   | 20    | 20    |
| Circ. Q (SUS)   | -18  | 18   | 36    | -36   | -261  | 261   | 319   | -319  |
| Long. Pm (SUS)  | 35   | 35   | 35    | 35    | 35    | 35    | 35    | 35    |
| Long. Pl (SUS)  | -8   | -8   | 14    | 14    | -11   | -11   | 15    | 15    |
| Long. Q (SUS)   | -61  | 61   | 119   | -119  | -77   | 77    | 95    | -95   |
| Shear Pm (SUS)  | 0    | 0    | 0     | 0     | 0     | 0     | 0     | 0     |
| Shear Pl (SUS)  | 1    | 1    | -1    | -1    | -1    | -1    | 1     | 1     |
| Shear Q (SUS)   | 5    | 5    | 5     | 5     | 5     | 5     | 5     | 5     |
| Pm (SUS)        | 70.0 | 73.0 | 70.0  | 73.0  | 70.0  | 73.0  | 70.0  | 73.0  |
| Pm+Pl (SUS)     | 62.0 | 65.0 | 82.0  | 85.0  | 56.0  | 59.0  | 90.0  | 93.0  |
| Pm+Pl+Q (Total) | 78.9 | 92.0 | 168.3 | 119.3 | 205.1 | 320.1 | 409.1 | 226.2 |

#### Vessel Stress Summation Comparison (N./mm<sup>2</sup>):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 73.00     | 138.00         | Passed |
| Pm+Pl (SUS)         | 93.02     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 409.14    | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm<sup>2</sup>):

| Type of Stress Int. Location | Stress Values at |     |      |      |     |      |      |      |
|------------------------------|------------------|-----|------|------|-----|------|------|------|
|                              | Au               | Al  | Bu   | Bl   | Cu  | Cl   | Du   | Dl   |
| Circ. Pm (SUS)               | 4                | 7   | 4    | 7    | 4   | 7    | 4    | 7    |
| Circ. Pl (SUS)               | -8               | -8  | 12   | 12   | -11 | -11  | 15   | 15   |
| Circ. Q (SUS)                | 0                | 0   | 0    | 0    | 0   | 0    | 0    | 0    |
| Long. Pm (SUS)               | 2                | 2   | 2    | 2    | 2   | 2    | 2    | 2    |
| Long. Pl (SUS)               | -2               | -2  | 2    | 2    | -1  | -1   | 1    | 1    |
| Long. Q (SUS)                | -1               | 1   | 1    | -1   | -5  | 5    | 5    | -5   |
| Shear Pm (SUS)               | 0                | 0   | 0    | 0    | 0   | 0    | 0    | 0    |
| Shear Pl (SUS)               | 0                | 0   | 0    | 0    | 0   | 0    | 0    | 0    |
| Shear Q (SUS)                | 0                | 0   | 0    | 0    | 0   | 0    | 0    | 0    |
| Pm (SUS)                     | 4.0              | 7.0 | 4.0  | 7.0  | 4.0 | 7.0  | 4.0  | 7.0  |
| Pm+Pl (SUS)                  | 4.0              | 1.0 | 16.0 | 19.0 | 8.0 | 5.0  | 19.0 | 22.0 |
| Pm+Pl+Q (Total)              | 4.0              | 2.0 | 16.0 | 19.0 | 7.0 | 10.0 | 19.0 | 24.0 |

#### Nozzle Stress Summation Comparison (N./mm<sup>2</sup>):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 7.00      | 118.00         | Passed |
| Pm+Pl (SUS)         | 22.00     | 177.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 24.00     | 353.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.



**Input Echo, Tubesheet Number 1, Description: U-Tubesheet**

**Shell Data:**

**Main Shell Description: Main shell**

|                                         |          |             |
|-----------------------------------------|----------|-------------|
| Shell Maximum Design Pressure           | Psd,max  | 25.51 bars  |
| Shell Maximum Operating Pressure        | Psox,max | 15.32 bars  |
| Shell Minimum Operating Pressure        | Psox,min | 0.00 bars   |
| Shell Thickness                         | ts       | 12.0000 mm. |
| Shell Internal Corrosion Allowance      | cas      | 3.1750 mm.  |
| Shell External Corrosion Allowance      | casext   | 0.0000 mm.  |
| Inside Diameter of Shell                | Ds       | 490.000 mm. |
| Shell Temperature for Internal Pressure | Ts       | 126.67 °C   |
| Shell Material                          |          | SA-516 70   |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                       |    |                           |
|---------------------------------------|----|---------------------------|
| Shell Material UNS Number             |    | K02700                    |
| Shell Allowable Stress at Temperature | Ss | 138.00 N./mm <sup>2</sup> |
| Shell Allowable Stress at Ambient     |    | 138.00 N./mm <sup>2</sup> |

**Channel Data:**

**Channel Description: CH.SHELL**

|                                      |          |             |
|--------------------------------------|----------|-------------|
| Channel Type:                        |          | Cylinder    |
| Channel Maximum Design Pressure      | Ptd,max  | 5.52 bars   |
| Channel Maximum Operating Pressure   | Ptox,max | 5.00 bars   |
| Channel Minimum Operating Pressure   | Ptox,min | 0.00 bars   |
| Channel Thickness                    | tc       | 12.0000 mm. |
| Channel Internal Corrosion Allowance | cac      | 3.1750 mm.  |
| Channel External Corrosion Allowance | cacext   | 0.0000 mm.  |
| Inside Diameter of Channel           | Dc       | 490.000 mm. |
| Channel Design Temperature           | TEMPC    | 76.67 °C    |
| Channel Flange Matl                  |          | SA-516 70   |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                         |    |                           |
|-----------------------------------------|----|---------------------------|
| Channel Material UNS Number             |    | K02700                    |
| Channel Allowable Stress at Temperature | Sc | 138.00 N./mm <sup>2</sup> |
| Channel Allowable Stress at Ambient     |    | 138.00 N./mm <sup>2</sup> |

**Tube Data:**

|                                                  |            |                           |
|--------------------------------------------------|------------|---------------------------|
| Number of Tube Holes                             | Nt         | 160                       |
| Tube Wall Thickness                              | et         | 2.7700 mm.                |
| Tube Outside Diameter                            | D          | 19.0500 mm.               |
| Total Straight Tube Length                       | Lt         | 6093.00 mm.               |
| Straight Tube Length (bet. inner tubsht faces) L |            | 6041.00 mm.               |
| Design Temperature of the Tubes                  |            | 126.67 °C                 |
| Tube Material                                    |            | SA-179                    |
| Tube Material UNS Number                         |            | K01200                    |
| Is this a Welded Tube                            |            | No                        |
| Tube Material Specification used                 | Smls. tube |                           |
| Tube Allowable Stress at Temperature             |            | 92.40 N./mm <sup>2</sup>  |
| Tube Allowable Stress At Ambient                 |            | 92.40 N./mm <sup>2</sup>  |
| Tube Yield Stress At design Temperature          | Syt        | 160.80 N./mm <sup>2</sup> |
| Tube Pitch (Center to Center Spacing)            | P          | 31.7500 mm.               |
| Tube Layout Pattern                              |            | Triangular                |
| Fillet Weld Leg                                  | af         | 2.7700 mm.                |
| Groove Weld Leg                                  | ag         | 2.7700 mm.                |
| Tube-Tubesheet Joint Weld Type                   |            | Full Strength             |
| Method for Tube-Tubesheet Jt. Allowable          |            | UW-20.3                   |
| Is the Tube Inset                                |            | No                        |
| Radius to Outermost Tube Hole Center             | ro         | 228.665 mm.               |
| Largest Center-to-Center Tube Distance           | Ul         | 35.0000 mm.               |
| Length of Expanded Portion of Tube               | ltx        | 0.0000 mm.                |
| Tube-side pass partition groove depth            | hg         | 6.0000 mm.                |

**Tubesheet Data:**

**Tubesheet TYPE: U-tube, Gasketed both Sides, Conf. d**

Tubesheet Design Metal Temperature T 126.67 °C  
 Tubesheet Material SA-266 4

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                           |      |                           |
|-------------------------------------------|------|---------------------------|
| Tubesheet Material UNS Number             |      | K03017                    |
| Tubesheet Allowable Stress at Temperature | S    | 138.00 N./mm <sup>2</sup> |
| Tubesheet Allowable Stress at Ambient     | Tt   | 138.00 N./mm <sup>2</sup> |
| Thickness of Tubesheet                    | h    | 52.0000 mm.               |
| Tubesheet Corr. Allowance (Shell side)    | Cats | 3.1470 mm.                |
| Tubesheet Corr. Allowance (Channel side)  | Catc | 3.1750 mm.                |
| Tubesheet Outside Diameter                | A    | 622.000 mm.               |

**Additional Data for Stepped Tubesheets:**

|                                                 |     |            |
|-------------------------------------------------|-----|------------|
| Is the Tubesheet Stepped?                       |     | Yes        |
| Is the Tubesheet Flat on Tubeside?              |     | No         |
| Step 1 Diameter on the Tubeside                 | dt1 | 521.00 mm. |
| Step 1 Depth on the Tubeside                    | ht1 | 6.00 mm.   |
| Step 2 Diameter on the Tubeside                 | dt2 | 487.00 mm. |
| Step 2 Depth on the Tubeside                    | ht2 | 6.00 mm.   |
| Is the Tubesheet Flat on Shellside?             |     | No         |
| Step 1 Diameter on the Shellside                | ds1 | 521.00 mm. |
| Step 1 Depth on the Shellside                   | hs1 | 6.00 mm.   |
| Step 2 Diameter on the Shellside                | ds2 | 487.00 mm. |
| Step 2 Depth on the Shellside                   | hs2 | 6.00 mm.   |
| Calculated Tubesheet Diameter per Div 2, 4.18.4 |     | 487.00 mm. |

Note: Tubesheet diameter is now: 487.000 mm. Div 2, 4.18.4.

|                                  |    |                       |
|----------------------------------|----|-----------------------|
| Dimension G for the Channel Side | Gc | 508.000 mm.           |
| Area of the Untubed Lanes        | AL | 253.0 cm <sup>2</sup> |

|                                      |                             |
|--------------------------------------|-----------------------------|
| Junction Stress Reduction option     | Perform Plastic Calculation |
| Perform Differential Pressure Design | No                          |
| Run Multiple Load Cases              | Yes                         |

**Additional Data for Gasketed Tubesheets:**

|                                            |                 |                          |
|--------------------------------------------|-----------------|--------------------------|
| Tubesheet Gasket on which Side             |                 | Both                     |
| Flange Outside Diameter                    | A               | 622.000 mm.              |
| Flange Inside Diameter                     | B               | 490.000 mm.              |
| Flange Face Outside Diameter               | Fod             | 524.000 mm.              |
| Flange Face Inside Diameter                | Fid             | 490.000 mm.              |
| Gasket Outside Diameter                    | Go              | 518.000 mm.              |
| Gasket Inside Diameter                     | Gi              | 498.000 mm.              |
| Small end Hub thk.                         | g0              | 12.0000 mm.              |
| Large end Hub thk.                         | g1              | 16.0000 mm.              |
| Gasket Factor,                             | m               | 3.75                     |
| Gasket Design Seating Stress               | y               | 62.06 N./mm <sup>2</sup> |
| Flange Facing Sketch                       | Code Sketch 1a  |                          |
| Gasket Thickness                           | tg              | 4.5000 mm.               |
| Full face Gasket Flange Option             | Program Selects |                          |
| Length of Partition Gasket                 | lp              | 498.000 mm.              |
| Width of Partition Gasket                  | wp              | 10.0000 mm.              |
| Partition Gasket Factor,                   | mPart           | 3.7500                   |
| Partition Gasket Design Seating Stress     | yPart           | 52.40 N./mm <sup>2</sup> |
| Partition Gasket Facing Sketch             | Code Sketch 1a  |                          |
| Partition Gasket Column for Gasket Seating | Code Column II  |                          |

**Bolting Information:**

|                                       |                    |                           |
|---------------------------------------|--------------------|---------------------------|
| Diameter of Bolt Circle               | C                  | 580.000 mm.               |
| Nominal Bolt Diameter                 | dB                 | 19.0500 mm.               |
| Thread Series                         | TEMA Thread Series |                           |
| Number of Bolts                       | n                  | 24                        |
| Bolt Material                         |                    | SA-193 B7                 |
| Bolt Allowable Stress At Temperature  | Sb                 | 172.00 N./mm <sup>2</sup> |
| Bolt Allowable Stress At Ambient      | Sa                 | 172.00 N./mm <sup>2</sup> |
| Weld between Flange and Shell/Channel |                    | 0.0000 mm.                |

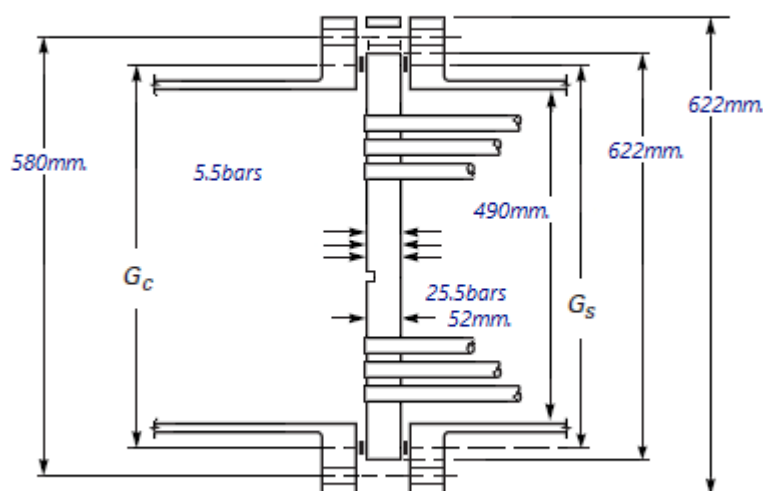


Alternate Flange Operating Bolt Load, Wm1 73293.32 Kg  
 Alternate Flange Seating Bolt Load, Wm2 63798.16 Kg  
 Alternate Flange Design Bolt Load, W 82012.02 Kg

Note: The full bolt load Sa\*Ab, was used for all load cases.

Tubesheet Integral with None  
 Tubesheet Extended as Flange Yes  
 Thickness of Extended Portion of Tubesheet Tf 32.0000 mm.  
 Is Bolt Load Transferred to the Tubesheet Yes

Tubesheet Gasketed w/Shell and Channel



Configuration d:

## ASME TubeSheet Results per Part UHX/4.18, 2025

### Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

|                                     |          |                  |
|-------------------------------------|----------|------------------|
| Elastic Mod. at Design Temperature  | 126.7 °C | 0.19650E+09 KPa. |
| Elastic Mod. at Ambient Temperature | 21.1 °C  | 0.20270E+09 KPa. |

Channel - TM-1 Carbon Steels with C<= 0.3%

|                                     |         |                  |
|-------------------------------------|---------|------------------|
| Elastic Mod. at Design Temperature  | 76.7 °C | 0.19952E+09 KPa. |
| Elastic Mod. at Ambient Temperature | 21.1 °C | 0.20270E+09 KPa. |

Tubes - TM-1 Carbon Steels with C<= 0.3%

|                                      |          |                  |
|--------------------------------------|----------|------------------|
| Elastic Mod. at Tubsht. Design Temp. | 126.7 °C | 0.19650E+09 KPa. |
| Elastic Mod. at Ambient Temperature  | 21.1 °C  | 0.20270E+09 KPa. |

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

|                                     |          |                  |
|-------------------------------------|----------|------------------|
| Elastic Mod. at Design Temperature  | 126.7 °C | 0.19650E+09 KPa. |
| Elastic Mod. at Ambient Temperature | 21.1 °C  | 0.20270E+09 KPa. |

### Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:  
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= (5.52 \cdot (19.05/2 - 0)) / (92.4 \cdot 1.0 + 0.4 \cdot 5.52)$   
 $= 0.0567 + 0.0000 = 0.0567 \text{ mm.}$



**Tube Required Thickness under External Pressure (Shellside pressure) :**

External Pressure Chart CS-1 at 126.67 °C  
Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

| TCA    | ODCA  | SLEN    | D/T  | L/D     | Factor A  | B     |
|--------|-------|---------|------|---------|-----------|-------|
| 2.7700 | 19.05 | 6269.67 | 6.88 | 50.0000 | 0.0232575 | 95.84 |

EMAWP =  $(2.167/(D/T)-0.0833)*B = 222.1421$  bars

Results for Req'd Thickness for Ext. Pressure (Tca):

| TCA    | ODCA  | SLEN    | D/T   | L/D     | Factor A  | B     |
|--------|-------|---------|-------|---------|-----------|-------|
| 0.5129 | 19.05 | 6269.67 | 37.14 | 50.0000 | 0.0007975 | 71.07 |

EMAWP =  $(4*B)/(3*(D/T)) = (4 * 71.0702 )/( 3 * 37.14 ) = 25.5128$  bars

**Summary of Tube Required Thickness Results:**

|                                                   |             |
|---------------------------------------------------|-------------|
| Total Required Thickness including Corrosion all. | 0.5129 mm.  |
| Allowable Internal Pressure at Corroded thickness | 304.06 bars |
| Required Internal Design Pressure                 | 5.52 bars   |
| Allowable External Pressure at Corroded thickness | 222.14 bars |
| Required External Design Pressure                 | 25.51 bars  |
| Required Thickness due to Shell Side pressure     | 0.5129 mm.  |

**Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)**

**Intermediate Calculations For Tubesheets Extended As Flanges:**

**ASME Code, Section VIII Division 1, 2025**

|                           |                     |             |
|---------------------------|---------------------|-------------|
| Gasket Contact Width,     | $N = (Goc-Gic) / 2$ | 10.000 mm.  |
| Basic Gasket Width,       | $b0 = N / 2$        | 5.000 mm.   |
| Effective Gasket Width,   | $b = b0$            | 5.000 mm.   |
| Gasket Reaction Diameter, | $G = (Go+Gi) / 2$   | 508.000 mm. |

**Results for ASME U-tube Tubesheet Calculations for Configuration d:**

**Per Edition 2025, Original Thickness :**

**ASME Section VIII-2 Part 4.18.7.4 Step 1:**

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot ro + dt \\ &= 2 \cdot 228.665 + 19.05 \\ &= 476.380 \text{ mm.} \end{aligned}$$

Basic ligament efficiency for shear [ $\mu$ ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (31.75 - 19.05) / 31.75 \\ &= 0.400 \end{aligned}$$

**ASME Section VIII-2 Part 4.18.7.4 Step 2:**

Ratio [ps]:

$$\begin{aligned} &= Gs / Do \text{ (Configurations d, e, f)} \\ &= 508.0 / 476.38 \\ &= 1.0664 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= Gc / Do \text{ (Configurations d)} \\ &= 508.0 / 476.38 \\ &= 1.0664 \end{aligned}$$

**ASME Section VIII-2 Part 4.18.7.4 Step 3:**

Ratio [ $\rho$ ]:

$$\begin{aligned} &= ltx / h \\ &= 0 / 52.0 \end{aligned}$$



= 0.000 ( must be  $0 \leq \rho \leq 1$  )

Effective tube hole Diameter [d\*]:

$$= \max( dt - 2t( Et/E ) \cdot ( St/S ) \cdot ( \rho ); dt - 2t )$$

$$= \max( 19.05 - 2 \cdot 2.77 ( 0.19650E+09 / 0.19650E+09 ) \cdot$$

$$( 92/137 ) \cdot ( 0 ); 19.05 - 2 \cdot 2.77 )$$

$$= 19.0500 \text{ mm.}$$

Effective tube pitch [p\*]:

$$= p / \sqrt{ ( 1 - 4 \cdot \min( A; 4 \cdot Do \cdot p ) / ( \pi \cdot Do^2 ) ) }$$

$$= 31.75 / \sqrt{ ( 1 - 4 \cdot \min( 252.98 ; 4 \cdot 476.38 \cdot 31.75 ) / ( \pi \cdot 476.38^2 ) ) }$$

$$= 34.2755 \text{ mm.}$$

Effective ligament efficiency for bending [ $\mu^*$ ]:

$$= ( p^* - d^* ) / p^*$$

$$= ( 34.2755 - 19.05 ) / 34.2755$$

$$= 0.4442$$

$h/p = 1.637795$  ;  $\mu^* = 0.444209$  ;  $E^* = 95607992 \text{ KPa.}$   
 $E^*/E = 0.486563$  ;  $v^* = 0.297378$

**Skip Step 4 for Configuration d :**

**ASME Section VIII-2 Part 4.18.7.4 Step 5:**

Diameter ratio [K]:

$$= A / Do$$

$$= 487.0 / 476.38$$

$$= 1.0223$$

Coefficient [F]:

$$= ( 1 - v^* ) / E^* \cdot ( E \cdot \ln(K) )$$

$$= ( 1 - 0.3 ) / 95607992 \cdot ( 0.19650E+09 \cdot \ln(1.02) )$$

$$= 0.0318$$

**ASME Section VIII-2 Part 4.18.7.4 Step 6:**

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= Do^2/16 \cdot [ (ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt ]$$

$$= 476.38^2/16 \cdot [ (1.066 - 1) \cdot (1.066^2 + 1) \cdot 25.511 -$$

$$(1.066 - 1) \cdot (1.066^2 + 1) \cdot 5.516 ]$$

$$= 40230.3086 \text{ bars} \cdot \text{mm.}^2$$

Moment acting on unperforated tubesheet rim [M\*]:

$$= Mts + W^* \cdot ( Gc - Gs ) / ( 2 \cdot \pi \cdot Do )$$

$$= 40230.3 + 82012.0 \cdot ( 508.0 - 508.0 ) / ( 2 \cdot \pi \cdot 476.38 )$$

$$= 40230.3086 \text{ bars} \cdot \text{mm.}^2$$

*Note: W\* is the maximum of the bolt loads between the shell and channel sides.*

**ASME Section VIII-2 Part 4.18.7.4 Step 7:**

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ( (M^*) - Do^2/32 \cdot F( Ps - Pt ) ) / ( 1 + F )$$

$$= ( (40230.31) - 476.38^2/32 \cdot 0.032 ( 25.51 - 5.52 ) ) / ( 1 + 0.03 )$$

$$= 34613.4648 \text{ bars} \cdot \text{mm.}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot ( 3 + v^* ) \cdot ( Ps - Pt )$$

$$= 34613.46 + 476.38^2/64 \cdot ( 3 + 0.297 ) \cdot ( 25.51 - 5.52 )$$

$$= 268398.7188 \text{ bars} \cdot \text{mm.}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max( | Mp | ; | Mo | )$$

$$= \max( | 34613.465 | ; | 268398.719 | )$$

$$= 268398.7188 \text{ bars} \cdot \text{mm.}^2$$

**ASME Section VIII-2 Part 4.18.7.4 Step 8:**

Tubesheet Bending Stress at original thickness:



$$\begin{aligned} &= 6 \cdot M / ( \mu \cdot (h - h_g')^2 ) \\ &= 6 \cdot 268398.719 / ( 0.4442 ( 52.0 - 6.0 )^2 ) \\ &= 171.3380 \text{ N./mm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [ $\sigma_{All}$ ]:

$$\begin{aligned} &= 2 \cdot S \\ &= 2 \cdot 138.0 \\ &= 275.99 \text{ N./mm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [ $\sigma$ ]:

$$\begin{aligned} &= 6 \cdot M / ( \mu \cdot (h - h_g')^2 ) \\ &= 6 \cdot 268123.0 / ( 0.4442 ( 42.2308 - 6.0 )^2 ) \\ &= 275.9093 \text{ N./mm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [ $H_{reqB}$ ]:

$$\begin{aligned} &= h + C_{ats} + C_{atc} \\ &= 42.2308 + 0 + 0 \\ &= 42.2308 \text{ mm.} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [ $H_{req}$ ]:

$$\begin{aligned} &= \max( H_{reqB}; H_{reqS} ) \\ &= \max( 42.2308 ; 5.3929 ) \\ &= 42.2308 \text{ mm.} \end{aligned}$$

#### ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [ $|P_s - P_t|$ ]:

$$\begin{aligned} &= |P_s - P_t| \\ &= |25.51 - 5.52| \\ &= 19.995 \text{ bars} \end{aligned}$$

Shear Stress check [ $\tau_{limit}$ ]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / D_o \cdot \min( 0.8 \cdot S; 0.533 \cdot S_y ) \\ &= 4 \cdot 0.4 \cdot 52.0 / 476.38 \cdot \min( 0.8 \cdot 137.997 ; 0.533 \cdot 222.731 ) \\ &= 19.281 \text{ N./mm}^2 \end{aligned}$$

$|P_s - P_t|$  is  $\leq$  the Shear Stress limit ( $\tau_{limit}$ ). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [ $H_{reqS}$ ]:

$$\begin{aligned} &= 1 / ( 4 \cdot \mu ) \cdot ( D_o / \min( 0.8 \cdot S; 0.533 \cdot S_y ) ) \cdot |P_s - P_t| + C_{ats} + C_{atc} \\ &= 1 / ( 4 \cdot 0.4 ) \cdot ( 476.38 / 110.398 ) \cdot |25.51 - 5.52| + 0 \\ &= 5.3929 \text{ mm.} \end{aligned}$$

Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [ $A_t$ ]:

$$\begin{aligned} &= \pi ( d_o - t ) t \\ &= \pi ( 19.05 - 2.77 ) 2.77 \\ &= 1.417 \text{ cm}^2 \end{aligned}$$

#### Tube Weld Size Results per UW-20.3:

Tube Strength [ $F_t$ ]:

$$\begin{aligned} &= \pi \cdot t ( d_o - t ) S \\ &= \pi \cdot 2.77 ( 19.05 - 2.77 ) 92.398 \\ &= 1334.810 \text{ Kg} \end{aligned}$$

Groove Weld Strength [ $F_g$ ]:

$$\begin{aligned} &= 0.85 \cdot \pi \cdot a_g ( d_o + 0.67 \cdot a_g ) S_w \text{ (but not } > F_t ) \\ &= 0.85 \cdot \pi \cdot 2.77 ( 19.05 + 0.67 \cdot 2.77 ) 92.4 \\ &= 1334.8102 \text{ Kg} \end{aligned}$$

Weld Strength Factor [ $f_w$ ]:

$$\begin{aligned} &= S / S_w \text{ where } S_w = \min( S \text{ and } S_t ) \\ &= 92.398 / 92.398 \\ &= 1.000 \end{aligned}$$

Design Strength Ratio [ $f_d$ ]:

$$= 1 \text{ for full strength welds}$$



**Fillet Weld Strength [Ff]:**

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 2.77 (19.05 + 0.67 \cdot 2.77) 92.4$$

$$= 942.7503 \text{ Kgf}$$

**Minimum Length of Weld [ar]:**

$$= 2 \left( \sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd)} - 0.75 \cdot do \right)$$

$$= 2 \left( \sqrt{((0.75 \cdot 19.05)^2 + 1.07 \cdot 2.77 \cdot (19.05 - 2.77) 1.0 \cdot 1.0)} - 0.75 \cdot 19.05 \right)$$

$$= 3.198 \text{ mm.}$$

Minimum Required Fillet Weld Leg [afr]: 1.6620 mm.

Minimum Required Groove Weld Leg [agr]: 1.6620 mm.

Tube-Tubesheet Jt allowable, 1334.81 is >= tube strength 1334.81 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

**Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):**

| Stress Description     | Actual   | Allowable                | Pass/Fail |
|------------------------|----------|--------------------------|-----------|
| Tubesheet bend. stress | 171.3 <= | 276.0 N./mm <sup>2</sup> | Ok        |
| Tubesheet shear stress | 11.4 <=  | 110.4 N./mm <sup>2</sup> | Ok        |

**Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):**

| Thickness (mm.)                  | Required | Actual | P/F |
|----------------------------------|----------|--------|-----|
| Tubesheet Thickness :            | 42.231   | 52.000 | Ok  |
| Tube-Tubesheet Fillet Weld Leg : | 1.662    | 2.770  | Ok  |
| Tube-Tubesheet Groove Weld Leg : | 1.662    | 2.770  | Ok  |

**U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025**

**Results for 6 Load Cases:**

| Case# | --Reqd. Thk. + CA<br>Tbsht Extnsn | ----  | Tubesheet<br>Bend Allwd | Stresses<br>Shear Allwd | Case<br>Type | Pass/<br>Fail |
|-------|-----------------------------------|-------|-------------------------|-------------------------|--------------|---------------|
| D1uc  | 24.987 ...                        | 47    | 276                     | 3 110                   | Ps+Pt D1     | Ok            |
| D2uc  | 47.023 28.013                     | 219   | 276                     | 15 110                  | Ps+Pt D2     | Ok            |
| D3uc  | 42.231 ...                        | 171   | 276                     | 11 110                  | Ps+Pt D3     | Ok            |
| D1c   | 28.109 ...                        | 54    | 276                     | 4 110                   | Ps+Pt-c D1   | Ok            |
| D2c   | 50.076 28.013                     | 252   | 276                     | 17 110                  | Ps+Pt-c D2   | Ok            |
| D3c   | 45.399 ...                        | 197   | 276                     | 13 110                  | Ps+Pt-c D3   | Ok            |
| Max:  | 50.0765 28.013 mm.                | 0.912 | 0.151 (Str. Ratio)      |                         |              |               |

**Load Case Definitions:**

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures  
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the  
 Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Note:

Because there was no net pressure differential across the tubesheet due to



the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

#### Summary of Thickness Comparisons for 6 Load Cases:

| Thickness (mm.)                         | Required | Actual | P/F |
|-----------------------------------------|----------|--------|-----|
| Tubesheet thickness :                   | 50.076   | 52.000 | Ok  |
| Tubesheet thickness flanged extension : | 28.013   | 32.000 | Ok  |
| Tube thickness :                        | 0.513    | 2.770  | Ok  |
| Tube-tubesheet fillet weld leg :        | 1.662    | 2.770  | Ok  |
| Tube-tubesheet groove weld leg :        | 1.662    | 2.770  | Ok  |

Note: This is a full strength Tube to Tubesheet Joint.

#### Tubesheet MAWP used to Compute Hydrotest Pressure:

| Stress / Force Condition   | Tubeside MAWP | Ø shellside Stress Rat. | Shellside MAWP | Ø tubeside Stress Rat. |
|----------------------------|---------------|-------------------------|----------------|------------------------|
| Tubesheet Bending Stress   | 27.970        | 1.000                   | 27.970         | 1.000                  |
| Tubesheet Shear Stress     | 169.359       | 1.000                   | 169.359        | 1.000                  |
| Tube Pressure Stress       | 304.060       | 1.000                   | 222.141        | 1.000                  |
| Tubesheet Extension Stress | No Calc       | No Calc                 | 28.598         | ...                    |
| Minimum MAWP               | 27.970        |                         | 27.970         |                        |

#### Tubesheet MAPnc used to Compute Hydrotest Pressure:

| Stress / Force Condition   | Tubeside MAPnc | Ø shellside Stress Rat. | Shellside MAPnc | Ø tubeside Stress Rat. |
|----------------------------|----------------|-------------------------|-----------------|------------------------|
| Tubesheet Bending Stress   | 32.207         | 1.000                   | 32.207          | 1.000                  |
| Tubesheet Shear Stress     | 192.798        | 1.000                   | 192.799         | 1.000                  |
| Tube Pressure Stress       | 304.060        | 1.000                   | 222.141         | 1.000                  |
| Tubesheet Extension Stress | No Calc        | No Calc                 | 28.598          | ...                    |
| Minimum MAPnc              | 32.207         |                         | 28.598          |                        |

(\*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

#### Tubesheet MDMT Calculations:

##### Determine the governing MDMT considering the governing condition:

Governing thickness on the shell side per figure UCS-66.3 (c):  
 = tubesheet thickness/4  
 = 52.0/4  
 = 13.000 mm.

Thickness Ratio = 0.892, Temperature Reduction per Fig. UCS 66.1 = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -21 °C  
 Min Metal Temp. at Required thickness (UCS 66.1) -27 °C  
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:  
 = max( pt/Tubeside MAPnc, ps/Shellside MAPnc ), must be ≤ 1  
 = max( 5.52/32.21, 25.51/28.6 )  
 = 0.892

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